Super ICT – Question Index

By Bit to exabyte

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Part A: Previous All questions (Chapter by Chapter)[1– 54]

No.	Question / topic	Pages
1	Previous All questions (Chapter By Chapter)	[1– 54]

Part B: Solved questions (1-120)

The solutions section holds 120 numbered questions, then descriptive, math, aptitude and BCS parts.

Qs. No.	Question / topic	
01	OSI model: layer functions and protocols; OSI vs TCP/IP	
02	Average Memory Access Time (AMAT) with cache hit rate	
03	SQL: highest salary, 2nd highest salary, duplicate rows	
04	RAID levels and how capacity is calculated	
05	ACID properties in DBMS	
06	C program: sum of digits	
07	Deadlock, four necessary conditions, prevention vs avoidance	
08	Banker's algorithm: safe state and safe sequence	
09	K-map simplification (4-variable and 3-variable)	
10	DoS vs DDoS attack	
11	Subnetting 192.168.10.0/24 into four subnets	
12	Private vs public IP; why NAT is needed	
13	C program: factorial using loop and recursion	

14	CPU functions; ALU vs CU vs registers; BIOS vs UEFI, firmware, cache vs RAM, DRAM vs SRAM notes	
15	Process vs program; process state diagram	
16	Normalization: 1NF, 2NF, 3NF, BCNF; functional, partial, transitive dependency	
17	DNS name resolution process	
18	Hub vs switch vs router	
19	Min heap construction (35, 12, 43, 8, 51, 20, 5); min vs max heap	
20	192.168.1.0/27: subnets, hosts, first-subnet addresses	
21	Polymorphism: method overloading vs overriding	
22	Waterfall vs Agile model	
23	C program: reverse a string without library functions	
24	2-D array address in row-major and column-major order	
25	Process vs program and process state diagram (repeat of 15)	
26	Cloud computing; IaaS, PaaS, SaaS	
27	SQL: Book and Author join with country and year filter	
28	C program: replace each alphabet with the next alphabet	
29	VLSM for 192.168.10.0/22 with five departments	
30	Truth table: tautology check	
31	Private vs public IP and NAT (repeat of 12)	
32	Linked list vs array; types of linked list; insertion pseudocode	
33	DBMS keys (primary, candidate, super, foreign, composite); indexing disadvantages	
34	Synchronous vs asynchronous counters; MOD-6 counter	
35	Functional vs non-functional testing; black box vs white box	
36	P, NP, NP-Complete, NP-Hard	
37	BFS and DFS traversal with example graph	

38	C program: array operations (max, min, sum, average, search, reverse)	
39	Network topologies compared; which is best	
40	Fastest sorting algorithm; Quick Sort trace and complexity	
41	BST construction and traversals; tree from traversals	
42	Waterfall vs Agile vs Spiral model	
43	C program: GCD using functions	
44	DFA vs NFA; regular vs context-free language	
45	ER diagram: library management system	
46	Microcontroller vs microprocessor	
47	Triggers in DBMS with SQL example	
48	ARP working procedure	
49	Call by value vs call by reference; swap with pointers	
50	API; REST vs SOAP; API request flow to database	
51	Structure vs union and memory management	
52	Inheritance; multiple inheritance in Java via interfaces	
53	Binary tree from inorder with postorder or preorder	
54	Universal gates; XOR using NAND and NOR	
55	C program: palindrome string, or matrix addition	
56	Linux commands: touch, mv, mkdir, cd, grep, chmod, ps, top, kill, ping and more	
57	Commercial bank network diagram and firewall protection	
58	Zero-day vulnerability vs exploit; bank's immediate response	
59	Max heap insertion of 88, 100, 65	
60	Time complexity of nested loops, O(n^2 log n)	
61	HTTP request; GET vs POST; which is more secure	
62	HTTP status codes 1xx to 5xx; why 404 is a client error	

63	Page vs frame; paging and fragmentation; page and frame bit calculation	
64	Virtual memory; page fault sequence	
65	Confusion matrix (TP 90, TN 80, FP 20, FN 10)	
66	8-bit 2's complement: 120 + (-45), overflow check	
67	Wi-Fi, Bluetooth, WiMAX; world without them	
68	Why packet switching suits the Internet	
69	B+ tree of order 3; B-tree vs B+ tree	
70	Man-in-the-Middle attack and two countermeasures	
71	Header files and library functions; C uppercase to lowercase	
72	Complexity; time complexity of double loop	
73	Types of ROM; SRAM vs DRAM	
74	Memory management with 8 GB RAM, 24 GB virtual memory, 14 GB demand	
75	2 GB file over 500 Mbps: transmission, propagation delay, RTT	
76	Maximum memory with a 20-bit address bus	
77	Preemptive vs non-preemptive priority scheduling	
78	Short notes: process vs thread, paging vs segmentation, virtual vs physical memory, deadlock vs starvation, multiprogramming vs multitasking, context switching, kernel	
79	CPU scheduling: FCFS, SJF, preemptive priority with Gantt charts	
80	Instruction pipelining; 5-stage pipeline; speedup calculation; graph concept comparisons	
81	AI agent; narrow, general and super AI	
82	CSRF, XSS, SQL injection and prevention	
83	Database vs data warehouse vs data mining	
84	Logical and physical address bits (512 pages, 128 frames, 2 KB page)	

85	OOP bank account program in C++ and encapsulation	
86	Version control (Git): commit vs push	
87	Ambiguous grammar for id + id * id	
88	NFA to DFA; DFA for exactly three 1s; DFA for substring 110; flip-flops for 10 states	
89	Software testing types: unit, integration, system, acceptance, smoke, regression, stress, performance	
90	Feasibility study; TELOS model	
91	Spiral model as risk-driven; Agile and Prototype models	
92	ATM transaction architecture; ACID in ATM transactions	
93	Social engineering; CERT; Bangladesh Bank heist	
94	Digital signature generation and verification; CA and PKI	
95	Type casting; output-tracing problems	
96	Handshaking; TCP three-way handshake; flow vs congestion control	
97	Database relationships; use case diagram; ER diagram for e-commerce	
98	Dynamic programming vs greedy algorithms	
99	Knapsack with greedy and DP; Tower of Hanoi for 3 disks	
100	Search algorithm choice for 1 million employee records	
101	Enterprise bank network design using the OSI model	
102	8086 addressing modes	
103	8086 effective and physical address calculation	
104	RISC vs CISC	
105	C program: star pyramid pattern	
106	C program: sum of series 1 + 11 + 111 + ... up to N	
107	Classful IP addressing (Class A to E)	
108	CRC remainder calculation; even parity error detection	

109	Loopback IP; accessing 127.0.0.1:8080; special IP addresses	
110	Binary tree representation using an array	
111	JVM vs JRE vs JDK	
112	Important protocols and port numbers (MCQ)	
113	Latch vs flip-flop	
114	Symmetric vs asymmetric encryption; number of 128-bit keys	
115	Bipartite graph and its condition	
116	C increment and decrement expression output	
117	Hadoop; architecture; HDFS block and replica calculations	
118	A* algorithm (in Bangla)	
119	TCP sequence number problems (10 questions)	
120	Linux file permissions and chmod	

Part C: Descriptive, Math and Aptitude sections

After question 120 the file continues with topic chapters, probability and word-problem math, and aptitude exam tasks.

Descriptive part

Chapter	Contents	
CHAP- 01	DDL, DML, DQL, DCL, TCL with examples and a quick comparison	
SQL command categories SQL basic to advanced Second SQL practice set SELF JOIN Memory management	34 numbered queries with outputs: create and insert, WHERE, AND/OR, BETWEEN, IN, LIKE wildcards, ORDER BY, DISTINCT, LIMIT, aggregates, GROUP BY, HAVING, joins, subqueries, correlated subqueries, CTE, RANK, second-highest salary, execution order	
	Department-wise count and average, INNER and LEFT JOIN, above-average salary, CASE expression, highest average department	

Chapter	Contents	
	Employee-manager query, manager earning more, same-salary pairs, SQL query pattern table, 10 advanced practice problems	
CHAP- 02 Logical and physical address (Bangla)	Logical vs physical address, fragmentation, paging, page table, TLB, page fault, demand paging, page replacement, thrashing, segmentation	
	Address translation, address bits, paging numericals, internal fragmentation, effective access time with TLB; the memory management answer repeated on 530–531	
CHAP- 03 Strings in C	Declaration, input and output, string.h functions, manual strlen, copy, reverse, palindrome, frequency analysis	
CHAP- 04 HTTP and HTTPS	Requests and responses, methods, status codes, TLS handshake, certificates, HTTP/1.1 vs 2 vs 3, caching, proxies	

Math

No.	Question / topic	
01	Bayes theorem: defective product from Machine B (70% A, 30% B)	
02	Same problem in Bangla with a 10,000-unit count method	
03	Probability a randomly chosen bag belongs to a Deputy Director	
04	Bayes theorem: failed transaction from Server B	
Practice	17 questions: probability (newspapers, football and cricket, exam failures, balls and containers), speed and distance, rectangle dimensions, work and wages, pipes and cistern, simple vs compound interest, mixtures	

Aptitude test questions

Qs. No.	Question / topic	
01	C file handling: read students.txt, filter CGPA above 3.50 into new.txt, average and count	
02	SQL on Department and Employee: inserts, join, department average, above-average employees	
03	Java inheritance: BankAccount, SavingsAccount (5% interest), CurrentAccount (overdraft)	
04	MS Excel: IF, SUMIFS, COUNTIFS, AVERAGEIFS, VLOOKUP, HLOOKUP, XLOOKUP, UNIQUE, sort and filter	

Part C: BCS question sets

The 50th BCS paper is fully solved in the file; the 47th and 46th BCS papers list questions only.

BCS Computer Science Questions Collection

No.	Topics	
1	50th BCS Computer Science (solved)	
2	47th BCS Computer Science	
3	46th BCS Computer Science (Solved in 1-120)	

QUESTIONS BANK (PREVIOUS ALL YEARS)

Previous Questions

Computer Fundamentals and Computer Architecture

1. What is Computer Generation? Describe the characteristics of different generations of computers? *[18th NTRCA School 2024]*
2. What is OCR? Write down the difference between OCR and OMR. *[NPCNL JAM 2022]*
3. Name and define the components of a computer system. Mention two optical input devices. *[BPSC ADA 2022]*
4. What do understand by the resolution of computer screen? *[BPSC AP 2022]*
5. What are the components of a Micro-computer system? *[BPSC ADA 2022]*
6. Explain Impact Printer Non-impact printer. *[Bank SO(AME) 2021]*
7. Distinguish between OMR and MICR. *[BB AME 2016]*
8. Suppose you are entering data into computer but facing some problem with your monitor. You need to buy a new monitor. What factor should you consider in case of monitor? *[BB AME 2016]*
9. Write down the difference between Serial Port and Parallel Port. *[DESCO SAE 2023]*
10. Differentiate among Core Intel i3, i5 and i7 processor. *[Combined Bank SO(IT) 2023]*
11. Difference between SAS and SATA. Which one is best server? *[PBL HE 2023]*
12. "Pentium processor has a superscalar architecture." Explain the meaning of statement. *[MM AP 2017]*
13. What is the difference between System Software and Application Software? *[NPCNL JAM 2022, 45th BCS (General) ICT 2024, MM AP 2017, BB AME 2016]*
14. একটি কম্পিউটারের BIOS সিস্টেমের এর মূল ভূমিকা কী? *[45th BCS (General), ICT 2024]*
15. Explain BIOS in Server. How does affect booting configuration in Hardware maintenance. *[RAKUB ANSE 2023]*
16. Describe the booting process in windows system. *[PBL HE 2023]*
17. What is BIOS? *[EGCB SAE(ICT) 2023]*

18. What is platform independent software discuss with example? *[Combined Bank SO(IT) 2023]*
19. Software বলতে কী বোঝায়? উদাহরণসহ System Software and Application Software-এর সংক্ষিপ্ত বর্ণনা দিন? *[40th BCS 2021]*
20. What is the function of GPU? *[BPSC AP 2022]*
21. Typical মাইক্রোকম্পিউটারে কী কী বাস থাকে। একটি মাইক্রোপ্রসেসর এর সাথে RAM, ROM এবং I/O এর কানেকশন বাস এর মাধ্যমে দেখাও। *[BPSC SAE 2021]*
22. What is computer bus? What is the unit of measurement of speed of computer bus? Describe about various types of computer bus. *[BKKB AP 2017]*
23. **Write down the necessary components of a USB bus with block diagram.** ***[ICT Ministry AP 2017]***
24. MS-Excel এর IF Function ব্যবহার করে A1, B1, C1 থেকে ছোট সেল খুঁজে D1 এর মধ্যে রাখার সমীকরণ লিখ। *[PGCB SAE (CSE) 2021]*
25. **What is pipelining? How does pipeline implementation make faster instruction execution?** ***[38th BCS]***
26. How does an interrupt can be enabled? What are the steps that occur when an interrupt is processed? *[38th BCS]*

Previous Questions — C Programming

1. What is the difference of the following: getc (), getchar (), getch () and getche ()? [*45th BCS, ICT 2024*]
2. What is the difference between Null and Void? [*BCC AP 2023*]
3. What is the difference between sizeof c+1 and sizeof (c+1)? [*BPSC AP 2023*]
4. **Local variable এবং Global variable এর মধ্যে পার্থক্য লিখুন। [*17th NTRCA(CSE) 2023*]**
5. Write about the syntax of function. [*BARC Programmer 2023*]

6. **Distinguish between Compiler and Interpreter with example. [*BPSC ADA, AP 2022; 40th BCS, PGCB SAE (CSE) 2021; 40th BCS 2020; BCC Programmer 2019*]**
7. Formatted Input/output Statement কাকে বলে? Key-Board থেকে কিভাবে input নেয়া যায়? %d এর অর্থ কী? [*BPSC ANE 2020*]
8. What are the main objectives of structured programming? Write two advantages of structured programming. [*SPARRSO 2020*]
9. Difference between getch () and getche (). [*BBL AM (ICT) 2019*]
10. Write down the role of a linker. [*BUET M.Sc. May-2016*]
11. **How can we initialize 1-D array? How can we pass the array to a function in C? [*45th BCS, CSE 2024*]**
12. if.....else statement এর format লিখ। [*BPSC ANE 2020*]
13. What is looping? Why use looping? Give a proper example. [*BCPSC AP 2018*]
14. What is the basic difference between looping and branching? [*BCPSC AP 2018*]
15. **Write a simple program in C to swap two integer values using 'Call by value'. [*BPSC AP 2023*]**
16. **Differences between 'Call by Value' and 'Call by Reference'. [*BPSC AP, 17th NTRCA(ICT) 2023; BPSC ADA, AP 2022; PSB AP 2018*]**
17. উদাহরণসহ recursion ব্যাখ্যা করুন ও Parameter Passing ব্যাখ্যা করুন। [*17th NTRCA(ICT) 2023*]
18. Difference between Array and Structure data type. [*BPSC AP 2022; BPSC AE 2022; BPSC AP, BOF AE(CSE) 2021*]
19. What do you mean by recursion? Calculate factorial function using recursion with C programming code. [*45th BCS ICT 2024; Milk Vita AM(CSE) 2023; BPSC AP 2022; BPSC AP, ANE 2020*]
20. Write a function to find the smallest element from an array. [*Combined Bank AP 2023*]
21. **What is the difference between array and pointer? [*BPSC AME 2021*]**
22. **Difference between ++i and i++.** [*17th NTRCA(ICT) 2023; NU AP 2020*]
23. What is file pointer in C? [*BBL AM (ICT) 2019*]

24. The maximum subarray is the task of finding a contiguous subarray with the largest sum within a given one dimensional array of numbers. Suppose the array is:

A: [-2, 1, -3, -1, 2, 1, -5, 4] [*BPDB AE(CSE) 2023*]

25. Find the output of following program: [*BGDCL AE (CSE) 2021*]

```
int i, j, p, sum;
sum = 0;
for(i = -1, p = 0; i <= 10; i = i + 2) {
    p = i * i;
    if(i > 5)
        break;
    sum = sum + (i + p);
    printf("i=%d, p=%d, sum=%d\n", i, p, sum);
}
printf("\n Outsite Loop, i=%d, p=%d, sum=%d\n", i, p, sum);
return 0;
```

26. Find the output of following program: [*PSB AP 2018*]

```
int main() {
    int n = 10;
    for(;;)
        printf("%d", n);
    return 0;
}
Or
int main() {
    int i = 2, j = 2;
    while(i ? --i : j++)
```

```
        printf("%d", i);
    return 0;
}
```

27. Find the output of following program: [*PSB Programmer 2018*]

```
int main() {
    char str[120] = "Digital Bangladesh";
    int n;
    n = strlen(str);
    str[4] = '\0';
    printf("%s", str);
    return 0;
}
Or
int main() {
    int i;
    for(i = 0; i < 5; i++) {
        if(i == 3)
            continue;
        printf("%d\n", i);
    }
    return 0;
}
```

28. After compilation and execution, what will be output in the following code: [*DPDC JAM (ICT) 2020*]

29. What will be the outcome of the following conditional statement if the value of variable s is 10?
30. What would happen to X in this expression: x+=15; (assuming the value of X is 5)
31. Find the output from the following: take input and looks the output: [*BGDCL, AM(CSE)2024*]

```
#include <stdio.h>
int reverseNumber(int num) {
    int reverse = 0;
    while (num != 0) {
        reverse = reverse * 10 + num % 10;
        num /= 10;
    }
    return reverse;
}
int main() {
    int input, output;
    printf("Enter a number: ");
    scanf("%d", &input);
    output = reverseNumber(input);
    printf("Input: %d\n", input);
    printf("Output: %d\n", output);
    return 0;
}
```

32. Write a C program to print Fibonacci series without using recursion and using recursion. [*17th NTRCA(CSE) 2023; BB AP 2016*]
33. Write a C program to find nth Fibonacci number using Recursion. [*KGDCL AE(CSE) 2021*]

34. Given two integers A and B as input write a program to compute the least common multiple of A and B. [*BB AP 2023*]
35. Write a c program to reverse given number.
36. Write a program to reverse string without using library function. [*BREB AP 2023; Combined 2 bank, ADA 2022*]
37. **Write a program in any language to find the sum of rows and columns of a m x n matrix, where m and n is taken input from the user. Give the output in the following format:** [*Combined Bank, SO(IT) 2024*]
38. Write a program in any language that takes two matrices A and B as inputs ensure your code handles matrices of different dimensions-

    A) Find matrices C that is multiplication A and B.

    B) Find average in A and B.

    C) Max from matrices C [*Combined Bank SO(IT) 2023*]
39. Write a C Program Find sum of the series: 1+2+4+7+11+.....+N [*Combined Bank, AP 2024*]
40. Write a program from the following series: $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots\ldots$ [*SB PCL, ADBA 2024*]
41. **Write a program in C++ to calculate the sum of the series: 1+(1+2)+(1+2+3)+....+(1+2+....+n).** [*MM AP 2017*]
42. Write a C program to find the length of a String without using function strlen(). [*BPSC AME 2021*]

Previous Questions — Object oriented Programming

1. What is Polymorphism? Discuss about different types of Polymorphism with example? [*Combined Bank, AP 2024*]

2. **Write the differences between structured programming and object-oriented programming.** [*45th BCS CSE 2024*, *Bank SO(AME) 2021*]
3. Write down the advantages of OOP over traditional structured programming language. [*Combined Bank Officer(IT) 2023*]
4. What is the diamond problem of multiple Inheritance in C++? [*BPSC AP 2023*]
5. কী কী ধারণার উপর ভিত্তি করে OOP প্রতিষ্ঠিত? ধারণাগুলো ব্যাখ্যা করুন। [*17th NTRCA(CSE) 2023*]
6. উদাহরণসহ Class এবং Object এর মধ্যে পার্থক্য ব্যাখ্যা করুন। [*17th NTRCA(CSE), (ICT) 2023*]
7. **Inheritance কী? উদাহরণসহ ব্যাখ্যা করুন। [*17th NTRCA(CSE) 2023*]**
8. **Write down the Principle of OOP. What is Polymorphism? Write the name of 3 OOP language.** [*DESCO SAE 2023*, *40th BCS 2021*]
9. How many classes can be used in Hybrid Inheritance? [*BCC AP 2023*]
10. What is virtual function with example? [*BITAC AP 2023*]
11. Define Method Overloading and Method Overriding with examples. [*BPSC AP, 17th NTRCA(ICT) 2023*, *Bank SO(AME)*, *RAKUB Programmer*, *BPSC AME 2021*, *NU AP 2020*]
12. What is Polymorphism? Java language এর আলোকে ব্যাখ্যা কর। [*BTCL, JAM 2022*]
13. What are the various Access Specification in C++? Explain their purpose with are example. [*BPSC, ADA 2022*]
14. Object Oriented Programming এর যেকোনো দুটি বৈশিষ্ট্য উদাহরণসহ ব্যাখ্যা করুন। [*BPSC AP 2021*]
15. Object Oriented Programming এ Static binding and Dynamic binding কি? ব্যাখ্যা করুন। [*BPSC AP 2021*]
16. Write the definition of Inheritance, Polymorphism with coding example. [*Combined Bank, AP 2021*]
17. Procedural Oriented ও Object Oriented Programming Languages এর মধ্যে পার্থক্য কি? উভয় Language এর ২টি করে উদাহরণ দিন। [*BPSC AP 2021*]
18. **Inheritance is one of important issues for any object oriented programming language. The main advantage of Inheritance is the ability to reuse the code. Explain in brief different types of Inheritance.** [*Combined Bank, Officer(IT) 2020*]
19. Briefly Describe Abstraction, Encapsulation. [*BCC Programmer 2019*]

20. What do you mean by Polymorphism and Inheritance in object oriented programming (OOP)? Give appropriate example. [*PBL SO 2018*]
21. Pointer কী? Pointer ব্যবহারের সুবিধাসমূহ লিখুন। [*17th NTRCA(CSE) 2023*]
22. What is constructor function? Write the properties of it. [*WZPGCL AE(CSE) 2023*]
23. পয়েন্টার কী? Malloc() এবং Calloc() এর মধ্যে পার্থক্য লিখুন। [*17th NTRCA(ICT) 2023*]
24. What is the constructor invoked in OOP? [*BPSC, AP 2022*]
25. Why are constructor and destructor functions used in object oriented programming? Give examples of each function in C++ or java language. [*BPSC SAE 2021*]
26. **What is main difference Destructor and constructor with example?** [*PSB Programmer 2018*]
27. Write the difference between throw and throws using Exception handling? [*PBL SO 2018*]
28. **What is Multithreading programming? Why Multithreading used in programming?** [*Combined Bank, AP 2024*]
29. **Write down the difference between Multitasking and Multiprocessing.** [*DESCO SAE 2023*]
30. How does dynamic programming relate with divide and conquer approach? [*BPSC AP 2023*]
31. Write down the difference between Divide and Conquer and Dynamic Programming. [*WZPGCL AE(CSE) 2023*]
32. What syntax is used for calling static methods in class? [*BCC AP 2023*]
33. What is the output of code snippet? [*BCIC, AP 2022*]
34. Both the algorithm the Divide and Conquer and Dynamic Programming solve a problem by breaking it into smaller problem instances and by solving them. What are the difference between their two techniques? [*BCC AP 2021*]
35. Object oriented programming- এর প্রেক্ষিতে function overloading বলতে কী বোঝায়? Function overloading এর একটি উদাহরণ দিন। [*40th BCS 2020*]
36. What is the difference among String, String Buffer and String Builder? [*DBBL PO 2017*]

37. Suppose you've a method name "totalAmount" and there three properties (transactionName, transactionType, amount). Write down the full code using JAVA where totalAmount method return total balance after debit or credited. [*PBL SO 2018*]
38. Write a program using any programming language that reads five numbers from keyboard and display the smaller, larger and average of those numbers. [*BDB SO (IT) 2017*]
39. Write a java program based on following conditions: There is one Circle class and one Cylinder Class. Circle class has an area function. Cylinder class has to calculate the circle area at the base and height of its own (Cylinder area = Circle area*height). Use inheritance to calculate Volumn of Cylinder. [*DBBL PO 2017*]
40. Write down a method to find out how many times an object has been instantiated in Java? [*BUET M. Sc. May 2016*]
41. Define types of processor for the following language: Python, C, C++, Java. [*BUET M. Sc. May 2016*]

Previous Questions

Cyber Security And Cloud Computing
Big Data, Data Mining

1. What is Cyber security? [*Combined Bank AME/AHE, SO(IT) 2023*]
2. What is CIA Triad? [*Teletalk AM(IT), EGCB SAE (ICT) 2023*]
3. What are the differences among Virus, Worms and Trojan horse? [*45th BCS, CSE 2024*]
4. Explain XSS and CSRF (how do you prevent these attacks). [*Titas Gas AE(CSE) 2024*]
5. Let you procure a microfinance application and host it in your office's data center. What kind of cyber-security threats should you be aware of and what steps would you take to mitigate the threats? [*Combined Bank, SO(IT) 2024*]
6. Write down the top 10 cyber-attack. [*Combined Bank AME/AHE, SO(IT), Officer(IT) 2023*]

7. Spyware এর কাজ কী? উদাহরণ দিন। [*45th BCS (General), ICT 2024*]
8. Discuss about Ransom ware and DDoS attack. [*Combined Bank SO(IT) 2023*]
9. Explain buffer overflow attack with an example. [*BTCL AME 2023*]
10. **Write the difference types of Web application attacks? [*MoL AME 2023*]**
11. **What is SQL Injection? How to Prevent against SQL Injection Attacks?** [*SGFL AE(IT) 2023; RAKUB Programmer, ME, SPCBL AME 2021, BTV AP 2019*]
12. Describe a man-in the middle attack on the Diffie-Hellman key exchange protocol in which the adversary generates two public key pairs for the attack. [*BB AP 2023*]
13. **Write two differences between SQL Injection and cross site scripting (XSS).** [*BCIC, AP 2022*]
14. What is DDoS and SQL Injection attack? [*SBL Officer(IT) 2021*]
15. একটি Virus ও Ransomware এর নাম লিখ? [*BTCL, JAM 2022*]
16. **Malware বলতে কী বুঝানো হয়? উদাহরণসহ সংক্ষেপে বর্ণনা করুন।** [*40th BCS 2021*]
17. If you downloaded the email, you will be able to face the problem. Which attack do you face? [*BCC AP 2023*]
18. Difference among Black Hat hacker, Grey hat hacker and white hat hacker. [*Combined Bank Officer(IT) 2023*]
19. **Difference between active and passive attack. [*BEPZA Programmer 2023*]**
20. **Difference between Conventional signatures & digital signatures?** [*45th BCS (General), ICT 2024*]
21. Difference between Digital signature and Digital certificate. [*Combined Bank Officer(IT) 2023*]
22. What is digital signature? Where is it used? [*BPSC AE 2022*]
23. What is a digital signature? Describe its role in digital security? [*BPSC AP 2022*]
24. Write down the purpose of Certification Authority (CA) in Digital Signature. [*BPSC SAE 2021*]
25. Write down the difference between Authentication and Authorization. [*40th BCS 2020*]
26. What is public key encryption? Explain digital signature with example. [*ICT Ministry AP 2017*]

27. Difference between symmetric and asymmetric key encryption. [*BEPZA Programmer 2023*]
28. **What is Symmetric and Asymmetric Encryption? Explain with example.** [*NPCBL ET(Soft); WZPGCL AE(CSE) 2023, BPSC SAE 2021*]
29. Distinguish between Symmetric Encryption and Asymmetric Encryption. Give some examples of encryption algorithm. [*Rupali Bank ANE 2023*]
30. What is meant by Encryption and Decryption? [*Combined Bank AME/AHE 2023*]
31. **What is Cryptography? Difference between Symmetric and Asymmetric encryption with example. Draw and design public key encryption using Hash function. Draw a diagram for e-commerce online transactions.** [*Combined Bank SO(IT) 2023*]
32. What are the different types of ciphers in cryptography? What are the factors to be considered for cryptographic strength? [*Rupali Bank ANE 2023*]
33. How do you define 2 factor authentication? Give example. [*BPSC AP 2023*]
34. অথেনটিকেশন বলতে কী বোঝায়? টু-ফ্যাক্টর অথেনটিকেশনের বর্ণনা দিন। [*45th BCS (General), ICT 2024*]
35. **Identify the type of algorithm? (i) MD5 (ii) AES (iii) RSA (iv) Diffie-Hellman** [*BAPEX AGM(ICT) 2023*]
36. Describe RSA Algorithm and how it works? [*Teletalk AM(IT) 2023*]
37. **Briefly describe the relationship among threat, zero days vulnerability and attack. Give examples.** [*BPSC AME 2021*]

Cloud computing

1. Cloud computing কী? উহার বৈশিষ্ট্য ও সুবিধা বর্ণনা করুন। [*17th NTRCA(CSE) 2023*]
2. What is Cloud Computing? Write its advantages and Disadvantages? [*NPCNL, JAM 2022*]
3. Cloud Computing এর সার্ভিসগুলো লিখুন। [*BPSC AP 2021*]
4. Write the three basic function of cloud services? [*BOF AE(CSE) 2021*]
5. Define cloud computing? What is the benefit of cloud computing? [*BLPU AP 2018*]
6. Write ten sentences on "Big Data". [*MRA SCO 2023*]
7. **Big data বলতে কি বুঝায়? Big data এর বৈশিষ্ট্যগুলো লিখুন।** [*BPSC AP 2021*]

8. Your bank wants to transform your full data center into a cloud. Now sketch your strategy and plan to implement this procedure. [*Combined Bank, AME/AE(IT) 2024*]
9. **Your bank wants to secure an e-banking online system and wants to configure a web server in your data center. What kind of tools and technology do you use for this?** [*Combined Bank, AME/AE(IT) 2024*]
10. What are the challenges in optimizing energy efficiency of data centers? Explain! [*BB AP 2023*]
11. Discuss server maintenance best practices, including routine tasks like cleaning, monitoring, and applying security patches. How do these practices contribute to server longevity and performance? [*Rupali Bank ANE 2023*]
12. **Define a virtual machine with a neat diagram, explain the working of VM. What are the benefits of a VM?** [*BB AP 2023*]
13. What do you mean by national data center? [*BPSC, AP 2022*]
14. Explain the component of Data Center. [*ABL SO (IT) 2017*]
15. To maintain a data center cooling system sometimes where you need a DC generator. Which type of generator do you prefer based on fuel type generator type, cost, energy, and budget? [*Combined Bank, AME/AE(IT) 2024*]

Previous Questions — Database and SQL

1. Discuss about different types of relations in DBMS. *[Combined Bank, AP 2024]*
2. সাধারণ ফাইলভিত্তিক সিস্টেমের চেয়ে DBMS এর সুবিধা কী কী? *[17th NTRCA(ICT) 2023]*
3. What is Database? *[EGCB SAE(ICT) 2023; BKKB AP 2017]*
4. What is the degree of relation in DBMS? *[BCC AP 2023]*
5. **What are the ACID properties of transaction to ensure data reliability and integrity?** *[Milk Vita AM(CSE), BPSC AP 2023; BBL AM (ICT) 2019]*

6. ডাটাবেস ট্রানসেকশনের ACID Properties সম্পর্কে লিখুন। *[17th NTRCA(ICT) 2023]*

7. Describe the advantages and disadvantages of DBMS-provided and application provided security. *[BPSC, AP 2022]*

8. What is distributed database? *[Combined 2 bank, ADA 2022]*

9. Write difference between relational database and NoSQL database. *[SBL Officer(IT) 2021]*

10. While converting E-R diagram into Tables, how is a Many-to-many relationship set between entities A and B is converted into database tables? *[BPSC SAE 2021]*

11. DBMS বলতে কী বোঝানো হয়? DBMS শ্রেণিভিন্যাস বর্ণনা করুন। *[40th BCS 2020]*

12. Explain the concept distributed DBMS. What are the features of DBMS? *[BB AME 2019]*

13. What is DBMS? Write the full meaning of DDL and DML. *[BKKB AP 2017]*

14. Product (pname,price,category,maker),
Purchase (buyer,seller,stock),
Company(cname,country),
Person(per-name, address, phone).
Here in Product table make is similar to cname, here buyer and seller in Purchase table is similar to per-name in Person table, Draw the ER diagram of from the above information:
    a. Select person name who bought American category product
    b. Find Sum of all product and Sum of All sales product according to country of origin. *[NWPGCL, AM(ICT) 2024]*

15. Query's: Employee & department table given:
    a. Write the employee name who got same salary named Rahim but not same job of Rahim.
    b. Write the employee's name who's average salary is more than company's average salary. *[BGDCL, AM(CSE)2024]*

16. From an Employee table. Write SQL statement statement according to the following question:

(i) Find out the employees who join the same date

(ii) Find those employees whose salary greater than 8,000 and Less than 25,000. *[BRDB AP 2023]*

17. Database Query:

Employee (ID, First_name, Last_name, Joining_date, Salary)

Employee_Info (Emp_ID, Designation)

a. Create both tables

b. Show department-wise min, max, average salary of employees

c. Show details of employees joined before February 2023. *[MRA SCO 2023]*

18. Suppose that we have a relational database with the following table. Underlined one **represent primary key:**

Movies (mid, title, year)

People (pid, name)

Genres (gid, genre)

HasRole (pid, mid, role)

Has Genre (gid, mid)

Write a SQL query to return the number of movies that are romantic comedies. *[BB AP 2023]*

19. Write following EMPLOYEE database table write an SQL query to find employee who work is a department where the average salary is lower than the average salary all the department..... *[Exam name/year omitted in source]*

20. Consider the two schema

employees (id, first_name, last_name, designation,
joining_date, salary, dept_id) and
department (dept_id, dept_name).

Where detp_id is forgeign key. Find the first_name and department name whose salary is maximum. *[BIWTA AE(CSE) 2023]*

21. Consider the Schema

employee(id, name, salary),
equipment(id, name, price),
hire(employee_id, equipment_id);

Write the SQL query to show the name of employee who borrow the maximum equipment? *[BAPEX AGM(ICT) 2023]*

22. Consider the employee tables: Create a SQL view that shows the details of Employee information who have the salary equivalent to the maximum, minimum and average salary of employee. *[Milk Vita AM(CSE) 2023]*

23. Analyze the following code:

```
SELECT department_name, AVG(salary) as average_salary
FROM employees
JOIN department d ON e.department_id = d.department_id
WHERE salary > (SELECT AVG(salary) FROM employees )
GROUP BY department_name
HAVING COUNT(*) > 2
ORDER BY average_salary desc
```

[Combined Bank Officer(IT), Rupali Bank ANE 2023]

24. Employee table হতে Employee_id, Employee কে খোঁজে বের করার SQL Command লিখ যাদের গড় salary 2000 উপরে। *[BTCL, JAM 2022]*

25. SQL Query for finding Dept names for departments Find out the employees whose salaries are greater than the salaries of their managers. *[NESCO AM(ICT) 2021*

26. Suppose, 'Employee' table (emp_id, emp_name, dept_id, salary) and 'Department' table (dept_id, dept_name, increment_dept). Create a tigger to increment the salary of the employee by 10% whose salary is above 30000. *[PGCB AE (CSE) 2021]*

27. Below tables are given, Employee (employee_id, name, salary, department) Leave (employee_id, date, reason, no_leaves) Holiday (Date, description):
    (i) Write query to show all employee's leave count.
    (ii) Write query to show employees who are in 'HR' department and have taken at least 5 leaves. *[RBL ANE 2021]*

28. Find the Query for the Instructor table:
    a) Find the average salary of instructors in each department.
    b) Find the names and average salaries of all departments whose average salary is greater than 42000.
    c) Find names of instructors with salary greater than that of some (at least one) instructor in the CSE department. *[NRCC AP 2021]*

29. Write a SQL query to find those employees who report that manager whose first name is 'abc'. Return first name, last name, employee ID and salary. *[PGCL SAE (CSE) 2021]*

30. emp [e_id, e_name, dept_id, salary, DOB], dept [dept_id, city, dept_name]; প্রত্যেকটি Department এর নাম এবং ঐ Department এর employee দের গড় Salary দেখার SQL Query লিখ। *[NESCO JAM(ICT) 2021]*

31. Given Table:

Project (Project_id, Project_name, Manager_name)

Location (location_id, Location_name, project_id)

Employee (Employee_id, Employee_Name, Location_id, Joning date, Salary)

Write a query to show project_name, Location_name, Total_salary of each projects employee who joined before 'January 2021' *[APSCL AE (ICT 2021)]*

32. Database table by name Loan Records is given below:

Borrower	Bank_Manager	Loan_Amount
Rahim	Shuvo	1000.00
Shamim	Rasel	500.00
Mamun	Shuvo	7000.00

What is the output of the following SQL query?

```
SELECT count (*) FROM (
(SELECT Borrower, Bank_Manager, FROM Loan_Records) AS S NATURAL JOIN
(SELECT Bank_Manager, Loan_Amount FROM Loan_Records) AS T);
```

[BAUST AP 2021]

33. Consider the following Schemas-

Movies (title, year, length, genre, studioName, producerCert)

starsIn (movieTitle, movieYear, starName, salary)

MovieStar (name, address, gender, birthdate)

MovieExec (name, address, producerCert, netWorth)

Studio (name, address)

Write SQL queries for the following:

(a) Find the address of a studio named MGM

(b) Find the number of movies that was not produced by "MGM" *[BPSC AME 2021]*

34. Given a database schema and worker table with fully code: Now writes SQL Query from the following questions.

- Write a query to get distinct name
- Write a query to get the first three letter of first_name column from worker table. *[DPDC 2020]*

35. SQL Query:

(i) Commission greater than 10% *[NU AP 2020]*

(ii) Remove duplicate data from table: *[NU AP 2020]*

36. Database Query: Given two table

COURSE (Course_ID, Course_name, Chief_ID) and

INSTRUCTOR (Instructor_ID, Course_ID, Salary, Joining_date).

Write down database query Average salary for different courses. *[NACTAR 2020]*

37. Design a database of student with the gpa of a university. Find the top 10% gpa holder from the different department of the university. *[5 Bank AME 2019]*

38. Write a query to select top 10% row from grade table. *[CB AP 2018]*

39. What is the SQL command used to fetch first 3 characters name from employees? *[BCPSC AP 2018]*

40. Write an SQL query to insert a tuple in the table: Employee (ID, Name, Designation, and Salary). *[NESCO AM (ICT) 2018]*

41. Suppose you've two table (Employee, Department) in a Database and Employee table three cell (emp_name, dept_id, salary) also Department table two cell (dept_id, dept_name), now update the salary 10% increase value from Department table. Give the appropriate example. *[PBL SO 2018]*

42. Given two tables are

employee (id, name, salary, dept_id) and

department (dept_id, dept_name),

write SQL to find MAX salary and average salary of specific department. *[DBBL PO 2017]*

43. What is the SQL query for showing only the duplicate lists in student (id, name, gpa) table? *[ABL Officer (ICT) 2017]*

44. নিচের Table টি তৈরি করার SQL কমান্ড লিখুন।

student_info (std_id, name, department, phone_number)

a. Table টি তে ২টি record (insert) প্রবেশ করার SQL কমান্ড লিখুন।

b. Table টি থেকে CSE বিভাগের ছাত্র/ছাত্রীদের নামের তালিকা বের করার SQL command লিখুন। *[BPSC AP 2021]*

45. What is a weak entity for data modeling using the entity relationship model find out any weak entity and it's identify relationship for the school database? Which of the following table?

Student (student_id, student_name, admission_year)

Teacher (teacher_id, teacher_name, teacher_joindate)

Course (course_id, subject_name, credit) *[BCC AP 2021]*

Previous Questions — Operating System, Deadlock Paging, linux

1. Define deadlock in operating system. Draw a proper diagram to show a deadlock situation and discuss in brief. *[45th BCS, CSE 2024]*
2. Give the necessary conditions for deadlock to occur. Is it possible to have deadlock involving only a single process? Explain your answer. *[Combined Bank, Officer (IT) 2024]*
3. When does deadlock occur and how to prevent it? *[45th BCS, ICT 2024]*
4. Explain Circular wait deadlock. *[Titas Gas AE(CSE) 2024]*
5. Deadlock এর চারটি শর্ত লিখ। *[BTCL, JAM 2024]*
6. What is deadlock in operating system (OS)? What are the four necessary and sufficient conditions behind deadlock? *[BPSC AP 2023, Milk Vita AM(CSE) 2023]*

7. A system has P processes each needing a maximum of m resources and a total of r resources available. Which conditions must hold to make the system deadlock free? *[BPSC AP 2023]*
8. Name and define characteristics properties of the Deadlock situation in a computer system. *[BPSC, AP 2022]*
9. Describe how deadlock is possible with the dining-philosopher's problem. *[BPSC AME 2021]*
10. What is deadlock? Describe the way of detecting and protecting deadlock. *[38th BCS]*
11. Banker's Algorithm: 5 processes P_0 through P_4; 3 resource types A (10 instances), B (5 instances), and C (7 instances). *[RAKUB ME 2021]*

- Snapshot at time T_0:

Process	Allocation (A B C)	Max (A B C)	Available (A B C)
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

The content of the matrix. Need is defined to be Max – Allocation.

Process	Need (A B C)
P_0	7 4 3
P_1	1 2 2

Operating System

1. Mention the difference between Multitasking and Multiprocessing of an operating system. *[45th BCS, ICT 2024]*

2. What is the difference between micro kernel and macro kernel in the context of OS? *[BPSC AP 2023, BB AME 2019]*
3. Operating System-এর Memory hierarchy সচিত্র বর্ণনা করুন। *[17th NTRCA(CSE) 2023]*
4. Process Synchronization এর ক্ষেত্রে Race condition ব্যাখ্যা করুন। *[17th NTRCA(ICT) 2023]*
5. Real Time System বলতে কী বোঝায় ব্যাখ্যা করুন। *[17th NTRCA(ICT) 2023]*
6. Explain context switching in Operating System. *[MGMCL, AM (ICT) 2022]*
7. What is Multithreading and multitasking? Difference between Multithreading and Multitasking? *[RAKUB ME 2021]*
8. What is multitasking and multithreading? *[BCC Programmer 2019]*
9. What are the sub components of I/O manager in Windows NT? *[BB AME 2019]*
10. What is operating System? What are the main components of operating System? *[BCC Programmer 2019]*
11. What are the relationship between operating system and computer hardware? *[38th BCS]*
12. Explain the functionalities of operating system. *[ICT Ministry AP 2017]*
13. Define: Socket, Kernel, Process, Multiprogramming, Context Switching; Explain Preemptive Priority Scheduling algorithm with illustration; Consider the following six processes each having its own unique processing time and arrival time. *[Combined Bank, AME/AE(IT) 2024 / BIWTA AE(CSE) 2023]*
14. Find average turnaround time using shortest job first scheduling algorithm. Find average turnaround time and average waiting time using round robin and FCFS algorithm? *[Teletalk AM(IT) 2023]*
15. Process কী? একটি Process এর বিভিন্ন ধাপগুলো লিখুন। *[17th NTRCA(ICT) 2023]*
16. Process Control Block (PCB) কী? এটি একটি Process সংক্রান্ত যে যে তথ্য রাখে সেগুলো লিখুন। *[17th NTRCA(ICT) 2023]*
17. CPU Scheduling কী? যে কারণে CPU Scheduling করতে হয় সেগুলো লিখুন। *[17th NTRCA(ICT) 2023]*
18. Explain the process state. *[EGCB SAE(ICT) 2023]*

19. Write down the name of four information stored in PCB (Process Control Block). *[RPGCL AM (ICT) 2022]*
20. Differentiate between thread and process. *[BPSC AME 2021]*
21. Operating System Gi Process State Transition Diagram অঙ্কন ও ব্যাখ্যা করুন। *[BPSC AP 2021]*
22. Consider the processes P1, P2, P3, P4 given in the below table, arrives for execution in the same order, with Arrival Time 0, and given Burst Time, let's find the average waiting time using the FCFS scheduling algorithm. *[RAKUB ME 2021]*
23. List four important process attributes that an operating system maintains to keep track of information of a thread. *[BPSC AME 2021]*
24. Define FCFS, SJF and RR algorithm (Quantum=20) *[NU AP 2020]*
25. Operating system (OS) scheduling is the key concept of multiprogramming. List and briefly define the major types of OS scheduling. *[Combined Bank, Officer(IT) 2020]*
26. Turnaround time of FCFS and SJF. *[NU AP 2020]*
27. A, B two resources. Two processes (P1 and P2) share these resources. When a process request for a resources, if that resource is free then it will be allocated with that resources. If the resources are not free then the process will halt. Now the scenario is: P1 request A, P2 request B, P1 request B, P2 request A.
    (i) Now A is allocated with which process?
    (ii) (ii) Now B is allocated with which process?
    (iii) (iii) What is the name of the situation that happens in this scenario? *[DPDC 2020]*
28. What is the difference between preemptive and non-preemptive scheduling? *[37th BCS]*
29. What are the advantage threads over process? *[BCC Programmer 2019]*
30. Difference between physical memory and virtual memory, also describe the advantages and disadvantages of virtual memory. *[Combined Bank AME/AHE 2023]*
31. Virtual Memory এর প্রয়োজনীয়তা কি ব্যাখ্যা করুন। *[BPSC AP 2021]*
32. What do you mean by virtual memory? *[BPSC AME 2021]*

33. Explain: Kernel, Cache, Virtual Memory and RAID. *[SPCBL AME 2021]*
34. Explain: Swapping, Virtual Memory. *[Bank SO(AME) 2021]*
35. Explain LRU and NRU Page Replacement algorithm. *[Combined Bank, AME/AE(IT) 2024]*
36. Consider a reference string 4,7,6,1,2,7,2 the number of frames in the memory is 3. Using page Replacement Algorithm (LRU), find the number of page fault. *[BPDB AE(CSE) 2024]*
37. Difference between Paging and Segmentation. *[BTCL, JAM 2024, 17th NTRCA(CSE) 2023]*
38. Consider a machine with 64MB physical memory and a 32-bit virtual address space. If the page size is 4KB, what is the approximate size of the page table? *[45th BCS, ICT 2024]*
39. Define paging and trashing in the context of OS. *[BPSC AP 2023]*
40. Find out total number of pages, when page size 4KB and address space 32 bit. *[SGFL AE(IT) 2023]*
41. Internal এবং External fragmentation এর মধ্যে পার্থক্য লিখুন। *[17th NTRCA(ICT) 2023]*
42. What is page fault in computing systems? What does it occur? *[BCIC, AP 2022]*
43. A system uses 3 page frames for storing process pages in main memory. It uses the Least Recently Used (LRU) page replacement policy. Assume that all the page frames are initially empty. What is the total number of page faults that will occur while processing the page reference string given below? 4, 7, 6, 1, 7, 6, 1, 2, 7, 2. *[BPDB AE(CSE) 2021]*
44. What is page and block? Draw simple block diagram of SSD, page and block. *[Combined Bank, SO(IT) 2020]*
45. DOS কী? অপারেটিং সিস্টেমের কাজ ও প্রকারভেদ ব্যাখ্যা করুন। *[18th NTRCA School 2024]*
46. Blocking এবং Buffering OS এর পার্থক্য লিখুন। *[17th NTRCA(CSE) 2023]*
47. What is access time and transfer time? *[BTV AP 2019]*
48. If main memory access time is 100ns, cache access time is 50 ns, cache hit rate is 90% then what is the average time to read from memory? *[BB AME 2017]*

49. Show last 10 lines of log file which is continuously updating in Linux command? *[Titas Gas AE(CSE) 2024]*
50. Write appropriate Linux command: (Show hidden files and directories, Delete a directory and its file, Prints last five lines of a text file, Download a file from an URL) *[Milk Vita AM(CSE) 2023]*

linux

51. UNIX command with example: File move, Change Directory and search from a specific line. *[NPCBL ET(Soft) 2023]*
52. Write Linux command to find out the following question:
    (a) To show current file directory.
    (b) To show 11th to 15th line from file name myfile.
    (c) To show permission for read, write and execution file name myfile. *[BTCL AME 2023]*
53. Linux Command in ownership and group permission. *[PBL HE 2023]*
54. Linux Command লিখ: (a) একটি ফোল্ডারের সকল ফাইল দেখানোর কমান্ড; (b) নতুন ডিরেক্টরি তৈরির কমান্ড; (c) ফাইল/ফোল্ডারের পারমিশন দেখানোর কমান্ড। *[BTCL, JAM 2022]*
55. Write down the Linux command: All hidden flies, remove a file, permission of a file, search for a string. *[MGMCL, AM (ICT) 2022]*
56. Unix command (directory listing with hidden files). *[Combined 2 bank, ADA 2022]*
57. Write Linux commands to: (i) Make a directory named PSC *[BPSC SAE (CSE) 2021]*
58. Linux এর ক্ষেত্রে User Creation এর কমান্ডটি লিখ? *[PGCB SAE (CSE) 2021]*
59. Linux command for showing all files including the hidden files inside the home directory. *[NESCO AM(ICT) 2021]*
60. Linux command for showing page size, disk space in a human-readable format. *[NESCO AM(ICT) 2021]*
61. Linux Command: Write a code for listing home directory files with all details and human readable size got to Home directory, list directory files with 10-15 are display only 10th to 15th lines of words of them. Write the instructions in a way that they execute together and shows the result. *[BGDCL AE (CSE) 2021]*

62. In Linux, History is a very useful command to show you all of the last commands that have been recently used. Grep is a Linux command-line tool used to search for a string of characters in a specified file. Write grep and history command to find previous commands in Linux. *[BCC AP 2021]*
63. Write Linux command for following question: a) Create a file apscl.txt in current location. b) Given permission to all read write and execute to the file apscl txt c) Read first 7 lines from apscl.txt file d) Delete the file apscl.txt *[APSCL AE(ICT) 2021]*
64. Copy a directory with all its Contents into a directory name/home/admin. *[BPSC SAE 2021]*

Previous Questions — Software Engineering

1. Write down the SDLC steps with Explain. [BRICM AME 2024, EGCB SAE(ICT) 2023, BPSC ANE 2020]
2. What are the characteristics of software? Define black box testing and white box testing. [45th BCS, ICT 2024]
3. What is SDLC? Briefly explain its phase. [NPCBL ET(Soft) 2023]
4. Describe difference between Cycle of SDLC. [PBL SQA 2023, BPSC, AP 20
5. What is SDLC? Write the name of 7 phase of SDLC? [DESCO AE 2023]
6. Software engineering এ রিকোয়ারমেন্টস এনালাইসিস পর্যায়টির বর্ণনা দিন। [BTCL, JAM 2022]
7. SDLC diagram সহ বর্ণনা করুন। SDLC এর মেজর phases গুলি কি? [BPSC SAE 2021]
8. What is SDLC? List the stages involved in the SDLC process. Which stages ensures the user acceptance of the system? [BTRC SAD 2021]
9. Explain the main step of Software development life cycle (SDLC). [40th BCS 2020]
10. What is Software Maintenance Life Cycle (SMLC)? Write Steps of Software Maintenance Life Cycle. [BUET M.Sc. October 2019]
11. Waterfall model বিস্তারিত বর্ণনা করুন। এই model এর সুবিধা এবং সীমাবদ্ধতাগুলো উল্লেখ করুন। [17th NTRCA(ICT) 2023]
12. Software Engineering এর ক্ষেত্রে Waterfall Model বর্ণনা করুন। [17th NTRCA(CSE) 2023]
13. Answer the following question: (a) Write down the steps of Waterfall model. [BARC Programmer 2023]

14. Which SDLC do you prefer between Agile and waterfall model explain with example. [Combined Bank Officer (IT) 2023]
15. How does agile methodology used in software development differ from that of waterfall methodology? Explain in brief. [BICIC, AP 2022]
16. Software development এর Agile Method সম্পর্কিত আলোচনা করুন। [BPSC AP 2021]
17. What is Agile? Mention its four values. [BPSC ME 2021]
18. Write the agile method components for Software development. [PBL SO 2018]
19. Explain briefly the waterfall Model? [BUET M.Sc. April 2017]
20. What do you understand about software quality assurance (SQA)? While purchasing a software system for your company, as a SQA team leader what aspects will you look into for a quality software. [Combined Bank, SO(IT) 2024]
21. Write the responsibilities of a software engineer to design a software. [45th BCS, CSE 2024]
22. Software Requirement Specification (SRS) বলতে কি বুঝায়? Software Development এর কোন ধাপে SRS তৈরি করা হয়? [BPSC AP, 40th BCS 2021]
23. Describe different techniques use the risk analysis of software development. [38th BCS]
24. What is software testing? Write the differences between following terms?
    1. Black-box testing and White-box testing,
    2. (ii) Verification and Validation. [45th BCS, CSE 2024]
25. Difference between Alpha tests, Beta test, gamma test in software development. [PGCB AE(CSE) 2024]
26. Match the table: (Testing method vs Topic) [BPDB AE(CSE) 2024]
27. Define black box testing and white box testing. [45th BCS, ICT 2024]
28. Difference between White Box testing and Black Box testing. Give examples of both type of testing. [BARC Programmer, Teletalk AM(IT), PBL SQA, 17th NTRCA(ICT) 2023, BPSC AP, BPSC SAE 2021, PSB Programmer, EPB Programmer 2018, ICB AP 2017]
29. Software testing কি? ইহার গুরুত্ব আলোচনা করুন। [17th NTRCA(CSE) 2023]
30. Write difference between Verification and Validation. [PBL SQA 2023]
31. What is software testing? Discuss effective and exhaustive testing. [BPSC, AE 2022]
32. How alpha testing is performed in software development? [BPSC, ADA 2022]
33. Unit testing, Integration testing এবং Beta testing বলতে কি বুঝায়? [BPSC AP 2021]
34. Software development এ Black Box Testing বলতে কি বুঝায়? [PGCB SAE (CSE) 2021]

35. Software Engineering-এর প্রেক্ষিতে পাঁচটি ভিন্ন ভিন্ন design pattern এর নাম লিখুন। [40th BCS 2020]
36. Integration testing and Beta testing সম্পর্কে সংক্ষেপে লিখুন। [40th BCS 2020]
37. What do you know about unit testing? How can you perform acceptance testing? [BUET M.Sc. October 2019]
38. Define software security? Challenge of software security. [BLPU AP 2018]
39. What is software testing and name of some testing tool? [IBBL AD 2017]
40. Example 03: You are asked to lead a team of software engineers to develop an application software system for your company and deploy it as fast as possible. You need to gather user requirements, design, develop, test and then deploy the system. Between Waterfall Approach and Incremental Approach, which software development approach will you take for your software project? Explain your answer. [Combined Bank, SO (IT) 2024]

Previous Questions — Discrete Math and Web Engineering

1. Propositional logic ও Predicate Logic উদাহরণসহ বর্ণনা করুন। *[BPSC ANE 2020]*
2. Find the sets X and Y if $X \cup Y = \{1,2,3,5,6,8,9,10\}$, $X \cap Y = \{1,5\}$ and $Y - X = \{2,6,9,10\}$. *[Combined 2 Bank, Officer(IT) 2024]*
3. Which tag is used for creating button in html? *[BCC AP 2023]*
4. HTML Element কী? উদাহরণসহ বর্ণনা করুন। *[17th NTRCA(CSE) 2023]*
5. উদাহরণসহ HTML webpage এর গঠন ব্যাখ্যা করুন। *[17th NTRCA(CSE) 2023]*
6. নিচের লিস্টটি তৈরি করার জন্য HTML কোড লিখুন: *[17th NTRCA(CSE) 2023]*
7. নিম্নোক্ত উপাদানগুলোসহ একটি HTML page লিখুন। Hyperlink, Ordered list, Unordered list, Form (Text box, Check box, Option Button). *[17th NTRCA(ICT) 2023]*
8. Write down the description of <header> <footer> <section> <article> tag of new HTML5. *[5 Bank AME 2019]*
9. How do you change the value of a HTML element using HTML DOM? *[5 Bank AME 2019]*
10. Write down the name of HTML5 media tag. *[CB AP 2018]*
11. What is local Storage and session Storage in HTML5? *[ABL Officer (ICT) 2017]*

12. What is canvas HTML? What is difference between canvas and SVG? *[ICB AP 2017]*
13. What is CSS? What is CSS framework? Write down 3 CSS framework name? *[BEPZA Programmer 2023]*
14. CSS Box Model এ 'Padding' এবং 'Margin' এরিয়ার মধ্যে পার্থক্য লিখুন। *[17th NTRCA(ICT) 2023]*
15. What is the function of CSS in web designing? *[NESCO AM(ICT) 2021]*
16. HTML ও CSS কী? *[BPSC ANE 2020]*
17. Which tag is used to write JavaScript in html? *[BCC AP 2023]*
18. Suppose you've a JavaScript code name as "bankScript" write the code for loading in HTML using JS. *[PBL SO 2018]*
19. What is closure in JavaScript? Explain with an example? *[ICB AP 2017]*

Previous Questions — Microprocessor

1. Mention the differences between RISC and CISC microprocessors. [*Combined Bank, SO(IT) 2024*]
2. Fill in the gaps RISC or CISC: [*Titas Gas AE(CSE) 2024*]
    a. (i) Pipelining is less efficient due to instruction complexity and variability_____
    b. (ii) Emphasis on hardware simplicity and efficiency_____
    c. (iii) Complex decoding due to variable instruction length_____
    d. (iv) Each instruction typically executes in a single clock cycle_____
3. What is super-scalar processor, why Pentium is a super-scalar processor but 80386 is not? [*BUET M.Sc. CSE October-2017*]
4. What is superscalar microprocessor? Explain briefly. [*BUET M.Sc. CSE April 2017*]
5. What are the advantages of the 80386 microprocessor over the 80286? [*BUET M.Sc. CSE October 2017*]
6. Why Pentium is a superscalar processor? [*BUET M.Sc. CSE October-2015*]

7. Draw a block diagram of Intel 8086 microprocessor. Write the features of 8086 microprocessor. [*45th BCS, CSE 2024*]
8. **What exactly is a microcontroller? What distinguishes a microprocessor from a microcontroller?** [*Combined Bank, SO(IT) 2024*]
9. Why is memory segmentation needed in 8086 microprocessor? What is the technique to generate 20-bit physical address in a 16-bit microprocessor? [*45th BCS, CSE 2024*]
10. What is the difference between a microprocessor and a microcontroller? Differentiate between 8085 and 8086 microprocessor. [*45th BCS, ICT 2024; BUET M.Sc. CSE April 2017*]
11. Difference between 32 bit Microprocessor and 64 bit Microprocessor with example. What is the meaning of 2.40GHz Microprocessor? [*Combined Bank SO(IT) 2023*]
12. **What is Microprocessor? Explain basic component of Microprocessor.** [*SPCBL AME 2021*]
13. Difference between Microprocessor and Microcontroller. [*SPCBL AME 2021*]
14. **What is the difference between 8-bit (8085) and 16-bit (8086) microprocessor?** [*PGCB SAE (CSE) 2021*]
15. Describe addressing mode of 8086 Microprocessors. [*BB AME 2017*]
16. Difference between microprocessor and micro-controller. [*MM AP 2017*]
17. Write down advantages of microcontroller over microprocessor. [*BUET M.Sc. October 2015*]
18. What is assembly language? Write its features. Write a program in assembly language to add and multiply two numbers. [*45th BCS, CSE 2024*]

Previous Questions — Data Structure

1. Discuss about different types of relations in DBMS. [*Combined Bank, AP 2024*]

2. ডাটা স্ট্রাকচার কী? Linear এবং non-linear data structures উদাহরণসহ ব্যাখ্যা করুন। [*17th NTRCA(ICT) 2023*]
3. Difference between linear and nonlinear data structure. [*PSB ADA 2018*]
4. Write the advantages and disadvantages of using linked list instead of using array. [*45th BCS, CSE 2024*]
5. **Is it possible to apply Binary search in a sorted linked list? Justify your answer with proper example.** [*45th BCS, CSE 2024*]
6. Differentiate between Linked list and Array? [*BPSC AP, 17th NTRCA(ICT) 2023; BPSC AP, ME 2021; BNPS AP 2017*]
7. Linked List কী? উদাহরণসহ বর্ণনা করুন। [*17th NTRCA(CSE) 2023*]
8. What is a linked list? Given the algorithm to create a linked list and show an example graphically. [*BPSC, AE 2022*]
9. For which data structure operations, Linked List is better than Array? (Insert, Delete, Search). [*NESCO AM(ICT) 2021*]
10. **Illustrate singly linked list, doubly linked list and circular linked list with diagram.** [*BNPS AP 2017*]
11. Difference between Stack and Queue. Write about 2 problems solved by stack and queue. [*Combined Bank AP 2024; PSC Lecturer 2017*]
12. Write down the difference between Stack and Queue. [*DESCO SAE 2023*]
13. Stack এর operation গুলি সংক্ষেপে বর্ণনা করুন। [*BPSC SAE 2021*]
14. Queue is an abstract data structure. A queue is open at both its ends. One end is always used to insert data (Enqueue) and the other is used to remove data (dequeue). Write the steps of Enqueue Operation of Queue. [*Combined Bank, Officer(IT) 2020*]
15. Why do we use stack? [*BNPS AP 2017*]
16. Example 02: A list of numbers 3, 5, 7, 11, 13, 17 is stored in sequence. Which of the following data structure will have the least data movement when the number 11 is removed from the list? FIFO, QUEUE, HEAP, Linked list and stack. Justify your answer. [*38th BCS 2018*]

17. **Suppose that in-order and preorder traversal of a binary tree are dbeafcg and abdecfg respectively. What will be the post-order traversal of the Binary tree?** [*45th BCS, CSE 2024*]
18. Proper binary tree is one more node is Internal node prove it. [*Titas Gas AE(CSE) 2024*]
19. Inserting data to BST. Print the tree in post order traversal. Delete one of the node and redraw the valid BST again. [*PGCB AE(CSE) 2024*]
20. How to represent binary tree using array? [*BGDCL AM(CSE)2024; MRA SCO 2023*]
21. **Draw a binary tree of 15 elements in (a) Preorder (b) In-order (c) Post order traversals.** [*BPSC AP 2023*]
22. B-tree data structure কী? এর প্রয়োগ ব্যাখ্যা করুন। [*17th NTRCA(CSE) 2023*]
23. Explain Binary Search Tree Operation. [*17th NTRCA(CSE) 2023; Combined Bank SO(IT) 2020*]
24. Decision Tree কী? উদাহরণসহ বর্ণনা করুন। [*17th NTRCA(CSE) 2023*]
25. Explain binary tree with example. [*WZPGCL AE(CSE) 2023*]
26. What is the minimum number of nodes in a binary tree? [*BCC AP 2023*]
27. Write an algorithm to find a node in a binary search tree. [*PSB AP 2018*]
28. **Example 03: Given item- 40, 45, 80, 90, 50, 70. Draw Heap and Binary search tree (BST).** [*SGFL AE(IT) 2023*]
29. Example 04: Construct a BST from
    Pre-order (1 5 8 7 4 9 3) and
    In-order (8 5 7 1 9 4 3). [*APSCL AE (ICT) 2021*]
30. Example 05: Maximum and Minimum no of Nodes for a binary tree of height 7 where the root is considered as height 0. [*NESCO AM(ICT) 2021*]
31. Example 10: Consider the two given arrays as pre[]={1,2,4,8,9,5,3,6,7} and post[]={8,9,4,5,2,6,7,3,1}; Draw a binary tree from above array. [*BPDB AE(CSE) 2024; BUET M.Sc. CSE April 2016*]
32. Example 11: Draw a B+ tree using the indexes- 12, 14, 19, 30, 21, 22, 13, 34, 18, 23, 38, 40 where n=5? [*BUET M.Sc. April 2017*]

33. **Example 14: Find the Minimum Spanning Tree of the following graph using Kruskal's algorithm.** [*RAKUB Programmer 2021*]
34. Define feasible solution and optimal solution. Write the greedy algorithm to generate the single-source shortest path of a graph. [*45th BCS CSE 2024*]
35. **What are differences between Breadth first search (BFS) and Depth first search (DFS)? For what sort of problems it is better to use BFS than DFS?** [*45th BCS CSE 2024; BPSC AP 2021*]
36. Why Depth-first search is not optimal? [*45th BCS ICT 2024*]
37. **Define the following terms with example: Backtracking, NP, NP-Hard, and NP-complete problem.** [*45th BCS, CSE 2024*]
38. Does greedy algorithm always achieve optimal solution? It not, when does greedy approach achieve optimal solution? [*BPSC AP 2023*]
39. Difference between depth first and breadth first search. [*17th NTRCA(CSE) 2023; BPSC AP 2022*]
40. Problem solved more efficiently in adjacency list representation then adjacency matrix representation and problem solved more effective in adjacency matrix adjacency list. [*Combined Bank AP 2023*]
41. What is a strongly connected graph? [*BPSC AME 2021*]
42. Explain BFS Traversal of a graph with an example. [*17th NTRCA(ICT) 2023; DBBL PO 2017*]
43. Draw a Weighted graph from following adjacency representation of graph. [*BUET M.Sc. April 2017, May 2016*]
44. Example 16: S1, S2, S3, S4, S5 are five nodes and a value on lines denotes the cost to transmit power. (i) Draw a graph to find the shortest path to transmit power. (ii) Calculates the average cost. [*NESCO AM (ICT) 2018*]
45. Example 18: Given an adjacency list representation for a complete binary tree on 7 vertices. Given an equivalent adjacency matrix representation. Assume that vertices are numbered from 1 to 7 as in a binary heap. [*BB AP 2023*]

46. Define time complexity of an algorithm. Mention in brief-why is the worst-case Complexity of Quicksort $O(n^2)$. [*45th BCS, CSE 2024*]
47. Write the Merge Sort algorithm to sort n elements using divide and conquer strategy. [*45th BCS, CSE 2024*]
48. Make a comparison between "Blind search" and "Intelligent search". What is heuristic function? [*45th BCS, ICT 2024*]
49. Linear Search এবং Binary Search এর মধ্যে পার্থক্য লিখুন। [*17th NTRCA(CSE) 2023*]
50. অ্যালগরিদমসের complexity বলতে কী বোঝায়? কয়েকটি Sorting algorithm এর complexity উল্লেখ করুন। [*17th NTRCA(CSE) 2023*]
51. Explain Algorithm of Binary search. [*BEPZA Programmer 2023*]
52. Write a liner algorithm two sorted item merge. Why this algorithm takes O(n) time complexity? [*SGFL AE(IT) 2023*]
53. Fastest sorting algorithms? [*BCC AP 2023*]
54. The complexity of merge sort is $T(n) = 2T(n/2)+n$. Explain how the above equation is derived? [*BPSC AP 2023; IBBL AD 2017*]
55. Bubble sort, Quick sort and Merge sort algorithm এর Worst case complexity নির্ণয় কর। [*BTCL, JAM 2022*]
56. Find out Best case, Worst case complexity of Binary search, Quick sort, Depth First Search. [*RPGCL AM (ICT) 2022*]
57. Write down the Pseudo code of quick sort algorithm through recursive algorithm. Express the arrange complexity off this algorithm. [*BPSC, ADA 2022*]
58. How the quick sort is implemented? What is the complexity of quick sort? [*BPSC AME 2021*]
59. Difference between Heap Sort and Merge Sort. [*BPSC AP 2021*]
60. Compare and contrast between Quick sort and Merge sort in terms of their time and space complexity. [*BPSC ME 2021*]
61. Bubble sort Algorithm লিখুন। এ অ্যালগরিদমটির Time Complexity বের করুন। [*BPSC AP 2021*]

62. Linear Search অ্যালগরিদম কী? এই অ্যালগরিদম এর best case এবং wrose case complexity বর্ণনা করুন। [*BPSC SAE 2021*]
63. There are no well-defined standards for writing algorithms. Efficiency of an algorithm depends on several factors. Similarly, complexity of an algorithm also depends of several factors. Describe the algorithm complexity factors. [*Combined Bank, Officer(IT) 2020*]
64. Advantage and complexity of Marge sort, Heap sort, Insertion sort. [*BBL AM (ICT) 2019*]
65. Write down time complexities of different sorting algorithm. [*DBBL PO 2017*]
66. Explain: Infix, Prefix, Postfix notation. [*BTCL, JAM 2022*]
67. Describe infix conversion method. [*BBL AM (ICT) 2019*]
68. What is maximum algorithm? Define 'Terminal' and 'Utility' in a game playing. Determine the limitation of the maximum algorithm. [*45th BCS, ICT 2024*]
69. Tower of Hanoi সমস্যাটি সমাধানের জন্যে একটি recursive অ্যালগরিদম লিখুন। [*17th NTRCA(ICT) 2023*]
70. Write down the Algorithm for determining Fibonacci number through dynamic programming. [*BPSC, ADA 2022; PSB Programmer 2018*]
71. Algorithm কি? Algorithm প্রকাশের তিনটি পদ্ধতির নাম লিখুন। [*BPSC AP 2021*]
72. What are the advantages of IDA* algorithm? [*BUET M.Sc. April 2017*]
    a. Randomized quicksort is an extension of quicksort where the pivot is chosen randomly. What is the worst-case complexity of sorting n numbers using randomized quicksort? [*Officer IT/ICT- 2019*]
    b. What is time complexity of Huffman coding? [*Officer IT/ICT- 2019*]
    c. The term push and pop are related to the- [*PKB Programmer- 2019*]
    d. The step-by-step instruction that solve a problem is called: [*BREB 2019*]
    e. The complexity of Binary search algorithm is- [*Combined Bank, IT/ICT-2018*]
    f. Given a sequence, S= {1, 2, 3, 8, 15, 10}; which of the following algorithms will be the fasted to sort this sequence in ascending order? [*Combined 4 Banks, AP 2020*]

g. Maximum how many nodes can be placed in a binary Tree of N levels? [*Combined 4 Banks, AP 2020*]
h. Minimum number of queues to implement stack is _____
i. What is an AVL tree?
j. The following given tree is an example for?
k. What is the traversal strategy used in the binary tree?
l. Given an array of element 5, 7, 9, 1, 3, 10, 8, 4. Which of the following is the correct sequences of elements after inserting all the elements in a min-heap?
m. What is the worst case complexity of bubble sort?
n. What is the average case time complexity of merge sort?
o. Which algorithm is used in graph traversal and path finding?
p. What is the time complexity of Kruskal's algorithm?

Previous Questions — Cache Memory

1. What is Cache Memory? Explain the principle of Locality of Reference (Spatial and Temporal locality) with examples. [*43rd BCS CSE; BPSC AP 2023*]
2. Differentiate between Direct mapping, Associative mapping, and Set-Associative mapping with proper diagrams. [*BPSC AE 2022; DESCO SAE 2021*]
3. Explain the differences between Write-Through and Write-Back cache write policies. Which one causes more bus traffic? [*Combined Bank SO(IT) 2020; BTCL JAM 2018*]
4. What is cache miss? Explain Compulsory, Capacity, and Conflict misses with techniques to eliminate them. [*BPDB AE(CSE) 2023; PSB ADA 2019*]
5. Explain different cache replacement algorithms (LRU, FIFO, and LFU). [*17th NTRCA(ICT) 2023; BBL AM(ICT) 2019*]

6. Why is SRAM used for cache memory and DRAM used for main memory? [*BPSC Lecturer 2019; NESCO AM(ICT) 2020*]
7. Example 01: A byte-addressable system has a 32-bit physical address space, a 64 KB cache, and a block size of 16 bytes. If direct mapping is used, determine the number of bits allocated to Tag, Line Index, and Block Offset fields. [*BPSC AP 2022; BUET M.Sc. 2019*]
8. Example 02: A 4-way set-associative cache memory has a total size of 32 KB with 64-byte blocks. If the physical address is 32 bits, find: (i) the number of sets in the cache, and (ii) the size of Tag, Set, and Offset fields. [*PGCB AE(CSE) 2023; Combined Bank AP 2020*]
9. Example 03: An L1 cache has an access time of 2 ns with a hit ratio of 95%. When a cache miss occurs, the miss penalty to access main memory is 60 ns. Find the Average Memory Access Time (AMAT). [*44th BCS CSE 2023; BREB 2019*]
10. Example 04: A system has two levels of cache: L1 cache with an access time of 1 ns and hit rate of 90%, and L2 cache with an access time of 5 ns and hit rate of 80%. If the main memory access time is 50 ns, calculate the effective average memory access time. [*Titas Gas AE(CSE) 2024; SGFL AE(IT) 2021*]
11. Example 05: A computer system has a direct-mapped cache of 8 blocks (numbered 0 to 7). Consider the block access sequence: 0, 4, 8, 2, 0, 4, 8, 2. If the cache is initially empty, determine the number of cache hits, cache misses, and the hit ratio. [*BPDB AE(CSE) 2024; BUET M.Sc. CSE 2016*]

Previous Questions — Digital Electronics

1. Write the decimal, octal and binary form of the number $(E)_{16}$. Write the next four numbers in the following hexadecimal sequences: 6FC, 6FE, ____, ____, ____, ____. *[45th BCS CSE 2024]*

2. Convert $39.B8_{16}$ into decimal. What is the difference between 1100101001-1100110110? *[45th BCS ICT 2024]*

3. Write the 2's Complement of $(+55)_{10}$. *[45th BCS, CSE 2024]*
4. Convert Decimal to Octal $(423)_{10}$ and Decimal to Hexadecimal $(3000)_{10}$. *[BPDB AE(CSE) 2024]*
5. $(CDAB)_{16}$ কে অকটাল এ রূপান্তর কর। *[BTCL, JAM 2024]*
6. ২-এর পরিপূরক পদ্ধতি কী? ২-এর পরিপূরক পদ্ধতি ব্যবহার করে $(-১৫)_{১০}$ থেকে $(+২১)_{১০}$ বিয়োগ করুন। *[18th NTRCA School 2024]*
7. $(\partial D.AB)_{১৬}$ ও $(৩৩৬.৫১)_{৮}$ যোগ করুন এবং ফলাফল বাইনারীতে প্রকাশ করুন। (১১০১০১) কোন সংখ্যা পদ্ধতির সংখ্যা হতে পারে বলে মনে করেন? *[18th NTRCA School 2024]*
8. Convert the following Octal number into Decimal and Hexadecimal: $(651)_8$ *[BPSC AME 2021]*
9. $(1AC)_{16} = (?)_{10}$ and $(?)_{10} = (?)_{16}$ *[DPDC 2020]*
10. Convert $(2345)_{10}$ to hexadecimal and $(ABCD)_{16}$ to octal number. *[ICT Ministry AP 2017]*
11. কোড কী? BCD এবং Binary কোডের মধ্যে পার্থক্য লিখুন। তিনভিত্তিক সংখ্যা পদ্ধতি সম্পর্কে ব্যাখ্যা করুন। $18^{th} NTRCA School 2024$
12. Explain parity method for error detection. Write down the bit strings of "Delta" using ASCII. $BBAP2023$
13. BCD Addition: $00010011 + 00100110$ *[NPCNL, JAM 2022]*
14. What is the Excess-3 code of 1010? $DPDC2020$
15. What is Boolean function? How can we simplify the following Boolean function?

$$F = A'B'C + A'BC + AB' \quad 45^{th} BCS, CSE2024$$

16. Determine the truth table of $(A \oplus B) \vee (\neg A) \vee (\neg C)$. *[45th BCS, CSE 2024]*
17. State De-Morgan's law with an appropriate example. $BPSCAP2023$
18. Write the function for which the output is 1 for that input. $BAPEXAGM(ICT)2023$
19. $AB + (A(BC))(AC + \overline{BC})$ *[NPCNL, JAM 2022]*
20. Simplify $Y = AB + (\overline{A} + B)C$ in digital logic design. *[BPSC, ADA 2022]*
21. নিম্নলিখিত Boolean Function টি সংক্ষিপ্ত আকারে লিখুন:

$$F(A,B,C,D) = \bar{A}\bar{B}\bar{C} + \bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C} \quad , BPSCS, AE2021$$

22. $A\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{D} + A\bar{B}CD$ convert it into minimum lateral. *[SGFL AGE 2021]*
24. Simplify the following Boolean expression: $(x + y + xy)(x + z)$ *[BPSC AME 2021]*
25. Simplify the Boolean expression as possible $A\bar{B}\bar{C}D + ABCD + A\bar{B}\bar{D}$ *[APSCL AE(ICT) 2021]*

26. Simplify the following expression using Boolean Algebra: $xyz + \bar{x}yz + x\bar{y}$ *[BPSC AP 2021]*
27. Simply the following function: $ABCD + \bar{A}BD + A\bar{B}\bar{C}D$ *[DPDC 2020]*
28. Draw a truth table that satisfies the above-mentioned equation.

$$Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BCD \qquad 38^{th}BCS$$

29. Draw the logic circuit of the Boolean: Expression, $Q = AB + BC(B + C)$; find Q as output where input $(A, B, C) = (1,0,1)$. *[Combined Bank, AME/AE(IT) 2024]*
30. Logic Circuit of Boolean algebra: $Q = \bar{C} + \bar{A}B + BC(B + C)$; Where output Q and input $Q(A, B, C) = (0,0,1)$? *[SB PLC, ADBA 2024]*
31. What is universal gates? Implement $F = AB + CD + E$ using NAND gate only. *[45th BCS, ICT 2024]*
32. What is Universal gate and how is constructed it? $BriCMAME2024$
33. Implement OR gate and AND gate using minimum number of NAND and NOR gate. $PGCBAE(CSE)2024$
34. Why NAND gate is called universal gate? Derive given Boolean expression using only NAND gate. An expression was given consisting of five variables. $MRASCO2023$
35. Derive given Boolean expression using only NAND gate.
$F = A'B'C'D' + AB'CD' + AB'C'D' + A'CD + A'B'CD'$ $,MRASCO2023$
36. Explain: NOR and NAND is a Universal gate. $NPCNL, JAM2022$
37. Define basic logical operations with examples. (AND, OR, NOT) $BPSC, ADA2022$
38. Implement X-OR gate using NAND gate. Maximum 4 NAND gate are using. $PGCBAE(CSE)2021$
39. Universal logic gate কি? 3-input এর একটি Universal logic gate এর Logic Symbol এবং Truth Table দেখান। $BPSCAP2021$
40. What is basic Logic gate? Which gate are called Universal gate and write down advantages of Universal gate? $SPCBLAME2021$
41. How can you Implement AND, OR and NOT gates using only NAND and NOR gates? $BPSCAP2021$
42. Make NAND gate using NOR gate. $BMAAE2021$
43. লজিক গেইট কী? মৌলিক লজিক গেইট কয়টি ও কী কী? সত্যক সারণিসহ আলোচনা করুন। $BPSCANE2020$
44. Truth Table from following circuit (2-bit input A, B full adder with carry bit C_{in}). *[SB PLC, ADBA 2024]*

45. Draw the logic diagram of Half-Adder the truth table of Full-Adder and use half Adder (S) and basic gates to build a Full-Adder. $BPSCSAE2021$
46. Draw a full adder that is made of two half adder and find out the function of output of that full adder. $PSCLecturer2017$
47. Show the differences between clocked J-K flip-flop and clocked D-flip-flop. $45^{th} BCS, CSE2024$
48. What is the difference between Latch and Flip-flop? $SPCBLAME2021, BPSCAP2021, BWDBAP2018, BBAME2017$
49. R-S Flip-flop এর সত্যক সারণি ও বৈশিষ্ট্য আলোচনা করুন। $BPSCANE2020$
50. Draw logic diagram of MOD-12 up counter using only T Flip flop. $BUETM.Sc.April2017$
51. Design a 5×32 decoder with four 3×8 decoders and a 2×4 decoder. Use block diagram construction. *[45th BCS, ICT 2024]*
52. একটি $2:4$ ডিকোডোর ও একটি OR গেট ব্যবহার করে একটি হাফ এডার ডিজাইন কর। *[BTCL, JAM 2024]*

16. Show minimal function using K-Map: $F(A,B,C,D) = \sum(2,8,9,11,13,15)$. *[BPDB AE(CSE) 2024]*

17. Show minimal function using K-Map: $F(A,B,C,D) = \sum(2,8,9,11,13,15)$. *[Teletalk AM(IT) 2023]*

18. Simplify the following Boolean function using K-map: $Y = ABCD + AB\overline{CD} + \overline{A}BC\overline{D} + \overline{A}BCD + ABCD + A\overline{B}C\overline{D} + ABCD + ABCD$ $BPSCAP2023$

19. Show minimal function using K-Map: $F(A,B,C,D) = \sum(2,8,9,11,13,15)$. *[BICIC, AP 2022]*

20. Simplify using K-Map: $F(W,X,Y,Z) = \sum(3,5,7,9,11,13,14,15)$ *[Bank SO(AME) 2021]*

21. Draw the k-map for the equation: $F = A'B'C'D' + A'B'CD' + A'BCD' + A'BCD + AB'C'D' + AB'CD' + ABCD' + ABCD$ *[BOF AE(CSE) 2021]*

22. Simplify $F(A,B,C,D) = ACD + AB + \overline{D} + AC\overline{D}$ using K-map and draw the logic circuits. *[BPSC, ADA 2022]*

23. $F = \overline{ABC} + A\overline{BC} + \overline{AB}C + \overline{A}BC + ABC$, Simplify using K-map with logic circuit. *[JBL,SO (ANE) 2020]*

24. Simplify the equation by using K-map: $f(A,B,C) = \overline{ABC} + \overline{A}BC + \overline{AB}C + ABC$ *[38th BCS]*

Previous Questions — IP Addressing and Subnetting

1. Given IP Address: 192.168.5.154/27, Calculate a) Network Address b) First valid host c) Last valid host d) Broadcast address e) Subnet mask *[NSDA, AME 2024]*
2. Write down the Public and Private IPv4 address for Class A, Class B and Class C. *[NSDA AME 2024]*
3. Given IP address 192.168.1.50, Subnet Mask: 255.255.255.240. Find the valid IP range. Also find Network address and Broadcast address. *[NWPGCL, AM(ICT) 2024]*
4. What is the primary motivation for classful IP address to classless IP addressing? *[SB PCL, ADBA 2024]*
5. Given IP address 192.168.2.0/24; Determine to network address and broadcast address. *[BRICM AME 2024]*
6. Find Network Address, Broadcast Address and Valid Host IP Range from given IP Address: 192.168.1.50 and Subnet Mask: 255.255.255.240 *[BPDB AE(CSE) 2024]*
7. What is a subnet? What benefits will you get using subnets for this office? *[Combined Bank, SO(IT) 2024]*
8. Give IP 192.168.1.50 and subnet mask 255.255.255.240; Find valid host, Network address, and Broadcast address. *[BIWTA AE(CSE) 2023]*
9. What do you mean by Subnet and Subnet Mask? The network address of 172.16.0.0/19 provides how many subnets and hosts? What is the function of OSPF? *[Rupali Bank ANE 2023]*
10. Write down the basic differences Public vs Private IP address *[Rupali Bank ANE 2023]*
11. Consider the following routing table at an IP router: *[BAPEX AGM(ICT) 2023]*

Network	Subnet mask	Outgoing Interface

172.168.164.0	255.255.254.0	Interface 0
172.168.166.0	255.255.254.0	Interface 1
172.168.168.0	255.255.254.0	Interface 2
172.168.170.0	255.255.254.0	Interface 3
0.0.0.0	Default	Interface 4

For each IP address in Group: I identify the correct choice of the outgoing from Group: II using the entries from the routing table above.

Group: I	Group: II
172.168.165.121	Interface 0
172.168.167.151	Interface 1
172.168.163.151	Interface 2
172.168.171.92	Interface 3
0.0.0.0	Interface 4

42. What is private IP range class A, B and C with maximum host of each class? *[BREB AP 2023]*
43. Find out the default mask, network address and broadcast address of the classful IPv4 address: 172.16.99.45 *[BPSC AP 2023]*
44. In HR department have 12 IP enable devices are available in our office and have a big IP block 172.16.5.0/24. To consider your HR department find a suitable IP block than also answer the following question. *[MoL AME 2023]* i. Subnet mask; ii. Number of usable IP address; iii. First and last IP Address of that block iv. Broadcast IP address
45. Given IP address 192.168.1.50, Subnet Mask: 255.255.255.240. Find the valid IP range. Also find network address and Broadcast address. *[BTCL AME 2023]*
46. Given IP address 200.1.2.0 make four equal subnets: *[MRA SCO 2023]* (i) First block address (ii) Total usable hosts (iii) Mention block wise IP address
47. Convert the decimal IP address 192.168.101.5 into binary IP address. Fill-up the following in tabular form: *[Rupali Bank ANE 2023]*
48. What is IP address? Explain the necessity of IP address in network? *[PBL HE 2023]*

49. Identify the class, network IP address, direct broadcast address and limited broadcast address of the following IP address: (i) 1.2.3.4 (ii) 130.1.2.3 (iii) 220.15.1.10 (iv) 200.1.10.100 *[BPSC AE 2022]*
50. What is the subnet mask in 10.2.1.3/22 network? *[BPSC AE 2022]*
51. Find network address, subnet mask, broadcast address and IP host range of 192.168.100.128/26 *[GTCL AE(CSE) 2022]*
52. Given IP Address: 192.168.5.154/26, Calculate network address and subnet mask. *[NSDA, AP 2022]*
53. Define IP 127.0.0.1, what is localhost? *[BMA AE 2021]*
54. What is static IP Address and dynamic IP Address? *[BMA AE 2021]*
55. Answer the following: *[BTCL AM (Technical) 2021]* (i) 192.168.10.0/23, How many usable address? (ii) 192.168.10.0/23, Find subnet mask. (iii) 192.168.10.0/23, Find Broadcast Address. (iv) 192.168.10.0/23, What is last usable host?
56. In a "Class A" network total 20 subnets are needed with maximum 260 hosts per subnets. Can 255.255.255.0 subnet mask be used in this? *[PGCB AE(CSE) 2021]*
57. CIDR কি? 192.168.100.9/26 IP address থেকে (a) Total subnets (b) Block size (c) Valid Hosts (d) Total hosts বের কর। *[BPSC AP 2021]*
58. Using the IP address 192.168.10.0/23 find out- *[SGFL AGE 2021]* (i) Subnet/First address (ii) Last Address (iii) Subnet mask
59. You are given a IP address 172.16.20.0/25 have four subnets. For each department find the following information. *[NRCC AP 2021]*
60. Find the Subnet mask from the following IP: 192.168.3.0/22 *[BOF AE(CSE) 2021]*
61. Difference between IPv4 and IPv6. *[WZPGCL AE(CSE) 2023, RAKUB ANSE 2023, BMA AE 2021, BPSC AME 2021, BREB AGM(IT) 2021]*
62. Suppose a network with IP address 192.16.0.0 is divided into 2 subnets, find number of hosts per subnet. Also for the first subnet, find- *[BAUST AP 2021]* (i) First Subnet address (ii) First host address (iii) Last host address (iv) Broadcast address
63. What is the usable number of host IP addresses available on a network that has a /26 mask? Write down the subnet mask of this network. Write down the first and the last

IP address that can be assigned to host PCs if the network address is 192.168.30.128/26. What address should be used for broadcast purpose in this Network? *[BPSC SAE 2021]*

64. Answer the following: *[BCC AP 2021]* (i) 192.168.10.2/28, Find subnet mask. (ii) 192.168.10.2/28, Find Network Address. (iii) 192.168.10.2/28, Find IP Address of the first host? (iv) 192.168.10.2/28, Find IP Address of the last host? (v) 192.168.10.2/28, Find Broadcast Address.
65. Find Network address, Valid Host, Subnet mask and Broadcast address from 172.16.128.120/25. *[APSCL AE (ICT 2021)]*
66. A network address is given 172.168.0.0/28, divide this network address into 4 subnets and find every subnet address, start address, subnet mask, broadcast address etc. *[RAKUB ME 2021]*
67. What is the subnet mask of 10.2.1.3/26 and what is the usable number of IP address on network that has a 26 mask? *[BPSC AP 2021]*
68. A network address is given 172.168.0.0/28, divide this network address into 4 subnets and find every subnet address, start address, subnet mask, broadcast address etc. *[RAKUB NSE 2021]*
69. What is the subnet mask of 10.2.1.3/26 and what is the usable number of IP address on network that has a 26 mask? *[BPSC AP 2021]*
70. A network address is given 172.18.10.0/23, divide this network address into 4 subnets and find every subnet address, start address, subnet mask, broadcast address etc. *[RAKUB NSE 2021]*
71. What is the range of class C IPv4 address? Suppose, Class C network has four subnets. How many usable PC needed each subnet? *[BGDCL AE (CSE) 2021]*
72. Given IP address is 172.168.10.0/24, administrator wants to create 32 subnets, then find out sub Net-mask, number of address of each subnet, first and last address of subnet 1, first and last address of subnet 32. *[JBL SO(ANE) 2020]*
73. Given IP address 172.16.128.120/25 what is the subnet mask, network address, broadcast address and total usable host in this network? *[NACTAR 2020]*
74. What is private IP? List the class B private IP. *[BTV AP 2019]*

75. Find subnet mask and number of host on each subnet mask at a class B IP is 172.16.2.1/23. *[PSB AP 2018]*
76. Find Network Address, Broadcast Address, Net mask, valid host of the IP: 192.16.13.0/30 *[NESCO AM (ICT) 2018]*
77. A host IP address in a network is written as both 192.68.25.16/28 and 3FFE:85B:IFIF::A9:1234. a) Which one is IPv6 address? How many bits are used for IPv6 address? How does the above IPv6 address completely with the addressing scheme? *[38th BCS]*
78. Given an IP address is 240.133.10.20/8 Find out network address, number of host and subnet mask. *[BWDB AP 2018]*
79. What do we need a static private IP address and what are the advantages? *[37th BCS]*

Previous Questions — Data Communication And Computer Networking

1. Differentiate between half-duplex and full duplex transmission. *[BPSC AP 2023]*
2. উদাহরণসহ Simplex, Half-duplex এবং Duplex কমিউনিকেশন সিস্টেমের পার্থক্য লিখুন। *[17th NTRCA(ICT) 2023]*
3. Distinguish between Synchronous and Asynchronous transmission. *[RAKUB ANSE 2023]*
4. Describe the data communication system with necessary diagram. *[BPSC AP 2023, BPSC AP 2022]*
5. How many types of modes are used in data transferring through networks? Briefly explain those modes. *[Combined Bank, SO(IT) 2024]*
6. What is data communication? Define Simplex, half duplex and full duplex. *[ICT Ministry AP 2017, BKKB AP 2017]*
7. What do you understand by Latency, throughput, and availability of a system? *[45th BCS ICT 2024]*

8. Draw Bit Error Rate vs Signal to Noise Ratio curve of QPSK and BPSK. *[NWPGCL, AM(ICT) 2024]*
9. কম্পিউটারের নেটওয়ার্কে ইনফরমেশন স্থানান্তরে ট্রান্সমিশন স্পিড/ব্যান্ডউইথ বলতে কী বোঝায়? এর একক কী? *[43th BCS 2023]*
10. Bandwidth এবং Through put এর মধ্যে পার্থক্য কী? *[17th NTRCA(ICT) 2023]*
11. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communication. The SNR is usually 3162. What will be the capacity for this channel? *[Combined Bank AP 2023]*
12. Suppose that a digitized TV picture is to be transmitted from a source that uses a matrix of 480 times 500 picture elements (pixels), where each pixel can take on one of 32 intensity values. Assume that 30 pictures are sent per second. (This digital source is roughly equivalent to broadcast TV standards that have been adopted.) Find the source rate R (bps). *[BB AP 2023]*
13. Using the Nyquist theorem, we can sample 12 million times/sec. Four-level signals provide 2 bits per sample, for a total data rate of 24 Mbps. *[NESCO AM(ICT) 2021]*
14. A PCM system have step resolution of 2V. Sinusoidal signal amplitude 10V. SNR=? And total number of bits=? *[BTCL AM (Technical) 2021]*
15. Two channels, one with a bit rate of 190kbps and another with a bit rate 180 kbps are to be multiplexed using g pulse stuffing TDM with no synchronization bits. Answer the following questions: *[RAKUB ME 2021]*

    (a) What is the size of a frame in bits? (b) What is the frame rate? (c) What is the duration of a frame? (d) What is the date rate?
16. Determine the Nyquist sampling rate and the Nyquist sampling interval for the signal $X(t) = \sin(2100\pi t)$. *[38th BCS]*
17. Consider a noiseless channel with a bandwidth of 3 KHz transmitting a signal with two signal levels. What is the maximum bit rate? *[MM AP 2017]*
18. What is Line coding? What are the different line coding techniques? *[SPCBL AME 2021]*

19. Assume we want to transmit the following binary string: 01001110. Show the resulting signal on the one using the following line coding techniques: (i) NRZ-L (ii) Manchester NRZ (iii) Unipolar RZ (binary string: 11011000100) *[BPSC, AE 2022]*
20. What is modulation? Why is it necessary? *[BPSC, AE 2022]*
21. What is modulation? Describe the pulse code modulation (PCM) technique. State the advantage and disadvantage of PCM. *[38th BCS]*
22. What is Modulation and demodulation? Write down five uses? *[BCSCL NE 2017]*
23. What is multiplexing in data communication? Write the differences between TDM and FDM. *[45th BCS, CSE 2024, BPSC AP 2021]*
24. What technique allows simultaneous transmission of multiple signals across a single data link? *[BCC AP 2023]*
25. FDM এবং TDM এর পার্থক্য লিখুন। *[17th NTRCA(CSE) 2023]*
26. Differentiate among TDM, FDM and WDM. How does working process in TDM? *[Combined Bank SO(IT) 2023]*
27. Describe the different types of Multiplexing. *[Combined Bank AME/AHE 2023]*
28. What is Multiplexing? Write about Time division Multiplexing. *[SPCBL AME 2021]*
29. Assume a TDMA based communication system having 8 transmission receiver pairs. Each source is sampled at 8 Khz. That generates 16bits per sample if two synchronization bits are added to each frame calculate the data rate of TDMA line. *[BTCL AM (Technical) 2021]*
30. Where are the low loss transmission windows of silicon based optical fiber and Which window is the most popular in communication and wave. Draw diagram of a long haul WDM Transmission system. *[BTCL AM (Technical) 2021]*
31. Figure shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1bit. Find (a) the input bit duration (b) the output bit duration (c) the output bit rate and (d) the output frame rate. *[BTCL AM (Technical) 2021]*
32. What are the different types of transmission media used for data communication? Explain their advantages and disadvantages. *[Combined Bank, SO(IT) 2024]*

33. Fiber optic cable, Twisted pair cable এবং Co-axial cable এর সুবিধাগুলো বর্ণনা করুন। *[17th NTRCA(ICT) 2023]*
34. ফাইবার অপটিক ক্যাবলের গঠন ও বৈশিষ্ট্য ব্যাখ্যা করুন। *[17th NTRCA(CSE) 2023]*
35. Difference between Guided and Unguided media. Difference between STP and UTP. Why using benefit UTP instead of STP? *[Combined Bank Officer(IT) 2023]*
36. Difference between TCP and UDP. Distinguish between Cat5 and Cat6. *[Combined Bank Officer(IT) 2023]*
37. What is the attenuation of transmission media? Distinguish between twisted pair, co-axial cable and fiber optics in tabular form. *[Rupali Bank ANE 2023]*
38. Why we used straight-through and cross cable with example? *[MoL AME 2023]*
39. Define the principle of data transmission through the fiber optic cable. *[BPSC, AP 2022]*
40. Write down the Working principle of Optical Fiber. *[MGMCL, AM (ICT) 2022]*
41. Which media transfer data with higher bandwidth? Advantages of this media. *[RAKUB NSE 2021]*
42. What is Topology in data communication? What are differences between Bus, Ring, Tree and Star topology? *[Combined Bank SO(IT) 2023]*
43. What is Network Topology? Distinguish between Bus, Ring, Tree and Star topology. Discuss how the Bus topology works. *[Rupali Bank ANE 2023]*
44. একটি নেটওয়ার্কে n সংখ্যক ডিভাইসের জন্য Bus, Mesh এবং Star টপোলজিতে তাদের সংযোগের সংখ্যা কত? *[17th NTRCA(ICT) 2023]*
45. What is network topology? Write the name all different topology used in computer networking with example, diagram and their activities. *[BPSC ADA 2022]*
46. Name and draw different type of network Topologies. *[BB IT 2020, BBL AM (ICT) 2019]*
47. Define network topologies with features. *[NU AP 2020]*
48. 3-way handshake protocol for TCP connection using diagram. *[BriCM AME 2024, BICIC, AP 2022, BGDCL AE (CSE) 2021, SBL Officer(IT) 2021]*

49. Define networking and Internetworking. What are the different types of network? Explain in details. *[BPSC ADA 2022, BPSC ANE 2020]*
50. Describe different types of computer network. *[SPARRSO 2020]*
51. Differentiate between client server network and peer-to-peer network. *[PSC Lecturer 2017]*
52. Briefly describe the following network devices: Repeater, Hub, Bridge, Switch and Router. *[Combined Bank, SO(IT) 2024, BCSCL NE 2017]*
53. Different types of network firewalls. Explain NGFW compared to traditional firewall. *[Combined Bank, AME/AE(IT) 2024]*
54. Write short note: Network, Protocol, link, gateway, Node. *[BREB AP 2023]*
55. দুটি ট্রানসিভলে Router ও Gateway এর মধ্যে কোনটি অধিকতর সুবিধাজনক-মতামত ব্যক্ত করুন। *[17th NTRCA(CSE) 2023]*
56. Difference among Switch, Bridge and Router. *[Combined Bank Officer(IT) 2023, PGCL SAE (CSE) 2021]*
57. What is Firewall? Discuss about different types of Firewall. *[Combined Bank Officer(IT) 2023]*
58. Draw a diagram of LAN including network Firewall. Why is firewall important in network security? List 5 major types of network firewalls. Differentiate between Traditional Firewall and Next Generation Firewall. *[Rupali Bank ANE 2023]*
59. What is gateway? Is router and gateway have any difference? *[BEPZA Programmer 2023]*
60. Write down the difference among Hub, Switch and Router. *[DESCO SAE 2023]*
61. What is purpose of routers? *[BPSC AP 2021]*
62. What is the function of a firewall? *[BCC AP 2023]*
63. Router and Gateway এর ফাংশন লিখুন। *[BPSC AP 2021]*
64. Router and Switch -এর মধ্যে পার্থক্য লিখুন। *[40th BCS 2021]*
65. Difference between Router and Modem. *[SPARRSO 2020]*
66. What is firewall? How you can ensure IT security in the computer system? *[SPARRSO 2020]*

67. Computer network এ router-এর কাজ কী? রাউটারের অপারেশনের ব্লক ডায়গ্রাম আঁকুন। *[40th BCS 2020]*
68. Write short note: Node, Hub, Backbone, Router and Gateway. *[BB AME 2019]*
69. Tabular representation of TCP/IP layer, functions of each layer, Associate protocols, device, and software in each layer. *[Combined Bank, AME/AE(IT) 2024]*
70. Using an explanation of the difference between flow-control and congestion control, discuss the impact of a stable end-to-end latency. *[Combined Bank Officer(IT) 2024]*
71. Differentiate between TCP and UDP. *[Combined Bank SO(IT) 2024, 45th BCS (CSE) 2024, BPSC AP 2021, 37th BCS]*
72. Explain TCP/IP model and its protocol and device. *[BriCM AME 2024, RAKUB NSE 2021, SPARRSO 2020]*
73. Write down the OSI model. *[BriCM AME 2024, NSDA AME 2024]*
74. Write the name of OSI layers protocol for every layers. *[NSDA AME 2024]*

Data Communication & Computer Networking

1. Congestion ? Network-এ কিভাবে Congestion নিরাসন করা যায়? আলোচনা করুন [*18th NTRCA College ICT 2024*]
2. List down the layers of OSI model in top-down manner. [*BPSC AP 2023*, *BARC Programmer 2023*]
3. Difference between OSI model and TCP/IP model. Relation between Data, Segment, Packet and Bit in OSI model. [*Combined Bank SO(IT) 2023*]
4. What is the TCP/IP Model and with layer? [*EGCB SAE(ICT) 2023*, *40th BCS 2021*]
5. Describe briefly the TCP/IP tunneling using appropriate diagram. [*BPSC AP 2023*]
6. What is the number for the Network layer and the support layer? [*BCC AP 2023*]
7. Which layer is used to link the network support layers and user support layers? [*BCC AP 2023*]
8. Fill up the following protocol table by work at which layer? [*BPDB AE(CSE) 2023*]
9. What is OSI and TCP/IP model and briefly explain? [*SPCBL AME 2021*]
10. What is OSI Model? Write all layer name sequence should be top to bottom or bottom to top. [*DESCO AE 2023*, *BTRC SAD 2021*, *RAKUB NSE 2021*]

11. Computer network এর OSI 7-Layer গুলো উদাহরণসহ লিখুন। [*BPSC AP 2021*]
12. What is OSI and TCP/IP model and briefly explain? [*SPCBL AME 2021*]
13. How congestion control works in the TCP? [*BPSC AP 2021*]
14. What is OSI model? Which layers are important for data transfer and user interaction? [*BWDB AP 2018*]
15. Discuss about ISO, OSI and TCP/IP model with figure. [*ICT Ministry AP 2017*]
16. Differentiate between Intranet and Extranet. [*BB AME 2016*]

Error Detection & Correction

17. CRC is a redundancy error technique used to determine the error. Suppose the original data is 11100 and divisor is 1001. [*Combined Bank AP 2023*]
18. Define HTTP and DNS. Suppose that mygov.bd is a domain name. How does this DNS work? [*45th BCS, CSE 2024*]
19. What is the DHCP and NAT in computer networking? [*BRICM AME 2024*]
20. Difference between HTTP and HTTPS. [*PGCB AE(CSE) 2024, RAKUB ANSE 2023, MGMCL AM (ICT) 2022*]
21. Write down the full form of NAT, DHCP, MAC and TCP-IP [*NSDA AME 2024*]
22. Why NAT is not required in IPv6? [*PGCB AE(CSE) 2024*]
23. Discuss the main role of Address Resolution Protocol (ARP) in the network layer of TCP/IP protocol suite. [*BPSC AP 2023*]
24. What is SSL, TLS, and HTTPS? [*MoL AME 2023, EGCB SAE(ICT) 2023*]
25. What is the purpose of IEEE 802.11 committee? [*Combined Bank SO(IT) 2023*]
26. Difference between: (i) SMTP and SNMP [*RAKUB ANSE 2023*]
27. Explain NAT? Explain with topological diagram. [*SGFL AE(IT) 2023*]
28. Explain how NAT works? [*RAKUB ANSE 2023, BPSC ME 2021*]
29. Write HTTP vs. HTTPS? How to recognize HTTPS working? If HTTPS is not working correctly, how can you solve it? [*JRBDB AP 2023*]
30. Explain how can you DHCP work? [*PBL HE 2023*]
31. What is VLAN? Difference between static and dynamic VLAN. [*RAKUB ANSE 2023*]

32. Which two services are required to enable a computer to receive dynamic IP address and access internet using domain names? [*BTCL JAM 2022*, *BB AME 2017*]
33. Write the port address of the following applications of data communications: (i) HTTP; (ii) HTTPS; (iii) FTP; (iv) SMTP; (v) POP [*BPSC ADA 2022*]
34. ব্যাখ্যা করুন: (a) 10Base5 (b) 10Base2 [*BPSC AP 2021*]
35. Explain IPTV and VOIP. How does a browser retrieve IP address from URL? [*BPSC ME 2021*]
36. What is DNS? "TCP/IP is used in DNS"-justify the statement. [*BPSC ME 2021*]
37. Why do we need NAT? What are its advantages? Draw a topology diagram to explain NAT. [*BPSC SAE 2021*, *RAKUB NSE 2021*]
38. Write the full form: TCP/IP, DHCP, CSMA/ [*BOF AE(CSE) 2021*]
39. Difference between broadband Wi-Fi and Wi-Max communication technology. [*BPSC AME 2021*]
40. Explain the difference between Stateless and Stateful protocols. Which type of protocol http is? [*BPSC AP 2021*]
41. Explain 10BaseT. [*RAKUB NSE 2021*]
42. Write down the difference between IPoE and PPPoE. [*RAKUB NSE 2021*]
43. What is APIPA and ARP? Briefly explain ARP. [*RAKUB NSE 2021*]

Data Communication & Networking

44. Write down the full meaning: DHCP, ICMP, ACNS and GARP [*RAKUB NSE 2021*]
45. What is PAT? How does a network PAT work? [*RAKUB NSE 2021*]
46. Which Protocol used for automatically assigning IP addresses? [*NESCO AM(ICT) 2021*]
47. What is NAT? Why used and how NAT translate? [*Bank SO(AME) 2021*]
48. Describe CSMA/CD protocol for medium access control? [*BPSC AME 2021*]
49. Wi-Fi Network সম্পর্কে সংক্ষিপ্ত বিবরণ দিন। Wi-Fi Sensor Network এবং Ad Hoc Network এর মধ্যে পার্থক্য লিখুন। [*BPSC AP 2021*]
50. HTTPS কি? এর ব্যবহারের প্রয়োজনীয়তা সংক্ষেপে বর্ণনা করুন? [*40th BCS 2021*]
51. For the following description of various IP networking protocols write down the protocol name and its full form in the following table: [*BTCL AM (Technical) 2021*]

52. What is the function of ARP in TCP/IP? What is the function of GGSN? [*Combined Bank SO(IT) 2020*]
53. What is main difference between Domain and Workgroup? [*BB AP 2016*]

Advanced Networking & Routing

54. What is Routing? Explain different types of Routing? Why using benefit of an Adhoc routing? Which routing algorithm is used in shortest path algorithm? [*Combined Bank Officer(IT) 2023*]
55. What are static and dynamic routing? Given their relative advantages. [*BPSC AE 2022*]
56. Define distance Vector and Link state routing protocols. [*BPSC AE 2022*]
57. What is Routing? Write down the difference between static routing and dynamic routing. [*RAKUB NSE 2021*]
58. Name of the Algorithm RIP, OSPF and EIGRP routing protocol. [*RAKUB NSE 2021*]
59. What is Autonomous system? What is the difference between Link state routing protocol and Distance vector routing protocol? [*RAKUB NSE 2021*, *SBL Officer(IT) 2021*]

Previous Questions — Banking IT

1. Make a list of banking software used in Bangladesh. List the essential features for successful Banking Software and Apps. *[Combined Bank, Officer (IT) 2020]*
2. How would you test an ATM in a banking system? *[BB, Combined 5 Bank AME 2019]*
3. Write down the use case diagram for ATM. *[CB AP 2018*

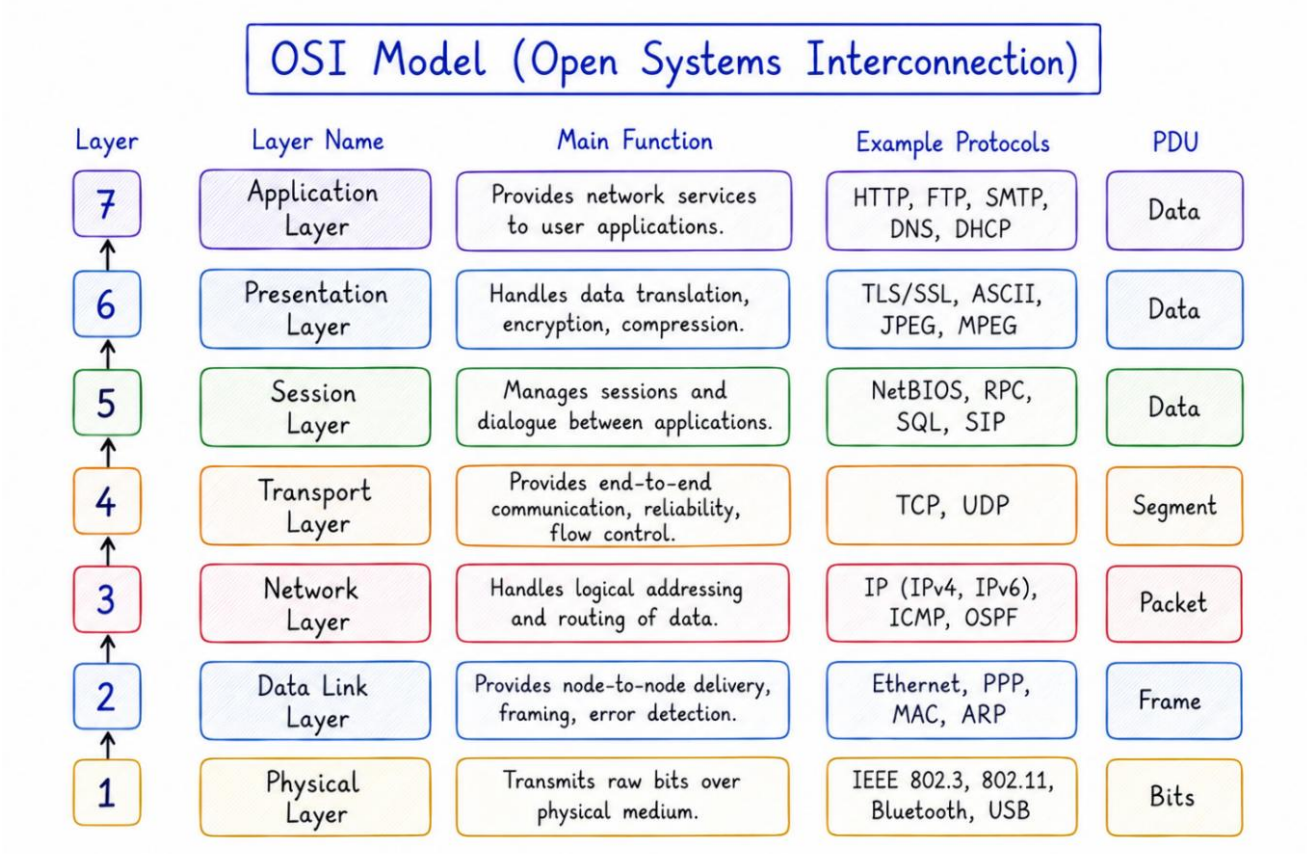
SUPER ICT 150++ QUESTIONS SOLUTION

Answers of Super ICT 150++ Questions

by Bit to Exabyte

01	Describe the OSI Model with Functions and Protocols of Every Layer. Differentiate Between the OSI Model and TCP/IP Model.

The **OSI (Open Systems Interconnection) Model** is a **7-layer conceptual networking model** developed by the **International Organization for Standardization (ISO)**. It standardizes how data is transmitted from one computer to another over a network. Each layer performs a specific function and communicates with the layers directly above and below it.



***see annex 01

পরীক্ষায় শুধু লেয়ারের নাম লিখে এক বা দুই শব্দে কাজ গুলো লিখবেন। অবশ্যই Protocols এর নাম শিখবেন। এমসিকিউ তে কাজে লাগবে।

Layer 7 – Application Layer

Provides network services directly to end users and application software. Interface between user applications and the network.

Responsibilities	Protocols
Email services Web browsing File transfer Remote login Network management	HTTP / HTTPS, FTP, SMTP,POP3, IMAP DNS, DHCP, SNMP Telnet, SSH

Example

Opening a website using a browser.

Layer 6 – Presentation Layer

Responsible for **data representation**.

Responsibilities	Protocols
Data encryption Data decryption Data compression Data decompression Character encoding Data format conversion	SSL/TLS JPEG,PNG, GIF, MPEG ASCII Unicode

Layer 5 – Session Layer

Creates, manages, and terminates communication sessions between applications.

Responsibilities	Protocols
Session establishment Session maintenance Session synchronization	NetBIOS RPC PPTP

□ Session termination	SIP

ex. Maintaining a video conference session.

Layer 4 – Transport Layer

Responsibilities	Protocols
□ Segmentation □ Flow control □ Error control □ Reliable delivery □ Multiplexing □ Port addressing	TCP (Connection oriented) UDP (Connectionless) SCTP

Provides reliable end-to-end communication.

Example : TCP guarantees reliable delivery of files.

Layer 3 – Network Layer

Responsible for routing packets from source to destination.

Responsibilities	Protocols
□ Logical addressing □ Routing □ Packet forwarding □ Path selection □ Fragmentation	IP (IPv4, IPv6) ICMP, ARP, OSPF RIP, BGP, EIGRP **Devices** Router, Layer-3 Switch

Layer 2 – Data Link Layer

Transfers data between directly connected devices.

Responsibilities	Protocols
□ Framing □ Physical (MAC) addressing □ Error detection □ Flow control □ Media access control	Ethernet (IEEE 802.3), Wi-Fi (IEEE 802.11) PPP, HDLC Frame Relay **Devices** Switch, Bridge

Layer 1 – Physical Layer

Transmits raw bits over the physical medium. Most of the time , Hacking taken place here. Because Attacker directly take raw data from this Layer.

Responsibilities	Protocols
□ Bit transmission □ Voltage levels □ Connectors □ Cables □ Signal encoding	**Standards** Ethernet cables, Fiber optic USB, Bluetooth Physical Layer RS-232 **Devices:** Hub, Repeater, Cables

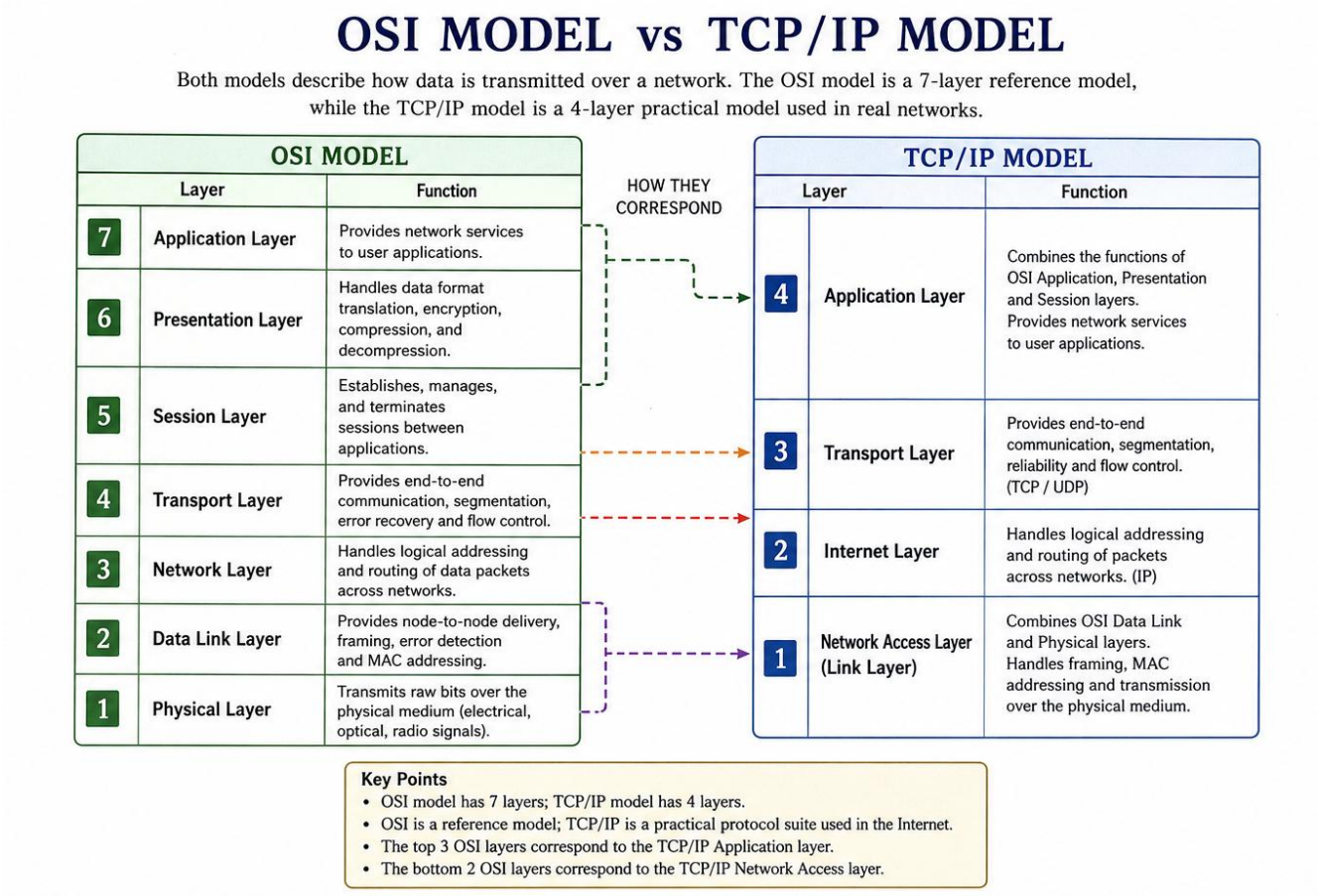
Protocols Used at Each Layer (For mcq)

Layer	Main Function	Common Protocols
Application	User services	HTTP, HTTPS, FTP, SMTP, DNS, DHCP
Presentation	Encryption, Compression	SSL/TLS, JPEG, MPEG
Session	Session management	NetBIOS, RPC, SIP
Transport	Reliable delivery	TCP, UDP
Network	Routing	IP, ICMP, ARP, OSPF, RIP
Data Link	Framing, MAC Address	Ethernet, PPP, HDLC
Physical	Bit transmission	Ethernet cable, Fiber, USB

Difference Between OSI Model and TCP/IP Model

Feature	OSI Model	TCP/IP Model
Developed By	ISO	DARPA (DoD)
Number of Layers	7	4
Nature	Reference model	Practical protocol suite/ In real, we use it.
Application Layer	Separate Application, Presentation, Session layers	Combined into one Application layer
Transport Layer	Transport	Transport
Network Layer	Network	Internet

Data Link + Physical	Separate layers	Combined into Network Access (Link) layer
Protocol Dependency	Protocol independent	Built around TCP/IP protocols
Usage	Teaching and design	Real Internet communication
Flexibility	More modular	Simpler and widely implemented



OSI vs TCP/IP Layer Mapping

OSI Layer	TCP/IP Layer
Application	Application
Presentation	
Session	
Transport	Transport
Network	Internet
Data Link	Network Access
Physical	

The OSI model provides a structured framework for understanding network communication through seven layers, each with specific functions and protocols. In contrast, the TCP/IP model is a practical four-layer protocol suite used on the Internet. While OSI is mainly used for learning and network design, TCP/IP forms the foundation of modern computer networks.

Practice Questions

a) A user accesses a website by typing a URL in a web browser. Explain how the data travels through all seven layers of the OSI Model from the sender to the receiver.
b) What is the relationship between the OSI Model and the TCP/IP Model?
c) MAC Address vs IP Address
d) Physical Address vs Logical Address
e) TCP vs UDP

02	A computer system has a cache access time of 10 ns, a hit rate of 90%, and a main memory access time of 100 ns. Calculate the Average Memory Access Time (AMAT).

[এই টাইপের মিনিমাম ৪/৫ টা ম্যাথ প্র্যাক্টিস করবেন। যেকোন ধরনের প্রশ্নের উত্তরের জন্য আমাদের ওয়েবসাইটে ভিজিত করুন। হোমপেইজ এ Request Answer বাটনে ক্লিক করে প্রশ্ন পাঠান অথবা Resource Browse করে প্রশ্নটি খুজে বের করুন]

Given:

Cache Access Time (Hit Time) = 10 ns

Hit Rate = 90% = 0.9

Miss Rate = 1 − 0.9 = 0.1

Main Memory Access Time (Miss Penalty) = 100 ns

formula 01:

$$AMAT = \text{Hit Time} + (\text{Miss Rate} \times \text{Miss Penalty})$$

$$AMAT = 10 + (0.1 \times 100)$$

$$= 10 + 10 = 20 \text{ ns}$$

(Formula 2: Probability Method)

$$AMAT = (\text{Hit Rate} \times \text{Cache Time}) + \text{MissRate} \times (\text{CacheTime} + \text{MainMemoryTime})$$

$$AMAT = (0.9 \times 10) + 0.1 \times (10 + 100)$$

$$= 9 + (0.1 \times 110)$$

$$= 9 + 11 = 20 \text{ ns}$$

95% of memory accesses are served directly from the **cache** in **2 ns**. Only **5%** of accesses miss the cache and require fetching data from **main memory**, increasing the average access time. Even though main memory is much slower (**100 ns**), the high cache hit ratio keeps the **Average Memory Access Time very low (7 ns)**.

Practice Questions

1. **Define the following terms:**

- Cache Hit
- Cache Miss
- Hit Ratio (Hit Rate)
- Miss Ratio (Miss Rate)
- Hit Time
- Miss Penalty

2. **Explain the following cache mapping techniques with suitable examples:**

    1. Direct Mapping
    2. Fully Associative Mapping
    3. Set-Associative Mapping

3. **A processor has the following cache hierarchy:**

    - L1 Cache: Hit Time = 1 ns
        - Hit Ratio = 95%
    - L2 Cache: Hit Time = 5 ns
        - Hit Ratio = 90%
    - Main Memory Access Time = 80 ns

Calculate the **Average Memory Access Time (AMAT)** considering both cache levels.

03	Read The following table [আইটির সকল পরীক্ষায় sql আসবেই। তাই কমন আনকমন সকল sql পড়ার ট্রাই করবেন। আমি যেগুলো দিয়েছি সেগুলো অবশ্যই পড়বেন। Descriptive Part এ আমি বেসিক থেকে Advanced পর্যন্ত sql এর ব্যাখ্যা সহ অধ্যায় দিয়েছি, পড়বেন]

emp_id	emp_name	salary
1	Rahim	50000
2	Karim	70000
3	Jamil	90000
4	Salma	70000

34 What is the difference between synchronous and asynchronous counters? Design a MOD-6 counter using flip-flops (explain logic steps).

A **counter** is a sequential digital circuit that counts clock pulses and changes its state according to the applied clock signal. Counters are widely used in digital clocks, timers, frequency dividers, and event-counting applications.

Counters are mainly classified into:

1. **Asynchronous (Ripple) Counter**
2. **Synchronous Counter**

Difference between Synchronous and Asynchronous Counters

Feature	Asynchronous Counter	Synchronous Counter
Clock Input	Only the first flip-flop receives the clock	All flip-flops receive the same clock simultaneously
Speed	Slower	Faster
Propagation Delay	High (ripple effect)	Low
Circuit Complexity	Simple	More complex
Accuracy	Lower for high-speed applications	Higher
Hardware Requirement	Less	More
Applications	Simple counters, low-speed circuits	High-speed digital systems

Design of a MOD-6 Counter

A **MOD-6 counter** counts from **0 to 5** and then repeats.

Step 1: Determine the Number of Flip-Flops

Number of flip-flops required:

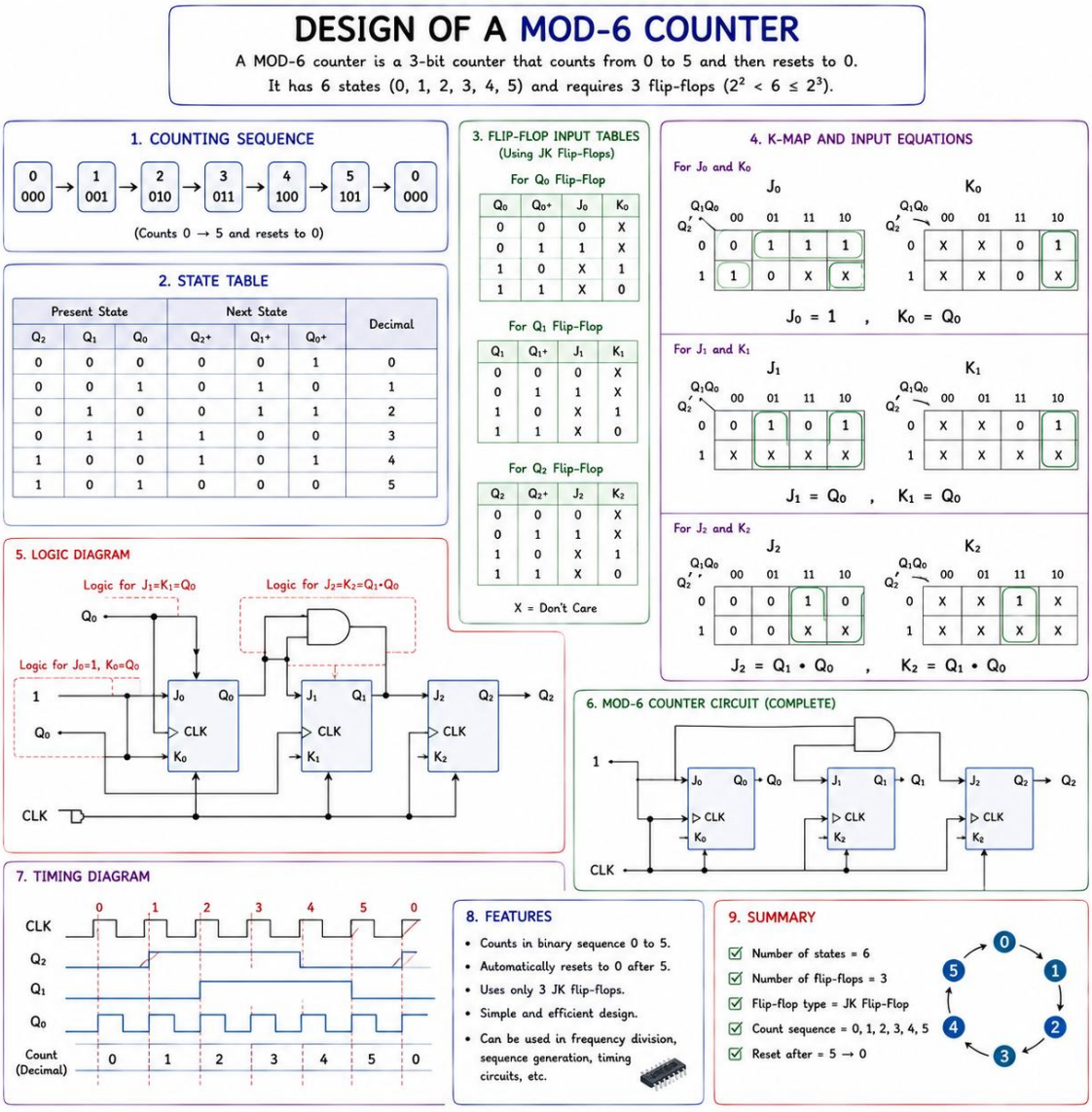
Number of flip-flops required:

$$2^n \geq 6$$
$$2^2 = 4 < 6$$
$$2^3 = 8 \geq 6$$

Therefore, **3 flip-flops** are required.

Step 2: Count Sequence

Decimal	Q2	Q1	Q0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1



***** You need to know only the logic diagram circuit.**

পরীক্ষায় শুধু লজিক ডায়াগ্রাম দিতে হয়। বাকি কিছু আপনার বুঝতে দেয়া হয়েছে।

When Q2 = 1 and Q1 = 1, the asynchronous clear resets all flip-flops to 000.

Circuit Components

- Three JK (or T) Flip-Flops
- Common clock (for synchronous design)
- AND gate for reset logic
- Clear input connected to all flip-flops

Advantages of Synchronous Counter

- High speed
- Low propagation delay
- Reliable
- Suitable for modern digital systems

A **MOD-6 counter** requires **3 flip-flops** because (2^3 = 8) is the smallest power of two greater than or equal to 6. It counts from 000 to 101 and then resets to 000, providing a counting sequence of six unique states. Compared to asynchronous counters, synchronous counters are faster and more suitable for high-speed digital applications.

35 Differentiate between Functional Testing and Non-Functional Testing with mentioning testing names. Compare Black Box Testing and White Box Testing with suitable examples.

Software Testing is the process of verifying and validating software to ensure that it meets specified requirements and works correctly. It helps identify defects, improve software quality, and ensure reliability before deployment.

Software testing can broadly be divided into **Functional Testing** and **Non-Functional Testing**.

Difference between Functional Testing and Non-Functional Testing

Feature	Functional Testing	Non-Functional Testing

Purpose	Verifies what the system does	Verifies how well the system performs
Based On	Functional requirements	Non-functional requirements
Focus	Features and business logic	Performance, security, usability, reliability
Checks	Correct outputs	Quality attributes
User Perspective	Functional behavior	System performance and quality

Common Functional Testing Types

1. Unit Testing
2. Integration Testing
3. System Testing
4. Acceptance Testing (UAT)
5. Smoke Testing
6. Sanity Testing
7. Regression Testing
8. Interface Testing

Common Non-Functional Testing Types

1. Performance Testing
2. Load Testing
3. Stress Testing
4. Volume Testing
5. Scalability Testing
6. Security Testing
7. Usability Testing
8. Compatibility Testing
9. Reliability Testing
10. Recovery Testing

Black Box Testing vs White Box Testing

Black Box Testing evaluates the software's functionality without examining its internal code. The tester provides inputs and verifies outputs based on the requirements.

White Box Testing examines the internal code, logic, control flow, and program structure. The tester has knowledge of the source code and verifies its correctness.

Difference between Black Box and White Box Testing

Feature	Black Box Testing	White Box Testing
Knowledge of Code	Not required	Required
Focus	Functionality	Internal code and logic
Performed By	Testers	Developers/Testers
Test Basis	Requirements and specifications	Source code and algorithms
Objective	Validate external behavior	Verify internal implementation
Techniques	Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, State Transition Testing	Statement Coverage, Branch Coverage, Path Coverage, Condition Coverage

Example of Black Box Testing

Suppose a login page requires:

- Valid Username
- Valid Password

Test Cases:

Username	Password	Expected Result
admin	1234	Login Successful
admin	9999	Invalid Password

wrong	1234	Invalid Username

The tester does not know how the login logic is implemented.

Example of White Box Testing

Consider the following C code:

```
if(mark >= 40)
    printf("Pass");
else
    printf("Fail");
```

The tester designs test cases to execute:

- mark = 50 → Pass branch
- mark = 30 → Fail branch

This ensures **100% branch coverage**.

Functional Testing verifies that software features work according to the specified requirements, while **Non-Functional Testing** evaluates quality attributes such as performance, security, usability, and reliability. **Black Box Testing** focuses on external functionality without knowledge of the source code, whereas **White Box Testing** examines the internal code structure and execution paths to ensure complete logic coverage. Together, these testing approaches help deliver reliable, secure, and high-quality software.

36 Compare P, NP, NP-Complete, and NP-Hard with suitable examples.(গুরুত্বপূর্ণ)

In **Computational Complexity Theory**, problems are classified based on the amount of time required by an algorithm to solve them. The four important complexity classes are **P, NP, NP-Complete, and NP-Hard.**

1. P (Polynomial Time)

A problem belongs to **P** if it can be **solved** by a deterministic algorithm in **polynomial time**, such as (O(n)), (O(n^2)), or (O(n^3)).

Characteristics

- Easy (efficiently solvable).
- Solution can be found quickly.
- Verification is also easy.

Examples

- Binary Search
- Merge Sort
- Breadth-First Search (BFS)
- Depth-First Search (DFS)
- Dijkstra's Algorithm (without negative weights)

2. NP (Non-deterministic Polynomial Time)

A problem belongs to **NP** if its proposed solution can be **verified** in polynomial time, even if finding the solution may be difficult.

Characteristics

- Solution verification is efficient.
- Solving may require exponential time.
- Every P problem is also an NP problem.

Examples

- Sudoku Verification
- Hamiltonian Path Verification
- Boolean Satisfiability (SAT)

3. NP-Complete

A problem is **NP-Complete** if:

1. It belongs to **NP**.
2. Every NP problem can be reduced to it in polynomial time.

These are the hardest problems within NP.

Characteristics

- Solution verification is polynomial.
- No known polynomial-time algorithm.

- If one NP-Complete problem is solved in polynomial time, then **P = NP**.

Examples

- Traveling Salesman Problem (Decision Version)
- SAT (Boolean Satisfiability)
- 3-SAT
- Vertex Cover (Decision)
- Clique Problem

4. NP-Hard

A problem is **NP-Hard** if it is at least as hard as every NP problem.
Unlike NP-Complete problems, an NP-Hard problem **does not have to belong to NP**, meaning its solution may not even be verifiable in polynomial time.

Characteristics

- Harder than or equal to NP-Complete problems.
- May not have polynomial-time verification.
- Can include optimization problems.

Examples

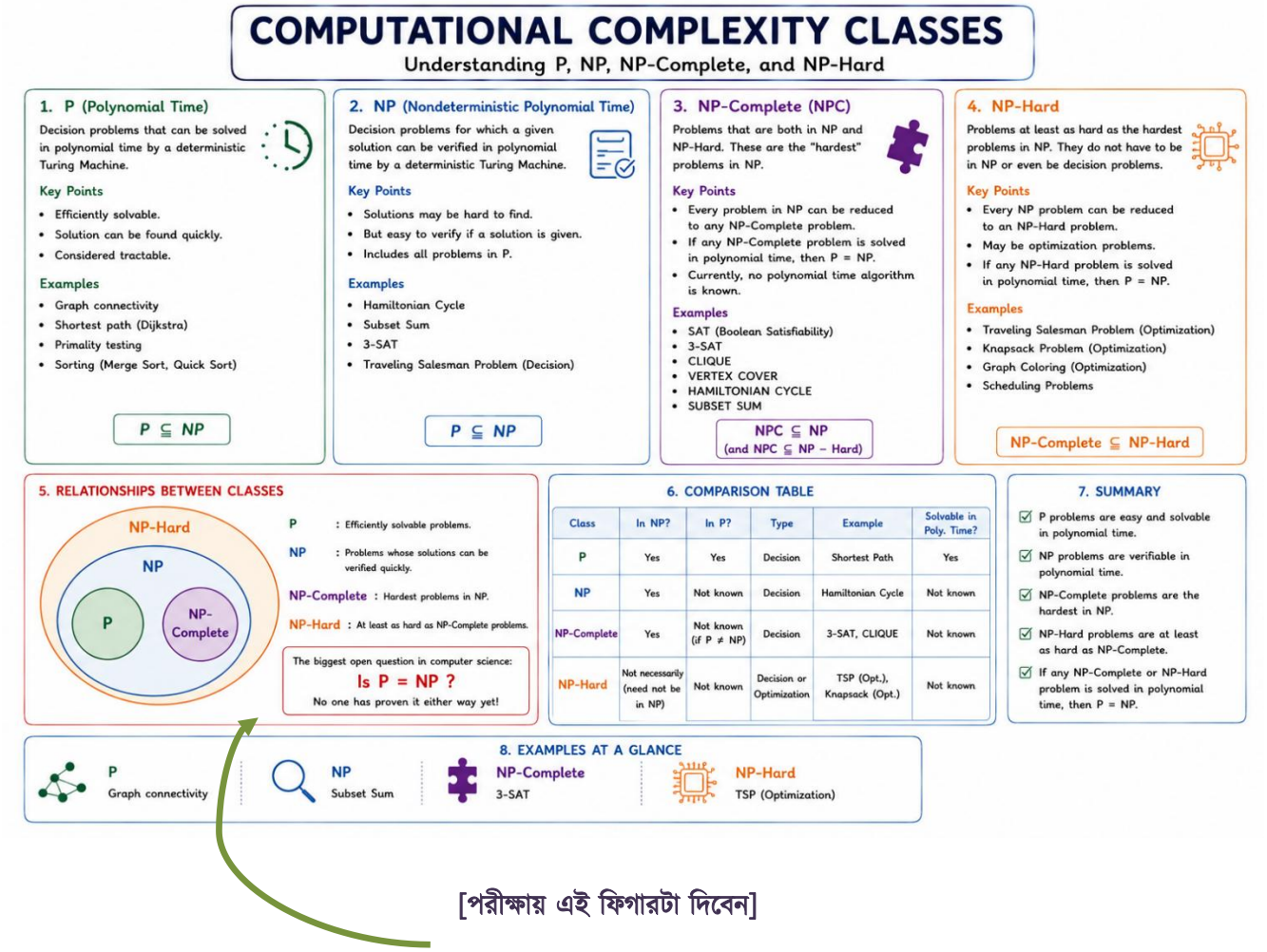
- Traveling Salesman Problem (Optimization Version)
- Halting Problem
- Job Scheduling Optimization

Comparison of P, NP, NP-Complete, and NP-Hard

Feature	P	NP	NP-Complete	NP-Hard
Solvable in Polynomial Time	Yes	Not known	Not known	Not known
Verifiable in Polynomial Time	Yes	Yes	Yes	Not always
Deterministic Algorithm	Yes	Not necessarily	No known	No known
Hardness	Easy	Harder	Hardest in NP	Hardest overall
Belongs to NP	Yes	Yes	Yes	Not always

Examples	BFS, DFS, Sorting	SAT Verification	3-SAT, Clique	TSP Optimization, Halting Problem

Relationship among Complexity Classes



[পরীক্ষায় এই ফিগারটা দিবেন]

- **P** contains problems that can be solved efficiently.
- **NP** contains problems whose solutions can be verified efficiently.
- **NP-Complete** contains the hardest problems in NP.
- **NP-Hard** contains problems that are at least as difficult as NP problems and may not even be verifiable in polynomial time.

37 Define Breadth-First Search (BFS) and Depth-First Search (DFS). State one application of each. Given the graph below, perform BFS and DFS traversal starting from vertex A.

Graph

Vertices:

{A, B, C, D, E, F, G}

Edges:

(A,B), (A,C), (B,D), (B,E), (C,F), (E,G), (F,G)

Breadth-First Search (BFS)

Breadth-First Search (BFS) is a graph traversal algorithm that visits all neighboring vertices first before moving to the next level. It uses a **Queue (FIFO)** data structure.

Application of BFS

- Finding the shortest path in an unweighted graph.

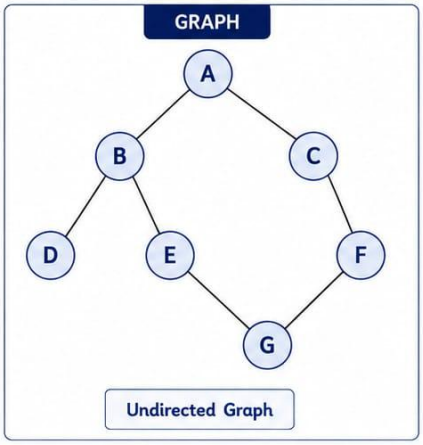
Depth-First Search (DFS)

Depth-First Search (DFS) is a graph traversal algorithm that explores one branch as deeply as possible before backtracking. It uses a **Stack (LIFO)** or **Recursion**.

Application of DFS

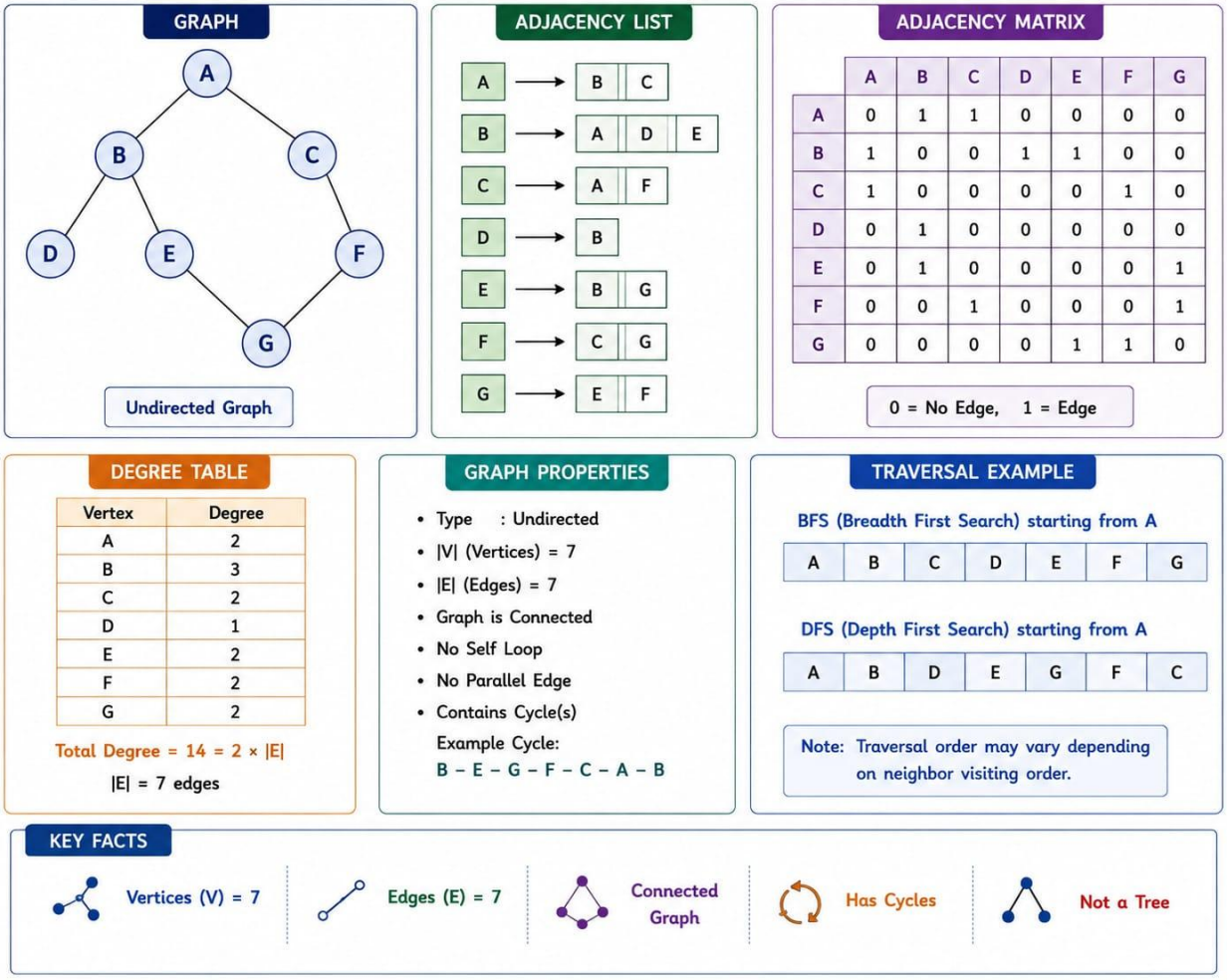
- Cycle detection in a graph.

Graph



Vertex	Adjacent Vertices
A	B, C

B	A, D, E
C	A, F
D	B
E	B, G
F	C, G
G	E, F



BFS Traversal (Starting from A)

Queue Operations

Step	Queue	Visited
Start	A	A
1	B C	A

2	C D E	A B
3	D E F	A B C
4	E F	A B C D
5	F G	A B C D E
6	G	A B C D E F
7	Empty	A B C D E F G

BFS Order

$$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$$

DFS Traversal (Starting from A)

Following the leftmost unvisited vertex first:

DFS Order

$$A \rightarrow B \rightarrow D \rightarrow E \rightarrow G \rightarrow F \rightarrow C$$

Difference between BFS and DFS

Feature	BFS	DFS
Full Form	Breadth-First Search	Depth-First Search
Data Structure	Queue	Stack / Recursion
Traversal	Level by level	Depth first
Shortest Path	Yes (Unweighted Graph)	No
Memory Usage	Higher	Lower
Application	Shortest path	Cycle detection, Topological sorting

BFS visits vertices level by level using a queue and is useful for finding the shortest path in unweighted graphs. **DFS** explores one branch completely before backtracking using a stack or recursion and is commonly used for cycle detection and graph analysis.

38 Write a C program that accepts N integers and performs the following operations:

- Display all elements
- Find the maximum and minimum values
- Calculate the sum and average

- Search for an element
- Reverse the array

The following program reads **N integers** into an array and performs all the required operations: displaying the elements, finding the maximum and minimum values, calculating the sum and average, searching for an element, and reversing the array.

Algorithm

1. Read the number of elements N.
2. Input the array elements.
3. Display all elements.
4. Find the maximum and minimum values.
5. Calculate the sum and average.
6. Read the search element and search it in the array.
7. Reverse the array.
8. Display the reversed array.

C Program

```
#include <stdio.h>

int main()
{
    int arr[100], n, i;
    int max, min, sum = 0;
    float avg;
    int search, found = 0, temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter %d integers:\n", n);

    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("\nArray Elements: ");
    for(i = 0; i < n; i++)
        printf("%d ", arr[i]);

    max = min = arr[0];
```

```
    for(i = 0; i < n; i++)
    {
        if(arr[i] > max)
            max = arr[i];

        if(arr[i] < min)
            min = arr[i];

        sum += arr[i];
    }

    avg = (float)sum / n;

    printf("\nMaximum = %d", max);
    printf("\nMinimum = %d", min);
    printf("\nSum = %d", sum);
    printf("\nAverage = %.2f", avg);

    printf("\nEnter element to search: ");
    scanf("%d", &search);

    for(i = 0; i < n; i++)
    {
        if(arr[i] == search)
        {
            found = 1;
            printf("Element found at index %d", i);
            break;
        }
    }

    if(found == 0)
        printf("Element not found");

    for(i = 0; i < n/2; i++)
    {
        temp = arr[i];
        arr[i] = arr[n-1-i];
        arr[n-1-i] = temp;
    }

    printf("\nReversed Array: ");

    for(i = 0; i < n; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

Sample Input

Enter number of elements: 5

Enter 5 integers:

10 20 30 40 50

Enter element to search: 30

Sample Output

Array Elements: 10 20 30 40 50
Maximum = 50
Minimum = 10
Sum = 150
Average = 30.00
Element found at index 2
Reversed Array:
50 40 30 20 10

This program demonstrates several fundamental array operations in C, including displaying elements, finding the maximum and minimum values, calculating the sum and average, searching for a specific element using linear search, and reversing the array. Each operation runs in **linear time O(n)**, making the program efficient for basic array processing tasks.

39 Briefly describe topology and classification. What topology is best? Compare Bus, Star, Ring, Mesh, Tree, and Hybrid Topologies with respect to Physical Structure, Cost, Scalability, Reliability, Fault Tolerance, Ease of Maintenance, Performance, and Suitable Applications.

A **Network Topology** is the physical or logical arrangement of computers, networking devices, and communication links in a computer network. It defines how devices are connected and how data flows between them.

The choice of topology affects the network's **performance, scalability, reliability, maintenance, and cost**.

Classification of Network Topology

Network topology is mainly classified into two types:

1. Physical Topology

- Describes the actual physical arrangement of cables, switches, routers, and devices.
- Example: Star, Bus, Ring.

2. Logical Topology

- Describes how data travels through the network, regardless of the physical layout.
- Example: Ethernet (Logical Bus), Token Ring (Logical Ring).

Types of Network Topology

1. Bus Topology
2. Star Topology
3. Ring Topology
4. Mesh Topology
5. Tree Topology
6. Hybrid Topology

Comparison of Bus, Star, Ring, Mesh, Tree, and Hybrid Topologies

Feature	Bus	Star	Ring	Mesh	Tree	Hybrid
Physical Structure	Single backbone cable	All devices connect to a central hub/switch	Devices connected in a closed loop	Every device connected to every other device	Hierarchical structure with parent-child branches	Combination of two or more topologies
Cost	Lowest	Moderate	Moderate	Highest	Moderate to High	High
Scalability	Poor	Excellent	Moderate	Poor (for large networks)	Excellent	Excellent
Reliability	Low	High	Moderate	Very High	High	Very High

Fault Tolerance	Low	High (failure of one cable affects only one device)	Moderate	Excellent	High	Very High
Ease of Maintenance	Difficult	Easy	Moderate	Difficult	Easy	Moderate
Performance	Decreases with more devices	High	Moderate	Excellent	High	Excellent
Suitable Applications	Small temporary LANs	Offices, schools, universities	Industrial and token ring networks	Military, banking, critical systems	Large organizations, campuses	Enterprise and data center networks

Advantages and Disadvantages

Bus Topology

Advantages

- Low installation cost.
- Requires less cable.
- Easy to install.

Disadvantages

- Backbone failure stops the entire network.
- Difficult fault detection.
- Performance decreases as devices increase.

Star Topology

Advantages

- Easy installation.
- Easy maintenance.
- High performance.
- Easy fault isolation.
- Scalable.

Disadvantages

- Central switch failure affects the entire network.
- Requires more cabling.
- Higher cost than bus.

Ring Topology

Advantages

- No data collisions.
- Equal network access for all devices.
- Predictable performance.

Disadvantages

- Failure of one node may disrupt the network.
- Difficult to troubleshoot.
- Adding/removing devices affects the network.

Mesh Topology

Advantages

- Very reliable.
- Excellent fault tolerance.
- Multiple communication paths.
- High security.

Disadvantages

- Very expensive.
- Large amount of cabling.
- Complex installation and maintenance.

Tree Topology

Advantages

- Highly scalable.
- Easy management.
- Easy expansion.
- Suitable for large organizations.

Disadvantages

- Backbone failure affects multiple branches.
- More cabling required.
- Configuration is relatively complex.

Hybrid Topology

Advantages

- Flexible.
- Highly scalable.
- Reliable.
- Combines strengths of multiple topologies.

Disadvantages

- Expensive.
- Complex design and maintenance.
- Requires skilled administrators.

Which Topology is Best?

There is **no single topology that is best for every situation**. The choice depends on the network size, budget, performance requirements, and reliability needs.

However, for **most modern LANs**, the **Star Topology** is considered the best because:

- Easy to install and maintain.
- High performance.
- Easy fault detection.
- Scalable.
- Widely used with Ethernet switches.
- Failure of one cable affects only the connected device, not the entire network.

For **mission-critical systems** (such as military networks or data centers), **Mesh Topology** provides the highest reliability and fault tolerance but at a much higher cost.

Network topology determines how devices are interconnected and communicate within a network. Among the various topologies, **Star Topology** is the most widely used due to its simplicity, scalability, reliability, and ease of maintenance. **Mesh Topology** offers the highest fault tolerance for critical applications, while **Hybrid Topology** is preferred for large enterprise environments requiring flexibility and scalability.

40 Write the name of the fastest sorting algorithm. What would be the complexity of an already sorted array if we use Quick Sort? OR Trace the complete execution of Quick Sort for the following array using the first element as the pivot. Show every partition and recursive call.

Array:

{50, 30, 70, 10, 90, 40, 80, 20}

Fastest Sorting Algorithm

There is **no single fastest sorting algorithm for all cases**, but for comparison-based sorting, **Quick Sort** is generally considered one of the fastest in practice due to its excellent average-case performance and cache efficiency.

Time Complexity of Quick Sort

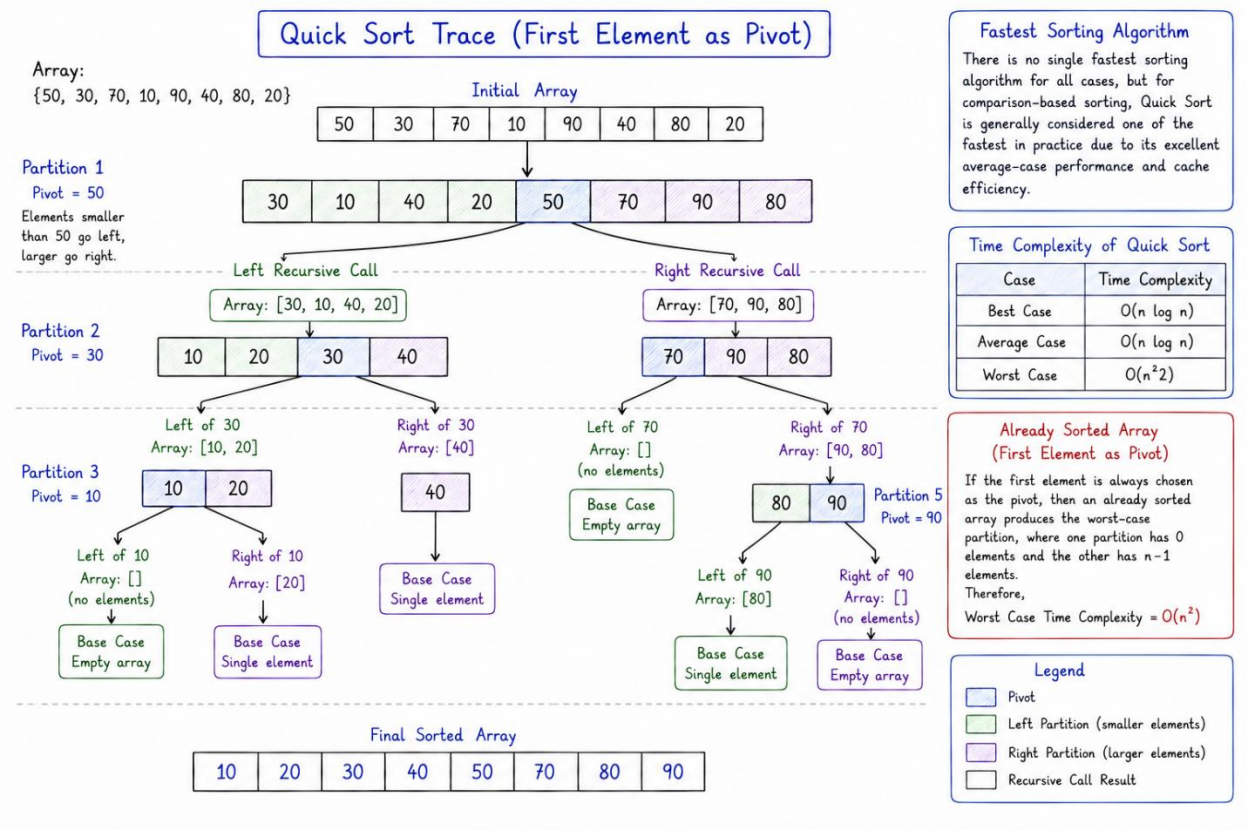
Case	Time Complexity
Best Case	O(n log n)
Average Case	O(n log n)
Worst Case	$O(n^2)$

Complexity for an Already Sorted Array (First Element as Pivot)

If the **first element is always chosen as the pivot**, then an already sorted array produces the **worst-case partition**, where one partition has **0 elements** and the other has **n−1 elements**. Therefore,

$$\text{Worst Case Time Complexity} = O(n^2)$$

Quick Sort Trace (First Element as Pivot)



Final Sorted Array

10 20 30 40 50 70 80 90

Quick Sort is one of the most efficient sorting algorithms with an average time complexity of **O(n log n)**. However, when the **first element is consistently chosen as the pivot** and the input array is already sorted, Quick Sort exhibits its **worst-case time complexity of O(n²)** due to highly unbalanced partitions.

41	Construct a Binary Search Tree (BST) by inserting the following keys in order: **50, 30, 70, 20, 40, 60, 80, 23, 77, 11, 90**

Then perform:

- **Inorder Traversal**
- **Preorder Traversal**
- **Postorder Traversal**

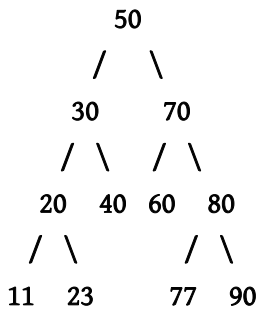
A **Binary Search Tree (BST)** is a binary tree in which:

- All keys in the **left subtree** are **smaller** than the root.
- All keys in the **right subtree** are **greater** than the root.
- Both left and right subtrees are also BSTs.

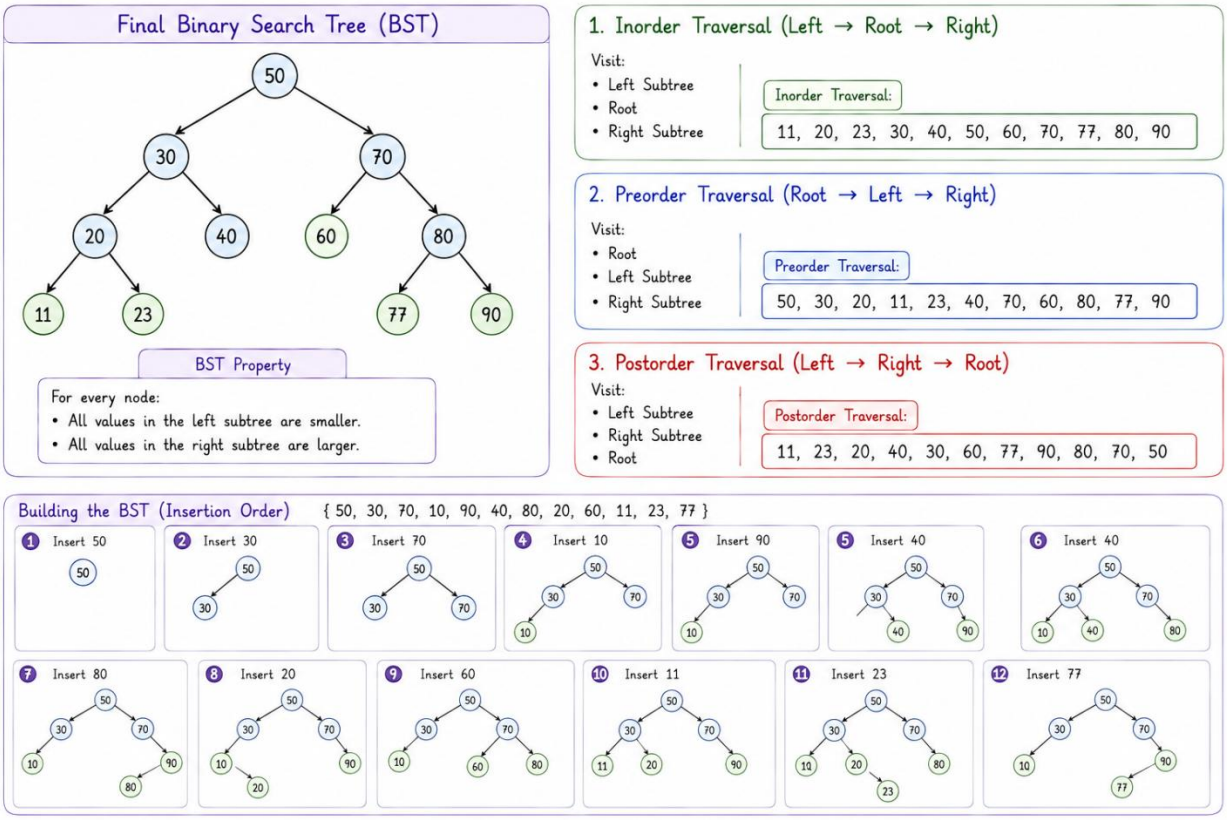
Step-by-Step Construction

Insert the keys in the given order:

Final Binary Search Tree



Inorder Traversal (Left → Root → Right)



Traversal	Output
Inorder	11, 20, 23, 30, 40, 50, 60, 70, 77, 80, 90
Preorder	50, 30, 20, 11, 23, 40, 70, 60, 80, 77, 90
Postorder	11, 23, 20, 40, 30, 60, 77, 90, 80, 70, 50

A **Binary Search Tree (BST)** organizes data such that the left subtree contains smaller values and the right subtree contains larger values than the root. After inserting the given keys, the constructed BST produces the following traversals: **Inorder** (sorted order) → **11, 20, 23, 30, 40, 50, 60, 70, 77, 80, 90**, **Preorder** → **50, 30, 20, 11, 23, 40, 70, 60, 80, 77, 90**, and **Postorder** → **11, 23, 20, 40, 30, 60, 77, 90, 80, 70, 50.**

Construct the Binary Tree from Given Traversals

1. **Preorder + Inorder**

   Construct the binary tree from:

   Preorder: A, B, D, E, C, F, G

   Inorder: D, B, E, A, F, C, G

2. **Postorder + Inorder**

   Construct the binary tree from:

   Postorder: 10, 30, 20, 60, 80, 70, 50

   Inorder: 10, 20, 30, 50, 60, 70, 80

   Then find:

   1. Preorder traversal
   2. Root node
   3. Leaf nodes
   4. Height of the tree

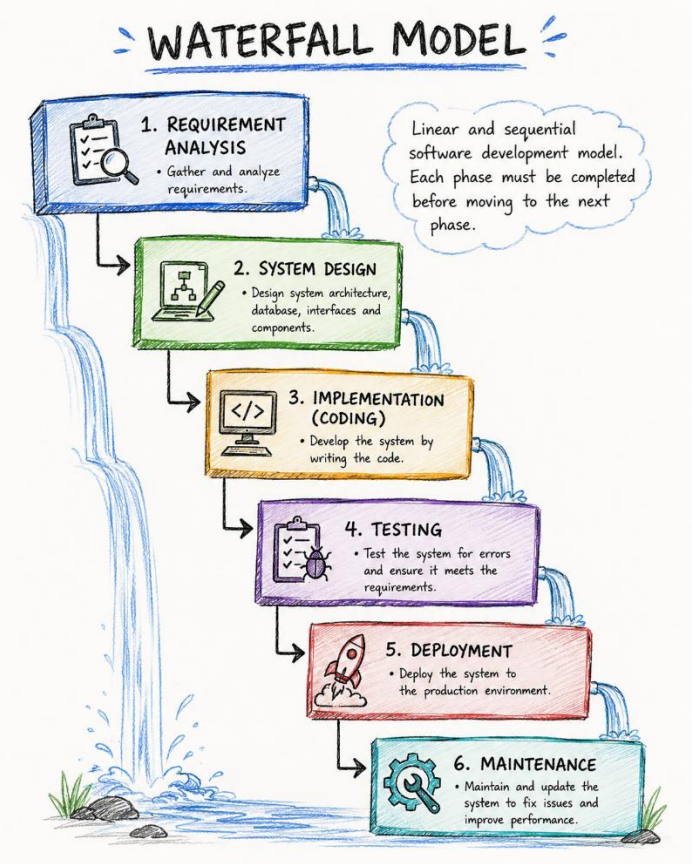
42 Differentiate between the Waterfall, Agile, and Spiral Model with suitable examples. When and why are the Waterfall and Agile models applicable?

A **Software Development Life Cycle (SDLC) model** is a structured approach used to plan, develop, test, deploy, and maintain software. Different SDLC models are suitable for different types of projects based on their requirements, complexity, and flexibility.

The three widely used SDLC models are:

1. Waterfall Model
2. Agile Model
3. Spiral Model

1. Waterfall Model



The **Waterfall Model** is a **linear and sequential** software development model. Each phase must be completed before moving to the next phase, and there is little or no overlap between phases.

Phases

Requirement Analysis
↓
System Design
↓
Implementation (Coding)
↓
Testing
↓
Deployment
↓
Maintenance

Developing a **payroll system** for a government office where requirements are fixed and rarely change.

Advantages

- Simple and easy to understand.
- Well-defined documentation.
- Easy project management.
- Suitable for small and fixed-scope projects.

Disadvantages

- Difficult to accommodate requirement changes.
- Testing begins only after coding is complete.
- High cost of correcting errors discovered late.

2. Agile Model

The **Agile Model** is an **iterative and incremental** development model. Software is developed in small iterations called **sprints**, allowing continuous feedback and frequent updates.

Example

Developing a **mobile banking application** where customer requirements change frequently and new features are added regularly.

Advantages

- Flexible to changing requirements.
- Faster delivery of working software.
- Continuous customer involvement.
- Early detection and correction of defects.

Disadvantages

- Requires experienced team members.
- Less predictable cost and schedule.
- Documentation may be less comprehensive.

3. Spiral Model

The **Spiral Model** combines features of the Waterfall and Prototyping models with a strong focus on **risk analysis**. Development proceeds through repeated cycles (spirals), each involving planning, risk assessment, engineering, and evaluation.

Phases of Each Spiral

1. Planning
2. Risk Analysis
3. Engineering (Development)
4. Customer Evaluation

Example

Developing an **air traffic control system** or **banking software**, where risks must be identified and minimized before implementation.

Advantages

- Excellent risk management.
- Suitable for large and complex projects.
- Easy to incorporate changing requirements.
- High-quality software.

Disadvantages

- Expensive.
- Complex to manage.
- Requires risk analysis expertise.

Difference among Waterfall, Agile, and Spiral Models

Feature	Waterfall	Agile	Spiral
Development Style	Sequential	Iterative & Incremental	Iterative with Risk Analysis
Requirement Changes	Difficult	Easy	Easy
Customer Involvement	Low	High	High
Risk Management	Low	Moderate	Excellent
Flexibility	Low	High	High
Testing	After development	Throughout development	Every iteration
Cost	Low	Moderate	High
Documentation	Extensive	Moderate	Extensive
Suitable Project Size	Small to Medium	Small to Large	Large and Complex

When and Why is the Waterfall Model Applicable?

Applicable When

- Requirements are clear, complete, and fixed.
- The project scope is unlikely to change.
- Technology is well understood.
- Small or medium-sized projects.
- Government or regulatory projects requiring detailed documentation.

Why Use Waterfall?

- Easy to manage.
- Simple planning and scheduling.
- Well-defined milestones.
- Comprehensive documentation.
- Predictable development process.

Examples

- Payroll Management System
- Library Management System
- Examination Result Processing System

When and Why is the Agile Model Applicable?

Applicable When

- Requirements change frequently.
- Continuous customer feedback is required.
- Fast delivery is important.
- Large teams work collaboratively.
- Rapid product releases are needed.

Why Use Agile?

- Adapts quickly to changing requirements.
- Delivers working software in short iterations.
- Improves customer satisfaction.
- Reduces project risk through continuous testing and feedback.

Examples

- E-commerce websites
- Mobile applications
- Social media platforms
- Online food delivery systems

The **Waterfall Model** is best suited for projects with stable and well-defined requirements, while the **Agile Model** is ideal for projects requiring flexibility, frequent customer feedback, and continuous delivery. The **Spiral Model** is preferred for large, complex, and high-risk projects because it emphasizes risk analysis throughout development.

43 Write a C program to find GCD of two numbers using functions. Let, num1 = 20; num2 = 15; Result (GCD) = 5.

The **Greatest Common Divisor (GCD)** or **Highest Common Factor (HCF)** of two numbers is the largest positive integer that divides both numbers without leaving a remainder.

Example: **GCD(20,15)=5**

Algorithm

1. Read two integers.
2. Call a function gcd(a, b).
3. Use the Euclidean algorithm:
    - If b == 0, return a.
    - Otherwise, return gcd(b, a % b).
4. Print the GCD.

C Program

```
#include <stdio.h>

   int gcd(int a, int b)
      {
          if(b == 0)
              return a;
          else
              return gcd(b, a % b);
      }

int main()
```

```
{
    int num1, num2;

    printf("Enter first number: ");
    scanf("%d", &num1);

    printf("Enter second number: ");
    scanf("%d", &num2);

    printf("GCD = %d", gcd(num1, num2));

    return 0;
}
```

Sample Output

Enter first number: 20

Enter second number: 15

GCD = 5

Working gcd(20,15)

= gcd(15,5)

= gcd(5,0)

= 5

The Euclidean algorithm is the most efficient method for finding the GCD of two numbers. It repeatedly replaces the larger number with the remainder until the remainder becomes zero. The last non-zero value is the GCD.

44 Differentiate between Deterministic Finite Automata (DFA) and Nondeterministic Finite Automata (NFA).

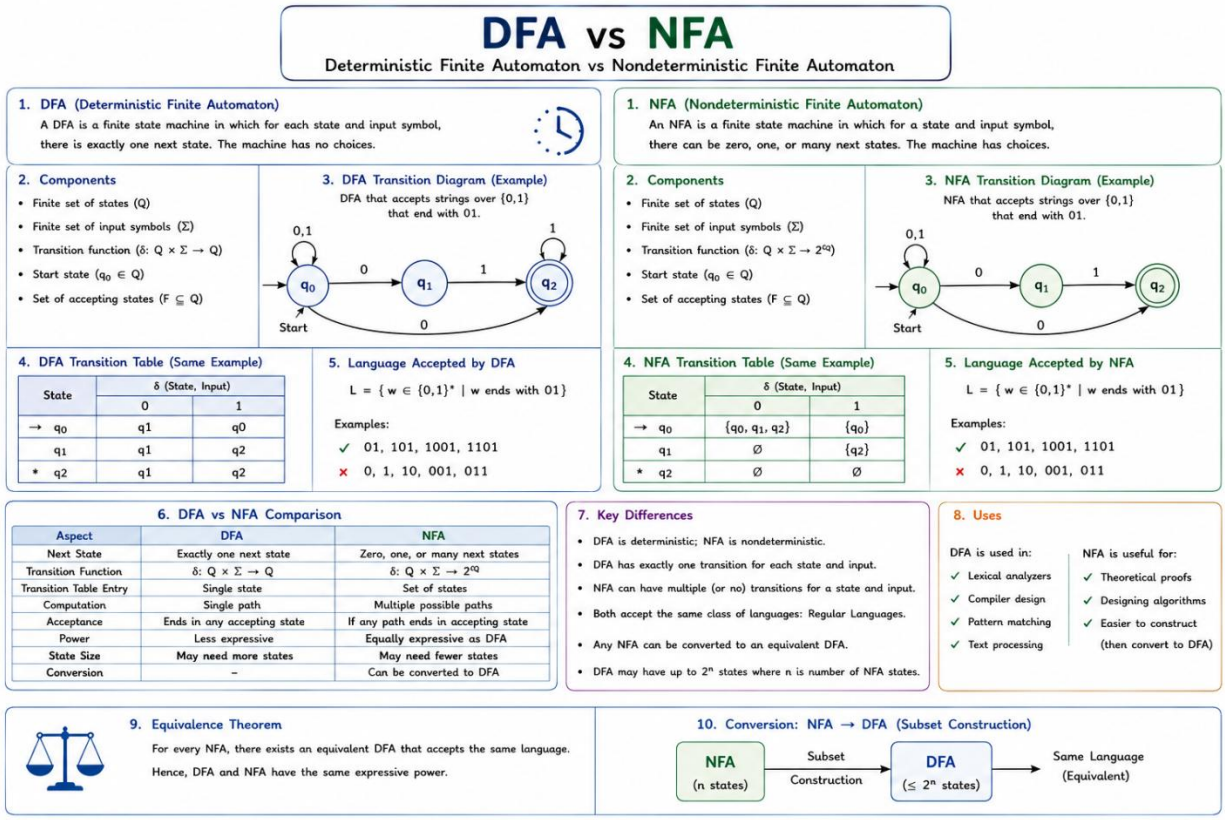
A **Finite Automaton (FA)** is a mathematical model used to recognize **regular languages.** It consists of a finite number of states and transitions between those states based on input symbols.

There are two main types:

1. Deterministic Finite Automata (DFA)
2. Nondeterministic Finite Automata (NFA)

Difference between DFA and NFA

Feature	DFA	NFA
Full Form	Deterministic Finite Automata	Nondeterministic Finite Automata
Next State	Exactly one next state for each input	Zero, one, or multiple next states
ε (Epsilon) Transition	Not allowed	Allowed
Transition Function	One transition per input symbol	Multiple transitions may exist
Computation	Single computation path	Multiple possible paths
Acceptance	Accepts if the single path ends in a final state	Accepts if **any** path reaches a final state
Design	Easier to implement	Easier to construct
Execution Speed	Faster	Conceptually explores multiple paths
Number of States	Usually more	Usually fewer



Applications

DFA

- Lexical analysis in compilers
- Pattern matching
- Input validation

NFA

- Regular expression engines
- Automata theory
- Language recognition

A **DFA** has exactly one transition for each input symbol from every state and does not allow ε-transitions, making it deterministic and straightforward to implement. An **NFA** may have multiple transitions (including ε-transitions) for the same input symbol, making it easier to design. Despite these differences, **DFA and NFA are equivalent in expressive power**, as both recognize exactly the **regular languages**.

OR Differentiate between Regular Language and Context-Free Language (CFL).

A **Regular Language (RL)** is a language that can be recognized by a **Finite Automaton (FA)** or described using a **Regular Expression**.

A **Context-Free Language (CFL)** is a language generated by a **Context-Free Grammar (CFG)** and recognized by a **Pushdown Automaton (PDA)**.

Difference between Regular Language and Context-Free Language

Feature	Regular Language (RL)	Context-Free Language (CFL)
Recognized By	Finite Automaton (DFA/NFA)	Pushdown Automaton (PDA)
Grammar Type	Regular Grammar	Context-Free Grammar
Memory	No extra memory	Uses Stack
Expressive Power	Less powerful	More powerful
Nested Structures	Cannot handle nested patterns	Can handle nested and recursive patterns
Examples	(0+1)*, strings ending with 01	Balanced parentheses, (a^n b^n)

Regular Language

$$L = \{ \text{Strings ending with 01} \}$$

Examples:

01

101

1101

1001

Recognized by a **Finite Automaton**.

Context-Free Language

$$L = \{ a^n b^n \mid n \geq 1 \}$$

Examples:

ab

aabb
aaabbb
aaaabbbb

This language **cannot** be recognized by a finite automaton because it requires counting, but it **can** be recognized by a **Pushdown Automaton (PDA)** using a stack.

A **Regular Language** is simpler and can be recognized by finite automata without additional memory, making it suitable for pattern matching and lexical analysis. A **Context-Free Language** is more expressive because it supports nested and recursive structures through the use of a stack, making it essential for parsing programming languages and representing hierarchical constructs such as balanced parentheses and syntax trees.

45 What is an ER Diagram? Draw an ER Diagram for a Library Management System with the following requirements: [এই টাইপের মিনিমাম ৪/৫ টা ER প্র্যাক্টিস করবেন]

a) **A library has many books.**
b) **Each book is written by one author.**
c) **Members can borrow multiple books.**
d) **Each borrowing record stores the issue date and return date.**

An **Entity-Relationship (ER) Diagram** is a graphical representation of a database that shows **entities**, **attributes**, and the **relationships** among them. It is used during database design to model the logical structure of a database before implementation.

Main Components of an ER Diagram

1. **Entity** – A real-world object (e.g., Book, Author, Member).
2. **Attribute** – A property of an entity (e.g., BookID, Title).
3. **Relationship** – Association between entities (e.g., Borrows, Writes).

Entities and Attributes

Output:

name	deptid	salary
Sakib	1	90000
Hasan	2	55000
Nila	3	80000
Tania	4	45000

For every employee, the correlated subquery calculates the maximum salary of that employee's department. The outer query keeps the employee whose salary equals that maximum.

CASE Expression

Suppose we want to classify employees according to salary.

```
SELECT
    name,
    salary,
    CASE
        WHEN salary >= 80000 THEN 'High'
        WHEN salary >= 60000 THEN 'Medium'
        ELSE 'Low'
    END AS salary_level
FROM Employee;
```

Output:

name	salary	salary_level
Rahim	60000	Medium
Karim	75000	Medium
Jannat	50000	Low
Hasan	55000	Low
Nila	80000	High
Rafi	65000	Medium
Tania	45000	Low

Sakib	90000	High

CASE works like conditional logic.

salary >= 80000 → High

salary >= 60000 → Medium

otherwise → Low

Advanced SQL Problem — Department with Highest Average Salary

```
SELECT d.deptname, AVG(e.salary) AS average_salary
FROM Employee e
JOIN Department d
ON e.deptid = d.deptid
GROUP BY d.deptid, d.deptname
ORDER BY average_salary DESC
LIMIT 1;
```

Output:

deptname	average_salary
ICT	75000

1. Join employees with departments.
2. Group employees by department.
3. Calculate average salary.
4. Sort averages in descending order.
5. LIMIT 1 returns the department with the highest average.

SQL problem using SELF JOIN with table, query, output, and explanation.

SELF JOIN

A **SELF JOIN** is a JOIN operation in which a table is joined with itself. It is useful when records within the same table have a relationship with each other. A common example is an employee-manager relationship, where the manager is also stored as an employee in the same table.

Employee Table

empid	name	managerid	salary
101	Rahim	NULL	90000
102	Karim	101	70000
103	Hasan	101	65000
104	Jannat	102	50000
105	Rafi	102	55000
106	Nila	103	48000

Here, managerid contains the empid of the employee's manager.

For example:

- Karim's managerid is 101 → Rahim is Karim's manager.
- Hasan's managerid is 101 → Rahim is Hasan's manager.
- Jannat's managerid is 102 → Karim is Jannat's manager.
- Rafi's managerid is 102 → Karim is Rafi's manager.
- Nila's managerid is 103 → Hasan is Nila's manager.
- Rahim has no manager, so his managerid is NULL.

Problem 1: Find the name of every employee along with the name of their manager.

SQL Query:

```
SELECT
    e.name AS employee,
    m.name AS manager
```

```
FROM Employee e
LEFT JOIN Employee m
ON e.managerid = m.empid;
```

Output:

employee	manager
Rahim	NULL
Karim	Rahim
Hasan	Rahim
Jannat	Karim
Rafi	Karim
Nila	Hasan

How the Query Works:

The same Employee table is used twice.

Employee e → represents the employee

Employee m → represents the manager

The aliases e and m allow SQL to distinguish between the two uses of the same table.

The important condition is:

ON e.managerid = m.empid

For example, Karim has:

Karim.managerid = 101

The query searches for the employee whose empid is 101:

empid = 101 → Rahim

Therefore:

Karim → Rahim

Similarly:

Jannat.managerid = 102

102 → Karim

Therefore:

Jannat → Karim

Why LEFT JOIN is Used:

LEFT JOIN is used because Rahim does not have a manager.

Rahim.managerid = NULL

Therefore, there is no matching manager record, but LEFT JOIN still keeps Rahim in the result.

employee	manager
Rahim	NULL

If INNER JOIN were used, Rahim would be excluded because he has no matching manager.

Problem 2: Find employees whose manager earns more than them.

SQL Query:

```
SELECT
    e.name AS employee,
    e.salary AS employee_salary,
    m.name AS manager,
    m.salary AS manager_salary
FROM Employee e
JOIN Employee m
ON e.managerid = m.empid
WHERE m.salary > e.salary;
```

Output:

employee	employee_salary	manager	manager_salary
Karim	70000	Rahim	90000
Hasan	65000	Rahim	90000
Jannat	50000	Karim	70000
Rafi	55000	Karim	70000
Nila	48000	Hasan	65000

How the Query Works:

First, the SELF JOIN establishes the employee-manager relationship:

ON e.managerid = m.empid

Then the WHERE condition compares their salaries:

WHERE m.salary > e.salary

For example:

Karim salary = 70000

Rahim salary = 90000

Since:

90000 > 70000

Karim appears in the result.

Similarly:

Jannat = 50000

Karim = 70000

Since:

70000 > 50000

Jannat also appears.

Problem 3: Find employees who have the same salary.

SELF JOIN can also be used to compare rows within the same table.

SQL Query:

```
SELECT
    e1.name AS employee1,
    e2.name AS employee2,
    e1.salary
FROM Employee e1
JOIN Employee e2
ON e1.salary = e2.salary
```

```
AND e1.empid < e2.empid;
```

If two employees have the same salary, they will appear as a pair.

The condition:

e1.empid < e2.empid

prevents duplicate results such as:

Rahim - Karim

Karim - Rahim

from being displayed separately.

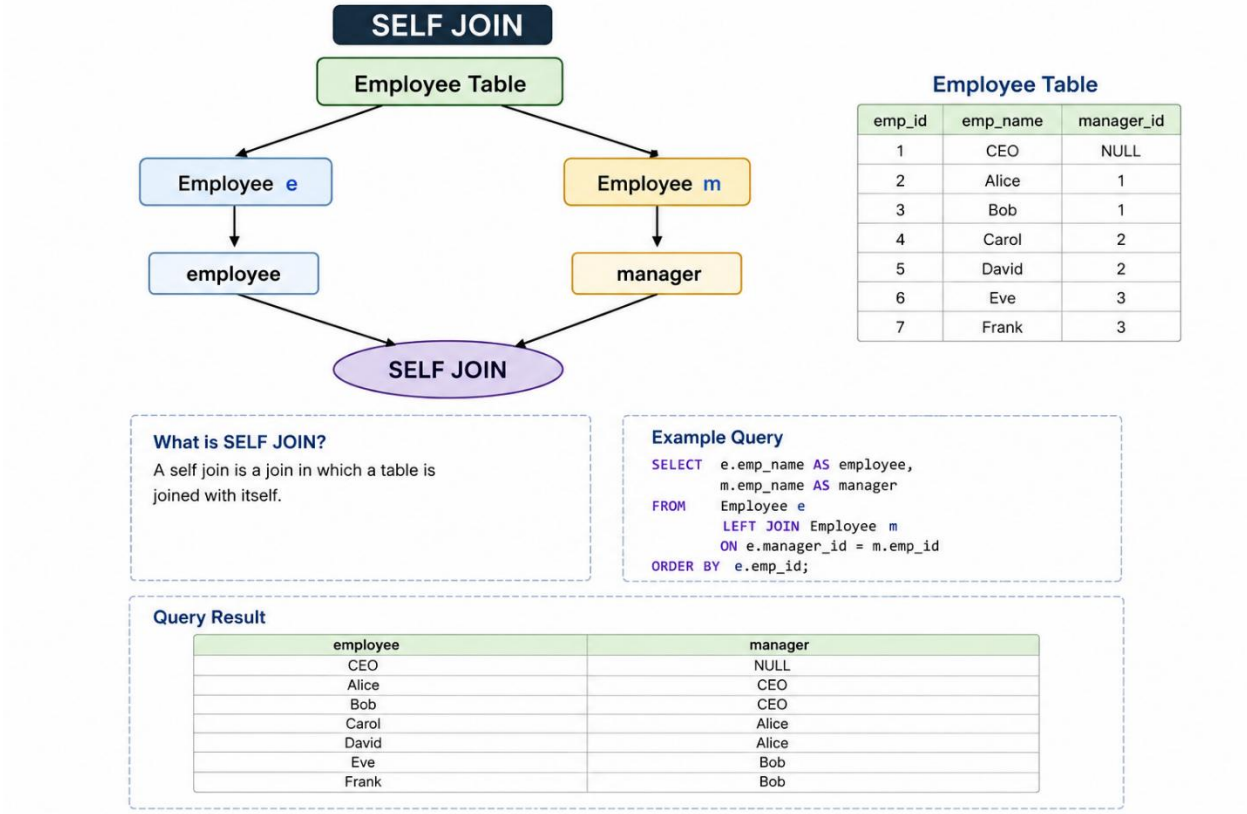
General Structure of SELF JOIN:

```
SELECT ...
FROM TableName A
JOIN TableName B
ON A.column = B.column;
```

Here, A and B are aliases representing two logical copies of the same table.

Important Point:

There is no special SQL keyword called SELF JOIN. It simply means that the **same table is referenced more than once using different aliases**.



Uses of SELF JOIN:

1. Employee-manager relationships.
2. Organizational hierarchy.
3. Parent-child relationships.
4. Category-subcategory relationships.
5. Finding duplicate records.
6. Comparing employees within the same table.
7. Comparing salaries between different employees.
8. Comparing dates or values between rows of the same table.

Key Concept:

SELF JOIN = Joining a table with itself using different aliases

For an employee-manager problem, the most important pattern to remember is:

```
SELECT e.name AS employee, m.name AS manager
FROM Employee e
JOIN Employee m
```

```
    ON e.managerid = m.empid;
```

This is one of the most common **SQL interview, viva, and competitive-exam problems involving SELF JOIN**.

Important SQL Query Patterns

Requirement	Common SQL technique
Filter rows	WHERE
Sort rows	ORDER BY
Remove duplicates	DISTINCT
Count rows	COUNT()
Find average	AVG()
Find maximum	MAX()
Find minimum	MIN()
Total	SUM()
Group records	GROUP BY
Filter groups	HAVING
Combine tables	JOIN
Find unmatched rows	LEFT JOIN / RIGHT JOIN
Nested query	Subquery
Correlated comparison	Correlated subquery
Ranking	Window functions
Conditional output	CASE
Top-N per group	ROW_NUMBER() / ranking functions
Recursive hierarchy	Recursive CTE
Temporary named result	CTE

Advanced Practice Problems

1: Find the employee who earns the **third-highest distinct salary**.

2: Find all employees whose salary is greater than the **average salary of their own department**.

3: Find the department having the **highest total salary**.

4: Find the **second-highest-paid employee from every department**.

5: Find employees whose salary is greater than **every employee in the HR department**.

6: Find employees whose salary is greater than **at least one employee in the Finance department**.

7: Find departments whose **average salary is greater than the overall company average salary**.

8: Find the employee with the **maximum salary without using MAX()**.

9: Find duplicate employee names.

10: Find the **top 3 salaries in each department**, including ties.

These problems cover the progression from **basic SQL → filtering → aggregation → joins → subqueries → correlated subqueries → window functions → advanced analytical queries**.

Chapter 02

Memory management including paging, frames, pages, page tables,

Memory Management

Memory management is an important function of an operating system that manages the computer's **main memory (RAM)**. It keeps track of which portions of memory are being used, which are free, and which processes are using particular memory locations.

The main objectives of memory management are to:

1. Allocate memory to processes.
2. Deallocate memory when a process finishes.
3. Keep track of free and occupied memory.
4. Provide protection between processes.
5. Support efficient utilization of RAM.
6. Translate logical/virtual addresses into physical addresses.
7. Support techniques such as **paging, segmentation, and virtual memory**.

For example, if three processes require memory:

Process P1 → 200 MB

Process P2 → 500 MB

Process P3 → 300 MB

The operating system decides where these processes will be placed in RAM and manages their memory throughout their execution.

2. Logical Address and Physical Address

Two important concepts in memory management are **logical address** and **physical address.**

Logical Address

A **logical address**, also called a **virtual address**, is the address generated by the CPU while a program is executing. The CPU generally does not directly use the actual RAM location.

Physical Address

A **physical address** is the actual address/location in **main memory (RAM)** where the data or instruction is stored.

The **Memory Management Unit (MMU)** performs the translation:

CPU

↓

Logical / Virtual Address

↓

MMU

↓

Physical Address

↓

RAM

For example:

CPU generates: 5000
↓
MMU
↓
Physical address: 15000
↓
RAM

The mapping depends on the memory-management technique being used.

Contiguous Memory Allocation

In **contiguous memory allocation**, each process is allocated a single continuous block of physical memory.

The major problem with contiguous allocation is **external fragmentation**.

Fragmentation

Fragmentation means memory is wasted because of the way memory is allocated.

There are two major types:

1. **Internal Fragmentation**
2. External Fragmentation

Internal Fragmentation

Internal fragmentation occurs when allocated memory contains some unused space inside the allocated block.

A process requires: **18 KB**

but the system allocates: **20 KB**

Then:

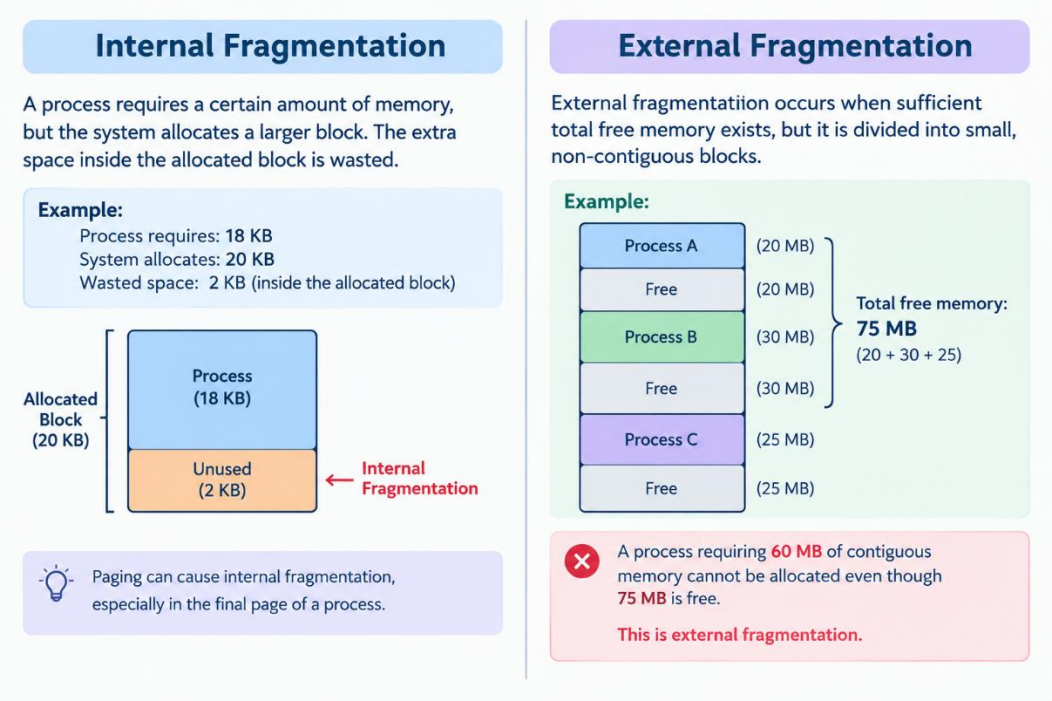
$$20 - 18 = 2 \text{ KB}$$

The 2 KB is wasted inside the allocated block.

Paging can cause internal fragmentation, especially in the final page of a process.

External Fragmentation

External fragmentation occurs when sufficient total free memory exists, but it is divided into small, non-contiguous blocks.



Total free memory:

$$20 + 30 + 25 = 75 \text{ MB}$$

A process requiring **60 MB of contiguous memory** cannot be allocated even though 75 MB is free.

This is **external fragmentation**.

5. Paging

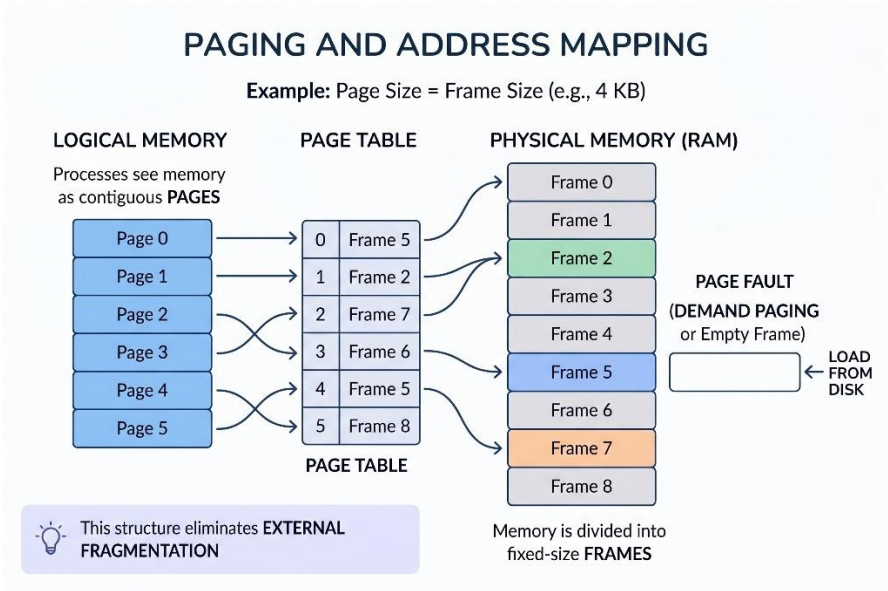
Paging is a non-contiguous memory-management technique that divides:

- Logical/virtual memory into fixed-size blocks called **pages**.
- Physical memory into fixed-size blocks called **frames**.

The size of a page and frame is always the same.

For example, if page size is: **4 KB**

then frame size is also: **4 KB**.



...

Pages of a process do **not need to be stored consecutively** in physical memory.

A page can be loaded into **any available frame** in RAM. Therefore, a process does **not need contiguous memory.**

Simple Example

Suppose:

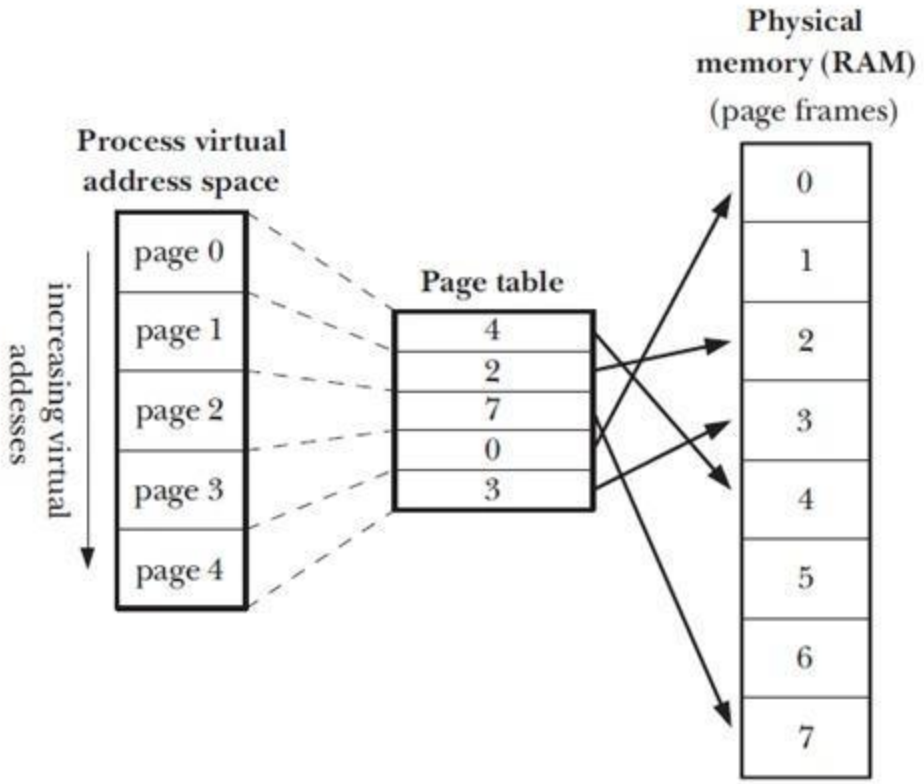
- Page size = **4 KB**
- A process requires = **18 KB**

The process needs: 18/4 =5 pages

So:

Page	Size
Page 0	4 KB
Page 1	4 KB
Page 2	4 KB
Page 3	4 KB
Page 4	2 KB used + 2 KB unused

The last page has **2 KB internal fragmentation.**



How Paging Works

The CPU generates a **logical address**, which is divided into:

Logical Address = Page Number + Page Offset

The **Page Number** is used to look up the corresponding **Frame Number** in the **Page Table.**

Then:

Physical Address = Frame Number + Page Offset

এই টেবল টি mcq এর জন্য important-

Term	Meaning
Page	Fixed-size block of logical/virtual memory
Frame	Fixed-size block of physical memory/RAM
Page Table	Maps page numbers to frame numbers
Page Number	Identifies a particular page
Offset	Position within a page
Page Fault	Occurs when a required page is not currently in RAM
TLB	Fast cache used to speed up page-table lookup

Internal Fragmentation	Wasted space inside allocated pages, usually in the last page

Advantages

1. Eliminates external fragmentation.
2. Allows **non-contiguous memory allocation**.
3. Supports **virtual memory**.
4. Makes memory allocation more flexible.
5. Allows processes to use memory larger than the currently available RAM through virtual memory.

Disadvantages

1. **Internal fragmentation** can occur.
2. Page tables require additional memory.
3. Address translation creates overhead.
4. A **page fault** can significantly slow execution.

Paging is a non-contiguous memory-management technique in which logical memory is divided into fixed-size pages and physical memory is divided into equal-size frames, with a page table mapping pages to frames.

Important:

Page = Virtual/Logical memory

Frame = Physical/RAM memory

6. Page

A **page** is a fixed-size block of **logical/virtual memory**.

If a process has:

16 KB of logical memory and page size is **4 KB**

then:

$$\frac{16}{4} = 4$$

So the process contains: **4 pages**

Process

Page 0 → 0–4095
Page 1 → 4096–8191
Page 2 → 8192–12287
Page 3 → 12288–16383

7. Frame

A **frame** is a fixed-size block of **physical memory (RAM)**.

If RAM is: **32 KB** and frame size is: **4 KB**

then:

$$\frac{32}{4} = 8$$

Therefore, RAM contains:

8 frames

Physical Memory

Frame 0
Frame 1
Frame 2
Frame 3
Frame 4
Frame 5
Frame 6
Frame 7

Page vs Frame

Page	Frame
Belongs to logical/virtual memory	Belongs to physical memory
Fixed-size block	Fixed-size block
Contains program data/instructions	Holds a page in RAM
Identified by page number	Identified by frame number

8. Page Table

A **page table** is a data structure maintained by the operating system that maps **page numbers to frame numbers**.

For example:

Page Number	Frame Number
0	5
1	2
2	7
3	1

This means:

Page 0 → Frame 5

Page 1 → Frame 2

Page 2 → Frame 7

Page 3 → Frame 1

Therefore, pages can be located in completely different frames.

9. Address Translation in Paging

A logical address is divided into two parts:

$$\boxed{\text{Logical Address} = \text{Page Number} + \text{Offset}}$$

The page number is used to find the corresponding frame using the page table.

Then:

$$\boxed{\text{Physical Address} = \text{Frame Number} + \text{Offset}}$$

The **offset remains unchanged** during translation.

Example

Suppose:

- Logical address = 13,000
- Page size = 4,000 bytes

Then:

$$Page\ Number = \left\lfloor \frac{13000}{4000} \right\rfloor = 3$$

and:

$$Offset = 13000 - (3 \times 4000) = 1000$$

Suppose page table says:

Page 3 → Frame 7

Then:

$$Physical\ Address = (7 \times 4000) + 1000$$
$$= \boxed{29000}$$

Therefore:

Logical Address
↓
Page 3 + Offset 1000
↓
Page Table
↓
Frame 7 + Offset 1000
↓
Physical Address = 29000

10. Number of Pages

If the logical address space is given, the number of pages can be calculated using:

$$\text{Number of Pages} = \frac{\text{Logical Address Space}}{\text{Page Size}}$$

For example:

Logical address = **16 bits**

Therefore:

$$2^{16} = 65536 \text{ bytes}$$

If page size = **1 KB = 1024 bytes**:

$$\frac{65536}{1024} = 64$$

Therefore:

$$\boxed{64 \text{ pages}}$$

11. Page Number and Offset Bits

If page size is:

$$2^n \text{ bytes}$$

then the number of **offset bits = n**.

For example:

Page size = 1 KB

$$1KB = 1024 = 2^{10}$$

Therefore:

$$\boxed{10 \text{ offset bits}}$$

If logical address size is 16 bits:

$$16 - 10 = 6$$

Therefore:

- Page number bits = **6**

- Offset bits = **10**

Number of pages:

$$2^6 = 64$$

12. Number of Frames

If physical memory size is known:

$$\boxed{\text{Number of Frames} = \frac{\text{Physical Memory Size}}{\text{Frame Size}}}$$

Example:

Physical memory = **64 KB**

Frame size = **4 KB**

$$\frac{64}{4} = 16$$

Therefore:

$$\boxed{16 \text{ frames}}$$

13. Advantages of Paging

Paging provides several important advantages:

28. **Eliminates external fragmentation.**
29. Allows non-contiguous memory allocation.
30. Makes efficient use of physical memory.
31. Supports virtual memory.
32. Allows a process to be divided into manageable fixed-size blocks.
33. Provides memory protection through page-table permissions.
34. Makes process relocation easier.

14. Disadvantages of Paging

35. Page tables require additional memory.
36. Address translation creates overhead.
37. Paging can cause **internal fragmentation**.
38. Page-table management can become complex for large address spaces.
39. Page faults can significantly reduce performance when required pages are not in RAM.

15. TLB — Translation Lookaside Buffer

A **TLB** is a small, fast associative memory used to store recently used **page-to-frame mappings**.

Without TLB:

CPU
↓
Page Table
↓
Frame
↓
RAM

With TLB:

CPU
↓
TLB
↓
If hit → Frame → RAM
↓
If miss
↓
Page Table
↓
Frame → RAM

TLB Hit- If the required page mapping is found in the TLB, it is called a **TLB hit**.

TLB Miss- If the mapping is not found in the TLB, it is called a **TLB miss**, and the page table must be accessed.

TLB improves memory-access performance by reducing the need to access the page table repeatedly.

16. Page Fault

A **page fault** occurs when a process tries to access a page that is currently **not present in physical memory (RAM)**.

The general process is:

CPU requests page
↓
Page in RAM?



Yes No
↓ ↓
Access Page Fault
↓
OS loads page from
secondary storage
↓
RAM
↓
Resume

A page fault is **not necessarily an error**. It is a normal mechanism of virtual memory, although handling it is much slower than accessing a page already in RAM.

17. Demand Paging

Demand paging is a virtual-memory technique in which a page is loaded into RAM **only when it is actually needed**.

Instead of loading the entire process into RAM:

Entire Process
↓
RAM

only required pages are loaded:

Process
↓
Required Page
↓
RAM

This allows programs larger than available physical memory to execute, provided the operating system can manage the required pages.

18. Page Replacement

If a page fault occurs and **no free frame is available**, the operating system must remove a page from RAM and load the required page.

This is called **page replacement**.

Common page replacement algorithms include:

40. **FIFO** $First - In, First - Out$
41. **Optimal Page Replacement**
42. **LRU (Least Recently Used)**
43. **Second Chance**
44. **Clock Algorithm**

The objective is generally to reduce the number of page faults.

19. Thrashing

Thrashing occurs when a system spends most of its time **swapping pages between RAM and secondary storage** rather than executing processes.

A simplified sequence is:

Too many page faults
↓
Frequent disk access
↓
Less CPU execution
↓
More page faults
↓
Performance collapses

Thrashing usually occurs when processes do not have enough physical frames to hold the pages they actively need.

20. Segmentation

Segmentation is another memory-management technique in which a program is divided into **logical segments of variable size**.

Typical segments include:

- Code
- Data
- Stack
- Heap

For example:

Program
* Code Segment
* Data Segment

* Stack Segment
* Heap Segment

A logical address in segmentation consists of:

Segment Number + Offset

Unlike paging, segments are **variable-sized**.

Paging vs Segmentation

Feature	Paging	Segmentation
Division	Fixed-size pages	Variable-size segments
Physical unit	Frame	Segment
Based on	Physical memory management	Logical program structure
Address	Page number + offset	Segment number + offset
Fragmentation	Internal fragmentation possible	External fragmentation possible
Programmer visibility	Generally transparent	More closely related to program structure
Main purpose	Efficient physical-memory allocation	Logical memory organization and protection

Segmentation with Paging

Some systems combine both techniques.

A process may first be divided into **segments**, and each segment may then be divided into **pages**.

The logical address can conceptually contain:

Segment Number + Page Number + Offset

This combines the logical organization of segmentation with the fixed-size allocation benefits of paging.

Important Terms at a Glance

Term	Meaning
Memory Management	OS management of main memory
Logical Address	Address generated by CPU
Physical Address	Actual address in RAM
MMU	Hardware that translates logical to physical addresses

Page	Fixed-size block of logical memory
Frame	Fixed-size block of physical memory
Page Table	Maps pages to frames
Offset	Position within a page/frame
TLB	Fast cache for page-table mappings
Page Fault	Required page is not currently in RAM
Demand Paging	Load pages into RAM when needed
Page Replacement	Replace a page when no free frame exists
Thrashing	Excessive paging causing severe performance degradation
Internal Fragmentation	Wasted space inside an allocated block
External Fragmentation	Free memory scattered into separate blocks
Segmentation	Variable-size logical memory division

Core Concept to Remember

The most important relationship in paging is:

$$\boxed{\text{Logical Address} = \text{Page Number} + \text{Offset}}$$

$$\boxed{\text{Page Number} \rightarrow \text{Page Table} \rightarrow \text{Frame Number}}$$

$$\boxed{\text{Physical Address} = \text{Frame Number} + \text{Offset}}$$

In short, **paging divides logical memory into pages and physical memory into equal-sized frames. The page table maps each page to a frame, while the offset identifies the exact location within that page/frame.** This mechanism allows processes to occupy non-contiguous areas of RAM and forms one of the fundamental techniques behind modern virtual memory.

Logical Address ও Physical Address — বিস্তারিত

Logical Address

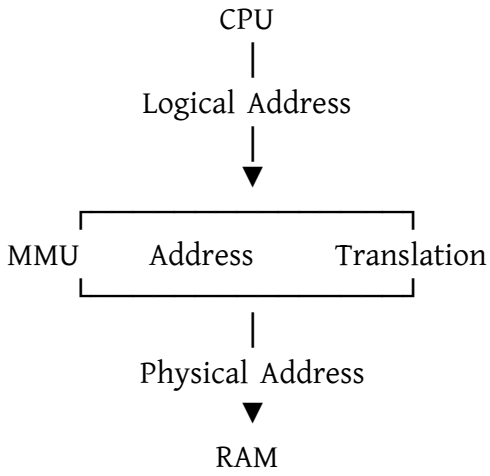
Logical Address হলো CPU কর্তৃক generated address, যা কোনো process-এর virtual address space-এর মধ্যে থাকে। এটি সরাসরি RAM-এর কোনো নির্দিষ্ট location নির্দেশ করে না।

Logical Address → MMU → Physical Address

2. Physical Address

Physical Address হলো Main Memory বা RAM-এর প্রকৃত address।
অর্থাৎ RAM-এর কোন memory location-এ data/instruction আছে, physical address সেটি নির্দেশ করে।

Address Translation



ধরা যাক:
Logical Address = 5000
MMU translation-এর পরে:
Physical Address = 12500
অর্থাৎ: **5000 → MMU → 12500**

Address Space

Logical Address Space

একটি process যে সমস্ত logical address ব্যবহার করতে পারে, তার collection হলো **Logical Address Space**।
যদি CPU-এর logical address size **32-bit** হয়:

$$Number\ of\ addresses = 2^{32}$$

অর্থাৎ:

$$2^{32} = 4{,}294{,}967{,}296$$

যদি প্রতিটি address 1 byte memory নির্দেশ করে:

$$Logical\ Address\ Space = 4GB$$

Physical Address Space

Physical memory-তে available address-এর range হলো Physical Address Space।
যদি physical address **20-bit** হয়:

$$2^{20} = 1{,}048{,}576$$

অর্থাৎ:

$$Physical\ Address\ Space = 1MB$$

গুরুত্বপূর্ণ Formula

Number of addresses

যদি address = **n bits**

$$\boxed{Number\ of\ addresses = 2^n}$$

Addressable Memory

যদি প্রতিটি address 1 byte নির্দেশ করে:

$$\boxed{Memory = 2^n\ bytes}$$

Example

16-bit address:

$$2^{16} = 65{,}536$$

অর্থাৎ:

$$65{,}536\ bytes = 64KB$$

32-bit address:

$$2^{32} = 4GB$$

64-bit address:

$$2^{64} = 16EB$$

তাই theoretical 64-bit byte-addressable address space:

$$\boxed{16\ Exabytes}$$

Paging-এর সঙ্গে Logical ও Physical Address

Paging-এ logical address দুই ভাগে বিভক্ত:

$$Logical\ Address = Page\ Number + Offset$$

Physical address:

$$Physical\ Address = Frame\ Number + Offset$$

এখানে:

- **Page Number (p)** → কোন page?
- **Offset (d)** → page-এর ভিতরে কোন location?
- **Frame Number (f)** → RAM-এর কোন frame?

Page Number বের করার Formula

যদি:

- Logical Address = LA
- Page Size = PS

তাহলে:

$$Page\ Number = \left\lfloor \frac{LA}{PS} \right\rfloor$$

এবং:

$$Offset = LA \bmod PS$$

Numerical Example — Paging

ধরি:

Logical Address = 2500

Page Size = 1000 bytes

তাহলে:

$$Page\ Number = [2500/1000]$$
$$= 2$$

এবং:

$$Offset = 2500 \bmod 1000$$
$$= 500$$

সুতরাং:

Logical Address

↓

Page = 2

Offset = 500

এখন Page Table-এ যদি:

$$\text{Page } 2 \rightarrow \text{Frame } 7$$

তাহলে physical address:

$$\text{Physical Address= (Frame * Page Size)+Offset}$$
$$=7*1000+500$$
$$= 7500$$

অর্থাৎ:

$$\boxed{Logical\ Address\ 2500 \rightarrow Physical\ Address\ 7500}$$

আরেকটি Exam Numerical

ধরি:

Logical Address = 4096

Page Size = 1024 bytes

Page number:

$$4096/1024 = 4$$

তাই:

$$Page = 4$$

Offset:

$$4096 \bmod 1024 = 0$$

ধরি Page Table:

$$\text{Page } 4 \rightarrow \text{Frame } 10$$

তাহলে:

Physical Address=10*1024+0 = 10240

Answer = 10240

Address Bits এবং Page Bits

এটি খুব গুরুত্বপূর্ণ numerical topic।

ধরি:

Logical Address = 32 bits

Page Size = 4 KB

প্রথমে:

$$4KB = 4096 = 2^{12}$$

তাই offset-এর জন্য প্রয়োজন:

$$\boxed{12\ bits}$$

কারণ:

$$2^{12} = 4096$$

Logical address = 32 bits।

তাই Page Number bits:

$$32 - 12 = 20$$

অর্থাৎ:

Logical Address = 32 bits

Page Number = 20b | Offset=12b |

Number of pages:

$$2^{20}$$

অর্থাৎ:

$$1{,}048{,}576\ pages$$

Physical Address-এর Numerical

ধরি:

Physical Address = 32 bits

Page size = 4 KB

Offset:

$$12\ bits$$

তাহলে Frame Number:

$$32 - 12 = 20\ bits$$

Number of frames:

$$2^{20}$$

প্রতিটি frame = 4 KB।

তাই physical memory:

$$2^{20} \times 2^{12}$$
$$= 2^{32}$$
$$= \boxed{4GB}$$

Internal Fragmentation

Paging-এর একটি গুরুত্বপূর্ণ disadvantage হলো **internal fragmentation**।

ধরি:

Page Size = 4 KB

Process Size = 18 KB

Process-এর জন্য প্রয়োজন:

$$18KB$$

Pages required:

$$\lceil 18/4 \rceil = 5$$

Allocated memory:

$$5 \times 4 = 20KB$$

Unused:

$$20 - 18 = 2KB$$

অর্থাৎ:

$$\boxed{Internal\ Fragmentation = 2KB}$$

Allocated = 20 KB

Process = 18 KB

Unused = 2 KB | ← **Internal Fragmentation**

Page Table

Paging-এ logical page-কে physical frame-এর সঙ্গে map করার জন্য **Page Table** ব্যবহৃত হয়।
উদাহরণ:

Page Number	Frame Number
0	5
1	2
2	8
3	1
4	7

যদি:

Logical Page = 3

তাহলে:

Page 3 → Frame 1

Offset অপরিবর্তিত থাকে।

Page Number পরিবর্তিত হয়ে Frame Number হয়; Offset পরিবর্তিত হয় না।

এটি খুব গুরুত্বপূর্ণ MCQ point।

TLB

প্রতিবার Page Table access করলে memory access time বেড়ে যেতে পারে।
এটি কমানোর জন্য **TLB (Translation Lookaside Buffer)** ব্যবহার করা হয়।

CPU
|
Logical Address
▼
TLB
|
Hit → Frame Number
|
Miss
↓
Page Table
↓
Frame Number

↓
RAM

TLB Hit

Page-to-frame mapping TLB-তে পাওয়া গেলে:

$$CPU \rightarrow TLB \rightarrow RAM$$

TLB Miss

Mapping না পাওয়া গেলে:

$$CPU \rightarrow TLB \rightarrow Page\ Table \rightarrow RAM$$

Effective Access Time (EAT)

TLB numerical-এ সাধারণত এই formula গুরুত্বপূর্ণ।
ধরি:

- TLB lookup time = t
- Memory access time = m
- TLB hit ratio = h

তাহলে সাধারণ single-level paging model-এ:

EAT=h(t+m)+(1-h)(t+2m)

কারণ TLB hit হলে:

$$TLB + Memory$$

TLB miss হলে:

$$TLB + Page\ Table + Memory$$

Example

ধরি:
TLB lookup = 10 ns
Memory access = 100 ns
Hit ratio = 80%
তাহলে:

EAT=0.8(10+100)+0.2(10+200)

$$= 0.8(110) + 0.2(210)$$
$$= 88 + 42$$
$$\boxed{130ns}$$

Logical বনাম Physical

Question: What is the difference between logical and physical address?

Answer:

A logical address is the address generated by the CPU and used by a process, whereas a physical address is the actual address in main memory. The MMU translates the logical address into a physical address. In paging, the logical address consists of a page number and offset, while the physical address consists of a frame number and the same offset.

Logical address is what the CPU/process sees; physical address is where the data actually resides in RAM.

Chapter 03 C Programming (String Management)

In C, a **string** is a sequence of characters stored in a character array and terminated by a special character called the **null character ('\0')**.

C does not have a separate built-in string data type like some other programming languages. Strings are generally represented using arrays of char.

Example:

```
char name[ ] = "Titon";
```

Internally, it is stored as:

T i t o n \0

The '\0' indicates the end of the string.

char name[6] = "Titon";

Here, 5 positions contain characters and the sixth position contains '\0'.

2. Character Array vs String

A character array is not necessarily a valid C string.

Example:

char a[5] = {'H', 'e', 'l', 'l', 'o'};

This array contains characters, but it does **not** contain '\0'.

Therefore, it should not be treated as a C string by functions such as printf("%s", a) or strlen(a).

A proper string is:

char a[6] = {'H', 'e', 'l', 'l', 'o', '\0'};

or:

char a[ ] = "Hello";

3. Declaring a String

A string can be declared using a character array:

char str[20];

This creates space for up to 19 visible characters plus the null character.

A string can also be initialized directly:

char str[] = "Hello";

The compiler automatically allocates enough space for the characters and '\0'.

Another form is:

char str[6] = "Hello";

Here:

H e l l o \0

4. Null Character ('\0')

The null character is represented by:

'\0'

Its ASCII value is 0.

It marks the end of a C string.

For example:

char str[] = "ABC";

Memory representation:

Index: 0 1 2 3

A	B	C	\0

Therefore: strlen("ABC")=3

but the array requires 4 characters of storage.

5. Taking String Input

A string can be read using scanf():

char name[50];

scanf("%49s", name);

%s reads characters until whitespace is encountered.

For example, if the input is:

Mofazzal Titon

only: Mofazzal will be read.

For strings containing spaces, fgets() is safer and more appropriate:

char name[50];

fgets(name, sizeof(name), stdin);

6. Printing a String

The %s format specifier is used to print a string.

char name[ ] = "Titon";

printf("%s", name);

Output:

Titon

7. Important String Functions

Most standard C string functions are declared in:

```
#include <string.h>
```

Important functions include:

Function	Purpose
strlen()	Finds string length
strcpy()	Copies a string
strncpy()	Copies up to a specified number of characters
strcat()	Concatenates strings
strncat()	Concatenates up to a specified number of characters
strcmp()	Compares two strings
strncmp()	Compares specified number of characters
strchr()	Finds a character
strrchr()	Finds the last occurrence of a character
strstr()	Finds a substring
strtok()	Tokenizes a string
strcspn()	Finds the length before characters from a specified set

8. strlen() — Finding String Length

strlen() returns the number of characters before '\0'.

Example:

```
#include <stdio.h>
#include <string.h>

int main(void)
{
    char str[] = "Computer";

    printf("Length = %zu\n", strlen(str));

    return 0;
}
```

Output:

Length = 8

Important:

strlen() does **not** count the null character.

Therefore:

"Computer"

8 characters

+ '\0'

= 9 bytes of array storage

9. Implementing strlen() Without Library Function

The same operation can be implemented manually:

```
#include <stdio.h>

int main(void)
{
    char str[] = "Computer";
    int i = 0;

    while (str[i] != '\0')
    {
        i++;
    }

    printf("Length = %d\n", i);

    return 0;
}
```

The loop stops when it reaches '\0'.

10. strcpy() — Copying Strings

strcpy() copies the source string into the destination.

```
#include <stdio.h>
#include <string.h>
int main(void)
```

```
{
    char source[ ] = "Computer";
    char destination[20];
    strcpy(destination, source);
    printf("%s\n", destination);
    return 0;
}
```

Output:

Computer

The destination array must have enough space to hold the source string including '\0'.

11. Manual String Copy

The copying process can be implemented without strcpy():

```
#include <stdio.h>

int main(void)
{
    char source[] = "Computer";
    char destination[20];
    int i = 0;

    while (source[i] != '\0')
    {
        destination[i] = source[i];
        i++;
    }

    destination[i] = '\0';
    printf("%s\n", destination);
    return 0;
}
```

The null character must also be copied. Otherwise, destination would not be a properly terminated C string.

12. strcat() — String Concatenation

strcat() joins the second string to the end of the first string.

```
#include <stdio.h>
#include <string.h>

int main(void)
{
    char first[30] = "Hello ";
    char second[] = "World";

    strcat(first, second);

    printf("%s\n", first);

    return 0;
}
```

Output:

Hello World

The destination array must have sufficient capacity for both strings and the terminating '\0'.

13. strcmp() — Comparing Strings

Strings should not normally be compared using:

str1 == str2

because that compares addresses when the operands are pointers.

Use strcmp() to compare string contents.

```
#include <stdio.h>
#include <string.h>

int main(void)
{
    char str1[] = "apple";
    char str2[] = "apple";
```

```
    if (strcmp(str1, str2) == 0)
    {
        printf("Strings are equal\n");
    }
    else
    {
        printf("Strings are different\n");
    }

    return 0;
}
```

strcmp() returns: 0

if the strings are equal.

A value less than zero if the first string is lexicographically smaller, and a value greater than zero if the first string is lexicographically greater.

It is important to test the **sign** of the result rather than relying on a particular nonzero value.

14. Searching for a Character — strchr()

strchr() searches for the first occurrence of a character.

```
#include <stdio.h>
#include <string.h>

int main(void)
{
    char str[] = "Computer";
    char *ptr;

    ptr = strchr(str, 'p');

    if (ptr != NULL)
    {
        printf("Character found: %s\n", ptr);
    }
```

```
    return 0;
}
```

Output:

Character found: puter

The returned pointer points to the first occurrence of the searched character.

15. Searching for a Substring — strstr()

strstr() searches for the first occurrence of one string inside another.

```
#include <stdio.h>
#include <string.h>

int main(void)
{
    char text[] = "I love computer science";
    char *ptr;

    ptr = strstr(text, "computer");

    if (ptr != NULL)
    {
        printf("Substring found: %s\n", ptr);
    }

    return 0;
}
```

Output:

Substring found: computer science

16. Converting Characters — <ctype.h>

The header:

#include <ctype.h>

provides useful character-processing functions.

Examples:

Function	Purpose
toupper()	Converts character to uppercase

tolower()	Converts character to lowercase
isalpha()	Checks alphabetic character
isdigit()	Checks digit
isalnum()	Checks alphabetic or digit
isspace()	Checks whitespace
ispunct()	Checks punctuation
islower()	Checks lowercase
isupper()	Checks uppercase

Example:

```
#include <stdio.h>
#include <ctype.h>

int main(void)
{
    char ch = 'a';

    printf("%c\n", toupper((unsigned char)ch));

    return 0;
}
```

Output: A

17. Traversing a String

A string can be processed character by character:

```
#include <stdio.h>

int main(void)
{
    char str[] = "Computer";
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        printf("%c\n", str[i]);
    }

    return 0;
}
```

This technique is the basis of many string-processing algorithms.

18. Counting Vowels and Consonants

Example:

```
#include <stdio.h>
#include <ctype.h>

int main(void)
{
    char str[100];
    int i, vowels = 0, consonants = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for (i = 0; str[i] != '\0'; i++)
    {
        char ch = (char)tolower((unsigned char)str[i]);

        if (ch >= 'a' && ch <= 'z')
        {
            if (ch == 'a' || ch == 'e' || ch == 'i' ||
                ch == 'o' || ch == 'u')
            {
                vowels++;
            }
            else
            {
                consonants++;
            }
        }
    }

    printf("Vowels = %d\n", vowels);
    printf("Consonants = %d\n", consonants);

    return 0;
}
```

19. Reversing a String

A string can be reversed using two indices:

```
#include <stdio.h>
#include <string.h>

int main(void)
{
    char str[100];
    int left, right;
    char temp;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    str[strcspn(str, "\n")] = '\0';

    left = 0;
    right = (int)strlen(str) - 1;

    while (left < right)
    {
        temp = str[left];
        str[left] = str[right];
        str[right] = temp;

        left++;
        right--;
    }

    printf("Reversed string: %s\n", str);

    return 0;
}
```

20. Palindrome String

A string is a **palindrome** if it reads the same from left to right and right to left.

Examples:

madam

level

radar

A palindrome can be checked by comparing characters from both ends.

21. String and Pointer Relationship

A character array can be accessed using a pointer.

```
char str[] = "Hello";
char *p = str;
```

Then:

p → H

p + 1 → e

p + 2 → l

p + 3 → l

p + 4 → o

p + 5 → '\0'

For example:

```
printf("%c\n", *(p + 2));
```

Output: l

24. Dynamic String Management

When the required string size is not known at compile time, dynamic memory allocation can be used.

Important functions are:

malloc()

calloc()

realloc()

free()

25. Common String Problems in C

Problem 1 — Count Vowels, Consonants, Digits and Spaces

Write a C program that reads a string and counts:

- Number of vowels

- Number of consonants
- Number of digits
- Number of spaces

Input:

CSE 2026 Exam

Output:

Vowels = 3

Consonants = 5

Digits = 4

Spaces = 2

26. Advanced String Problem — Frequency Analysis

A common interview/exam problem is to calculate the frequency of every character.

Example:

```
#include <stdio.h>

int main(void)
{
    char str[200];
    int freq[256] = {0};
    int i;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for (i = 0; str[i] != '\0'; i++)
    {
        unsigned char ch = (unsigned char)str[i];

        if (ch != '\n')
        {
            freq[ch]++;
        }
    }
```

```
    for (i = 0; i < 256; i++)
    {
        if (freq[i] > 0)
        {
            printf("%c = %d\n", i, freq[i]);
        }
    }

    return 0;
}
```

This technique is useful for solving problems involving duplicate characters, frequency comparison, and anagram detection.

Chapter 04

HTTP and HTTPS ports, advantages, disadvantages, important technical concepts from basic to advanced.

What is HTTP?

HTTP (HyperText Transfer Protocol) is an application-layer protocol used to transfer resources such as:

- HTML pages
- CSS files
- JavaScript files
- Images
- Videos
- JSON data
- API responses

between a **client** and a **server**.

For example:

```
Client/Browser
      |
      | HTTP Request
      ↓
   Server
      |
      | HTTP Response
      ↓
Client/Browser
```

HTTP is one of the fundamental protocols of the **World Wide Web**.

What is HTTPS?

HTTPS (HyperText Transfer Protocol Secure) is HTTP protected by **TLS (Transport Layer Security)**.

In simplified form:

HTTPS = HTTP + TLS

TLS provides security for data exchanged between the client and server.

Example:

Client

|

| Encrypted HTTPS communication

↓

Server

HTTPS is used for:

- Online banking
- E-commerce
- Login systems
- Social media
- Email services
- APIs
- Government websites
- Any application handling sensitive information

HTTP Default Port

HTTP normally uses: Port 80

http://example.com

The browser normally connects to port 80 unless another port is explicitly specified.

HTTPS Default Port

HTTPS normally uses: Port 443 https://example.com

The browser normally connects to port 443.

Why is HTTPS Needed?

Suppose a user sends:

Username: titon

Password: 123456

over an unencrypted HTTP connection.

An attacker capable of observing the network traffic may be able to read the information.

With HTTPS, the application data is protected by TLS encryption.

Conceptually:

HTTP:

Username=titon&Password=123456
↓
Network
↓
Attacker may read

HTTPS:

Username=titon&Password=123456
↓
TLS encryption
↓
Encrypted data
↓
Network
↓
Server

Three Major Security Properties of HTTPS

HTTPS primarily provides:

1. **Confidentiality**
2. **Integrity**
3. **Authentication**

Confidentiality

Prevents unauthorized parties from reading the protected application data.

Integrity

Helps detect whether protected data has been modified in transit.

Authentication

The TLS certificate helps the browser authenticate the server's identity, assuming the certificate is valid and the trust chain is properly established.

Therefore: **HTTPS □ rightarrow Confidentiality + Integrity + Authentication**

HTTP Request

When a browser requests a web resource, it sends an HTTP request.

GET /index.html HTTP/1.1

Host: example.com

A request can contain:

- □ HTTP method
- □ URL/path
- □ HTTP version
- □ Headers
- □ Optional body

HTTP Response

The server sends an HTTP response.

HTTP/1.1 200 OK

Content-Type: text/html

Content-Length: 1250

followed by the response body.

The response contains:

- □ Status code
- □ Headers
- □ Optional body

Common HTTP Methods

Method	Purpose
GET	Retrieve data
POST	Submit/create data
PUT	Replace a resource
PATCH	Partially modify a resource
DELETE	Delete a resource
HEAD	Retrieve headers without response body

OPTIONS	Discover supported communication options

For example:

GET /students

may request a list of students.

POST /students

may submit data to create a new student.

HTTP Status Codes

HTTP status codes indicate the result of a request.

They are divided into five classes.

Class	Meaning
1xx	Informational
2xx	Successful
3xx	Redirection
4xx	Client error
5xx	Server error

Important examples:

200 → OK

201 → Created

301 → Moved Permanently

302 → Found

304 → Not Modified

400 → Bad Request

401 → Unauthorized

403 → Forbidden

404 → Not Found

405 → Method Not Allowed

500 → Internal Server Error

502 → Bad Gateway

503 → Service Unavailable

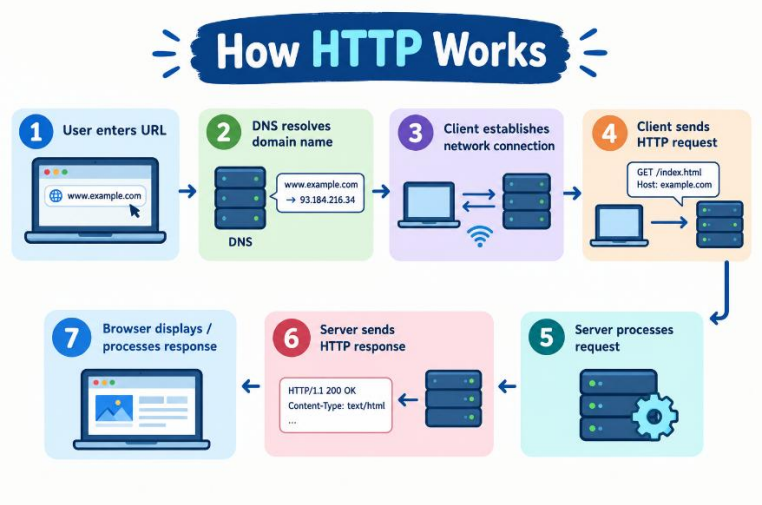
HTTP vs HTTPS — Basic Difference

Feature	HTTP	HTTPS
Full form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Default port	80	443
TLS	No	Yes
Encryption	No TLS encryption	Yes, through TLS
Confidentiality	Not provided by HTTP itself	Provided for protected TLS traffic
Integrity protection	No TLS integrity protection	Yes
Server authentication	No TLS certificate authentication	Yes, through TLS certificates
Security	Lower	Higher
Suitable for passwords	No	Yes

How HTTP Works

The simplified process is:

1. User enters URL
↓
2. DNS resolves domain name
↓
3. Client establishes network connection
↓
4. Client sends HTTP request
↓
5. Server processes request
↓
6. Server sends HTTP response
↓
1. Browser displays/processes response



For traditional HTTP/1.1 over TCP:

Browser
|
| TCP connection
↓
Server
|
| HTTP Request
↓
Server
|
| HTTP Response
↓
Browser

How HTTPS Works

HTTPS adds TLS before protected HTTP application data is exchanged.

Simplified process:

Browser
|
| TCP connection
↓
Server

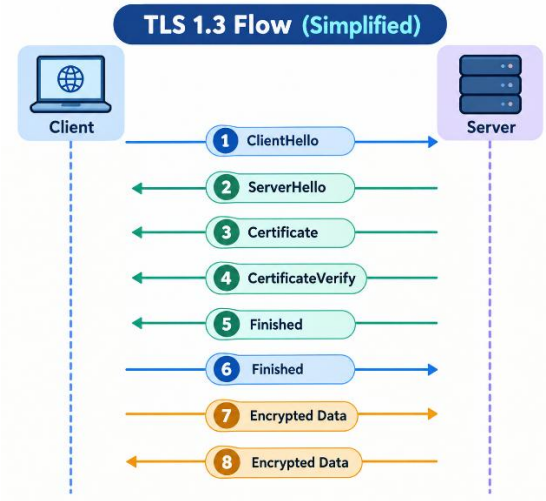
|
| TLS handshake
↓
Secure TLS session
|
| Encrypted HTTP request
↓
Server
|
| Encrypted HTTP response
↓
Browser

With HTTP/3, HTTPS uses **QUIC over UDP** instead of TCP, so the exact transport-level process differs.

TLS Handshake

The TLS handshake establishes the cryptographic parameters needed to protect communication.

A simplified modern TLS 1.3 flow is:



The exact messages and details can vary depending on configuration and TLS version.

What is a Digital Certificate?

A digital certificate binds a domain identity to a public key and is issued by a trusted

Certificate Authority (CA).

A certificate can contain information such as:

- Domain names
- Public key
- Certificate issuer
- Validity period
- Digital signature
- Certificate extensions

Example concept:

example.com

|

↓

Public Key

|

↓

Digital Certificate

|

↓

CA Signature

The browser verifies the certificate and its trust chain before accepting the server's identity.

What is a Certificate Authority?

A **Certificate Authority (CA)** is an entity trusted to issue and sign digital certificates.

Examples include certificate authorities such as:

- Let's Encrypt
- DigiCert
- GlobalSign

The operating system or browser maintains a set of trusted root certificates.

The trust model can be represented as:

Trusted Root CA

↓

Intermediate CA

↓

Server Certificate

↓

example.com

Symmetric Encryption in HTTPS

After the TLS handshake establishes the required keys, application data is protected using **symmetric cryptography**.

Symmetric encryption uses the same secret key for encryption and decryption.

Conceptually:

Plaintext

↓

Symmetric Encryption + Secret Key

↓

Ciphertext

↓

Symmetric Decryption + Secret Key

↓

Plaintext

Symmetric encryption is efficient and suitable for protecting large amounts of data.

Public-Key Cryptography in HTTPS

TLS also uses public-key cryptography during authentication and key establishment.

A key pair consists of:

- Public Key
- Private Key

The public key can be distributed, while the private key must remain secret.

In modern TLS, public-key cryptography is primarily used for authentication and establishing shared secrets, while symmetric cryptography protects the actual application data.

Why Not Encrypt Everything Using Public-Key Cryptography?

Public-key cryptographic operations are generally more computationally expensive than symmetric encryption.

Therefore, HTTPS uses a **hybrid cryptographic approach**:

Public-key cryptography

↓

Authentication / key establishment

↓

Shared symmetric keys

↓

Symmetric encryption

↓

Actual application data

This provides both security and efficiency.

HTTP is Stateless

HTTP is fundamentally a **stateless application protocol**.

This means each request is treated independently at the protocol level.

For example:

Request 1 → Login

Request 2 → Profile

Request 3 → Logout

HTTP itself does not inherently remember that the requests belong to the same user.

Applications therefore use mechanisms such as:

- Cookies
- Session IDs
- Authentication tokens
- JWTs

to maintain application state.

Cookies with HTTP/HTTPS

A server may send:

Set-Cookie: sessionid=abc123; Secure; HttpOnly

The browser can then send the cookie in subsequent requests.

Important cookie attributes include:

Secure

The cookie should only be sent over a secure HTTPS connection.

HttpOnly

Helps prevent JavaScript from directly accessing the cookie.

SameSite

Controls when cookies are sent in cross-site contexts and can help reduce certain cross-site request attacks.

HTTP vs HTTPS and Password Security

Consider: http://bank.example

If sensitive credentials are transmitted through an unencrypted HTTP connection, the network traffic is not protected by TLS.

With: https://bank.example

TLS protects the application data during transmission.

However, HTTPS **does not guarantee that the website itself is trustworthy**. A malicious website can also obtain a valid TLS certificate for its own domain.

Therefore: HTTPS --□ Guaranteed, Trustworthiness

It provides secure communication with the authenticated domain, not proof that the organization behind that domain is legitimate.

HTTP and HTTPS URL Structure

HTTP: http://www.example.com/index.html

HTTPS: https://www.example.com/index.html

General structure:

scheme://host/path

where:

http/https → scheme

example.com → host

/index.html → path

A URL may also contain:

scheme://host:port/path?query#fragment

HTTP/1.0

HTTP/1.0 commonly used a separate TCP connection for individual requests unless connection persistence was supported through mechanisms introduced around that era. This could increase connection overhead.

HTTP/1.1

HTTP/1.1 introduced persistent connections as a standard feature and supports mechanisms such as:

- Persistent connections
- Chunked transfer encoding
- Host header
- Caching controls
- Range requests

However, HTTP/1.1 can suffer from **head-of-line blocking at the application request/response level**, particularly when multiple requests share a connection.

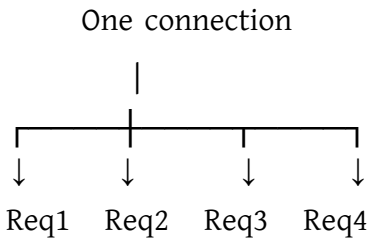
HTTP/2

HTTP/2 improves efficiency through features such as:

- Binary framing
- Multiplexing
- Header compression
- Stream prioritization mechanisms

Instead of treating an HTTP connection as a sequence of independent textual messages, HTTP/2 breaks communication into frames belonging to multiple streams.

Conceptually:



This allows multiple streams to share a connection.

HTTP/3

HTTP/3 is based on **QUIC**, which runs over UDP.

Conceptually:

HTTP/3
↓
QUIC
↓
UDP
↓
IP

Whereas traditional HTTP/1.1 and HTTP/2 commonly use:

HTTP
↓
TCP
↓
IP

HTTP/3 integrates TLS 1.3 into QUIC and provides stream multiplexing without TCP's connection-level head-of-line blocking.

HTTP Versions Comparison

Feature	HTTP/1.1	HTTP/2	HTTP/3
Common transport	TCP	TCP	QUIC/UDP
Binary framing	No	Yes	Yes
Multiplexing	Limited	Yes	Yes
Header compression	No dedicated mechanism	HPACK	QPACK
TLS	Optional	Commonly used	Integrated with QUIC
Connection-level TCP HOL blocking	Possible	Possible	Avoided by QUIC architecture

HTTP Caching

Caching allows browsers and intermediary systems to reuse previously retrieved resources.

Example:

Cache-Control: max-age=3600

This can reduce:

- Network traffic
- Server workload
- Page loading time

HTTP caching is important for performance.

HTTP Proxy

A proxy is an intermediary between a client and server.

Client
↓
Proxy
↓
Server

A proxy can be used for:

- Caching
- Filtering
- Access control
- Monitoring
- Traffic management

Reverse Proxy

A reverse proxy sits in front of servers.

Client
↓
Reverse Proxy
↓
Web Server

It can provide:

- Load balancing
- TLS termination
- Caching
- Security filtering
- Routing
- Compression

Examples include **Nginx** and **HAProxy**.

HTTPS and TLS Termination

In many architectures:

Client
|
HTTPS
↓
Reverse Proxy
|
HTTP or HTTPS
↓
Application Server

The reverse proxy may terminate the TLS connection and then forward the request internally.

However, whether the internal connection uses HTTP or HTTPS depends on the architecture and security requirements.

HTTP Security Problems

HTTP itself does not provide TLS-based confidentiality or integrity protection.

Potential risks include:

- Eavesdropping
- Traffic modification
- Session theft
- Credential exposure

- Content injection
- Man-in-the-middle attacks

HTTPS addresses the transport-security portion of these problems by using TLS.

HTTPS Does Not Protect Everything

HTTPS does **not** automatically protect against:

- SQL injection
- Cross-Site Scripting (XSS)
- Weak passwords
- Vulnerable server software
- Broken authentication
- Malware
- Poor application authorization
- Compromised servers

For example:

HTTPS + vulnerable SQL query

can still result in SQL injection.

HTTP vs HTTPS — Exam-Oriented Comparison

Parameter	HTTP	HTTPS
Protocol	HTTP	HTTP over TLS
Port	80	443
Encryption	No TLS encryption	Yes
Authentication	No TLS server authentication	Certificate-based server authentication
Data integrity	Not provided by HTTP itself	TLS provides integrity protection
Confidentiality	No	Yes
Certificate	Not required	Normally required
Performance	Historically simpler	TLS adds overhead, though modern TLS and HTTP versions are highly optimized
Security	Lower	Higher

Use	Public/non-sensitive communication	Sensitive and modern web communication

Important Interview Questions

Q1. Is HTTPS a separate application protocol from HTTP?

Not in the basic sense. HTTPS is HTTP carried through a TLS-secured connection.

HTTPS=HTTP+TLS

Q2. Why does HTTPS use port 443?

Port 443 is the standard registered port commonly used for HTTPS.

Q3. Does HTTPS encrypt the URL?

The full URL is not completely hidden from all observers. In traditional HTTPS, the path and query are protected after the TLS connection is established, but the destination domain has historically been visible through DNS and TLS metadata. Modern technologies such as encrypted DNS and ECH can reduce some metadata exposure.

Q4. Does HTTPS prevent hacking?

No.

It protects communication between the client and server against many network-level attacks, but it does not eliminate application vulnerabilities.

Q5. Which is faster: HTTP or HTTPS?

Historically, HTTP had less connection overhead because it lacked TLS. However, modern HTTPS implementations are highly optimized, and HTTP/2 and HTTP/3 can provide substantial performance benefits. Security should not normally be sacrificed for the small overhead of TLS.

Key Points to Remember

1. HTTP stands for **HyperText Transfer Protocol**.
2. HTTPS stands for **HyperText Transfer Protocol Secure**.
3. HTTP normally uses **port 80**.
4. HTTPS normally uses **port 443**.
5. HTTPS uses **TLS** to secure HTTP communication.

6. TLS provides **confidentiality, integrity, and authentication**.
7. Digital certificates help authenticate the server's domain identity.
8. Symmetric cryptography protects application data efficiently.
9. Public-key cryptography is used for authentication and key establishment.
10. HTTP is fundamentally **stateless**.
11. HTTP/1.1 commonly uses TCP.
12. HTTP/2 commonly uses TCP and supports multiplexing.
13. HTTP/3 uses **QUIC over UDP**.
14. HTTPS does not automatically make an application secure.
15. HTTPS is essential when protecting sensitive web communication.

Math

MATH- 01

A factory produces 70% of its products from Machine A and 30% from Machine B. The defect rates of Machine A and Machine B are 2% and 8%, respectively. If a randomly selected product is found to be defective, what is the probability that it was produced by Machine B?

This problem is solved using Bayes' Theorem.

Probability that a product is produced by Machine A: P(A)=70

Probability that a product is produced by Machine B: P(B)=30

Probability that a product is defective given that it is produced by Machine A:

$$P(D|A)=2$$

Probability that a product is defective given that it is produced by Machine B:

$$P(D|B)=8$$

Where,

A = Product is produced by Machine A.

B = Product is produced by Machine B.

D = Product is defective.

We are asked to find the probability that a defective product came from Machine B,

$$P(B|D)$$

Bayes' Theorem states:

$$\mathbf{P(B|D)= \frac{P(D|B)\times P(B)}{P(D)}}$$

Since (P(D)) (the probability that a product is defective) is not given directly, we calculate it using the Law of Total Probability.

$$\mathbf{P(D)=P(D|A)\times P(A)+P(D|B)\times P(B)}$$

Step 2: Calculate the Probability of a Defective Product

$$\mathbf{P(D)=(0.02\times 0.70)+(0.08\times 0.30)}$$

$$\mathbf{=0.014+0.024}$$

=0.038

Step 3: Calculate the Numerator

Now calculate

P(D|B)×P(B)

=0.08×0.30

=0.024

Now

P(B|D)=0.024/0.038

=0.6316

= 63.16%

MATH- 02

প্রশ্ন

একটি কারখানায়—

মোট পণ্যের 70% তৈরি হয় Machine A দিয়ে।

মোট পণ্যের 30% তৈরি হয় Machine B দিয়ে।

Machine A-এর defect rate = 2%

Machine B-এর defect rate = 8%

একটি পণ্য randomly selected করা হলো এবং দেখা গেল সেটি defective।

প্রশ্ন: সেই defective পণ্যটি Machine B-তে তৈরি হওয়ার সম্ভাবনা কত?

তাহলে,

$$P(A) = 70$$
$$P(B) = 30$$

Machine A থেকে তৈরি পণ্য defective হওয়ার সম্ভাবনা:

$$P(D|A) = 2$$

Machine B থেকে তৈরি পণ্য defective হওয়ার সম্ভাবনা:

$$P(D|B) = 8$$

আমাদের বের করতে হবে:

$$\boxed{P(B|D)}$$

অর্থাৎ, পণ্যটি defective জানা গেলে সেটি Machine B-তে তৈরি হওয়ার সম্ভাবনা কত?

পণ্যটি defective হওয়ার সম্ভাবনা $P(D) = P(A) \times P(D|A) + P(B) \times P(D|B)$

Bayes' Theorem:

$$P(B|D) = \frac{P(B) \times P(D|B)}{P(A) \times P(D|A) + P(B) \times P(D|B)}$$

$$P(B|D) = \frac{(0.30)(0.08)}{(0.70)(0.02) + (0.30)(0.08)}$$

$$P(B|D) = \frac{0.024}{0.014 + 0.024}$$

$$= \frac{0.024}{0.038}$$

$$= \frac{24}{38}$$

$$= \frac{12}{19}$$

$$\approx 0.6316$$

ধরি, কারখানা 10000টি পণ্য তৈরি করল।

Machine A

70% পণ্য A তৈরি করে: $10000 \times 70\% = 7000$

A-এর defect rate 2%:

$$7000 \times 2\%$$

অর্থাৎ Machine A থেকে 140টি defective।

Machine B

30% পণ্য B তৈরি করে: $10000 \times 30 = 300$ %

B-এর defect rate 8%:

$$300 \times 8\% = 240$$

অর্থাৎ Machine B থেকে 240টি defective।

এখন মোট defective পণ্য

$$140 + 240 = 380$$

এর মধ্যে Machine B-এর defective পণ্য:

$$240$$

সুতরাং,

$$P(B|D) = \frac{240}{380}$$

$$= \frac{24}{38} = \frac{12}{19}$$

***P(Machine B | Defective) = $\frac{\text{B থেকে defectiveB থেকে defective}}{\text{A থেকে defective + B থেকে defectiveB থেকে defective}}$

MATH- 03

In a Bangladesh Bank office, there are 6 Assistant Directors (ADs) and 4 Deputy Directors (DDs). Every Assistant Director brings one office bag, whereas only half of the Deputy Directors bring a bag. If one bag is selected at random from all the bags brought to the office, what is the probability that the selected bag belongs to a Deputy Director (DD)?

Number of Assistant Directors (ADs) = 6
Number of Deputy Directors (DDs) = 4
Every AD brings one bag.
Only half of the DDs bring a bag.

each AD brings one bag,

$$\text{Bags brought by ADs} = 6$$

only half of them bring bags.

$$\text{Bags brought by DDs} = \frac{4}{2} = 2$$

Calculate the total number of bags

$$\text{Total bags} = 6 + 2 = 8$$

Find the required probability

The probability that a randomly selected bag belongs to a DD is

$$P(\text{DD's bag}) = \frac{\text{Number of DD bags}}{\text{Total number of bags}}$$
$$= \frac{2}{8}$$
$$= \frac{1}{4}$$

$$= 0.25$$

$$= \mathbf{25\ \%}$$

MATH- 04

A bank processes 60% of its online transactions through Server A and 40% through Server B. The transaction failure rates for Server A and Server B are 2% and 6%, respectively. If a randomly selected transaction is found to have failed, what is the probability that it was processed by Server B?

$$P(A) = 0.60, \quad P(B) = 0.40$$

$$P(F|A) = 0.02, \quad P(F|B) = 0.06$$

We need: $P(B|F)$

$$\boldsymbol{P(F) = P(A)P(F|A) + P(B)P(F|B)}$$

Using Bayes' theorem:

$$P(B|F) = \frac{P(B)P(F|B)}{P(A)P(F|A) + P(B)P(F|B)}$$

$$= \frac{(0.40)(0.06)}{(0.60)(0.02) + (0.40)(0.06)}$$

$$= \frac{0.024}{0.012 + 0.024} = \frac{0.024}{0.036} = \frac{2}{3}$$

Answer: 66.67%

Practice Math Questions

1. In a survey of 500 people, it was found that 50 do not read The Observer and 25 do not read Ittefaq. Moreover, 10 people read neither of the two newspapers. If a person is chosen at random, what is the probability that the person reads Ittefaq but does not read The Observer? *[36th BCS]*

2. **Among 100 students of class eleven in a college, 30 play football, 40 play cricket, and 20 play both football and cricket. If one person is selected at random, what is the probability that the student plays football given that he plays cricket?**

3. **Out of 250 examinees in Mathematics and Statistics, 25 failed in Statistics and 45 failed in Mathematics. 15 students failed in both subjects. If one examinee is selected at random, what is the probability that the examinee passed in Statistics but failed in Mathematics?**

4. A box contains 5 red, 7 blue, and 3 green balls. If 3 balls are drawn at random one after another from the box, what is the probability that the drawn balls will be red, blue, and green in that order:
   (a) if the balls are drawn with replacement?
   (b) if the balls are drawn without replacement?

5. One container contains 2 white and 3 black balls, and another container contains 3 white and 4 black balls. If one ball is drawn from each container, what is the probability that:
   (a) the balls are of the same color?
   (b) the balls are of different colors?

6. **One vessel contains 5 red and 4 white balls, and another vessel contains 3 red and 6 white balls. If one ball is drawn from each vessel, find the probability that at least one of the two balls is red.**

7. A person walked from Gulistan to Gulshan at a speed of 4 km/h. After some time, he returned to Gulistan walking at a speed of 5 km/h. On the return journey, it took him half an hour less. What is the distance from Gulistan to Gulshan? *[10th BCS]*

8. If the length of a rectangle is decreased by 5 meters and the breadth is increased by 3 meters, its area remains unchanged. Again, if the length is increased by 5 meters and the breadth is decreased by 2 meters, its area also remains unchanged. Find the length and breadth of the rectangle. *[Class 9–10 Board Book (General Mathematics - 2002 Edition), Chapter 8.7 (Exercise No. 14)]*

9. The area of a rectangle is 24 square meters. The length and breadth of another rectangle are 4 meters and 1 meter more than the length and breadth of the first rectangle respectively, and its area is 50 square meters. Determine the length and breadth of the first rectangle. *[Class 9–10 Board Book (Higher Mathematics), Chapter 5.5 (Exercise No. 05)]*

10. 5 men earn as much as 9 women, and 6 women earn as much as 10 boys. If a boy earns 12.50 taka per day, how much will 10 men earn daily?

11. Working 8 hours a day, 50 people can complete a piece of work in 12 days. Working how many hours a day can 60 people complete the same work in 16 days? *[Class 9–10 Board Book (General Mathematics - 1993 Edition), Chapter 2 (Exercise No. 24)]*

12. **If 8 men working 7 hours a day can complete a piece of work in 12 days, in how many days can 14 boys finish it working 6 hours a day? It is given that the work done by 1 man = work done by 2 boys.** *[Class 9–10 Board Book (General Mathematics - 1993 Edition), Chapter 2 (Exercise No. 25)]*

13. A cistern has three pipes. The cistern is filled by the 1st and 2nd pipes in 12 and 18 minutes respectively, and emptied by the 3rd pipe. When all three pipes are opened together, the cistern is filled in 36 minutes. In how much time will the full cistern be emptied by the 3rd pipe alone? *[Class 9–10 Board Book (General Mathematics - 1993 Edition), Chapter 2 (Exercise No. 35)]*

14. What will be the difference between the simple interest and compound interest on 5,000 taka for 3 years at an annual interest rate of 10%? *[Class 8 Board Book (General Mathematics), Chapter 2.2 (Exercise No. ...)]*
15. A certain principal amounts to 28,000 taka with simple interest in 3 years, and 30,000 taka with simple interest in 5 years.

(a) Write the formula for finding the principal along with descriptions of the symbols.
(b) Determine the rate of interest.
(c) How much money needs to be deposited in the bank at the same rate so that the principal with interest amounts to 4,800 taka in 5 years? *[Class 8 Board Book (General Mathematics), Chapter 2.1 (Exercise No. ...)]*

16. **16. In a 60-liter mixture of milk and water, the concentration of milk is 70%. How much water must be added to reduce the concentration of milk to 40%?**

17. A chemist has two acid solutions: Solution A contains 20% acid and Solution B contains 50% acid. How many liters of each solution must be mixed to obtain 60 liters of a solution with a 30% acid concentration?

APTITUDE TEST QUESTIONS

Aptitude qs- 01

Programming (File Handling) – Complete Question (15 Marks)

A text file named **students.txt** contains student records in the following format:

101,Rahim,CSE,3.75

102,Karim,EEE,3.20

103,Nila,CSE,3.90

104,Rafi,ICT,2.95

105,Ayesha,CSE,3.65

106,Sumaiya,EEE,3.55

.....

Each record contains: *Student ID ,Student, Name, Department ,CGPA*

Write a C program(any structured) to perform the following tasks:

a) Read all student records from students.txt.
b) Display all student information on the screen.
c) Find all students whose **CGPA is greater than 3.50**.
d) Store the selected students in a new file named **new.txt**.
e) Calculate and display the **average CGPA** of all students.
f) Display the **total number of students** in the file.

Sol^{n}

A text file named **students.txt** contains student records in the following format:

101,Rahim,CSE,3.75

102,Karim,EEE,3.20

103,Nila,CSE,3.90

104,Rafi,ICT,2.95

105,Ayesha,CSE,3.65

106,Sumaiya,EEE,3.55

Each record contains **Student ID, Student Name, Department, and CGPA**. The following C program reads all records from the file, displays all student information, finds students whose CGPA is greater than **3.50**, stores those students in **new.txt**, calculates the average CGPA, and displays the total number of students.

```
#include <stdio.h>
#include <stdlib.h>

struct Student
{
    int id;
    char name[50];
    char dept[20];
    float cgpa;
};

int main()
{
    FILE *fp, *newfp;
    struct Student s;
    float totalCGPA = 0.0, averageCGPA;
    int count = 0;

    fp = fopen("students.txt", "r");

    if$fp == NULL$
    {
        printf("Error opening students.txt\n");
        return 1;
    }

    newfp = fopen("new.txt", "w");

    if$newfp == NULL$
    {
        printf("Error creating new.txt\n");
        fclose(fp);
        return 1;
    }

    printf("Student Information\n");
    printf$"-----------------------------------------\n"$;
    printf("ID\tName\t\tDepartment\tCGPA\n");
```

```
    printf$"----------------------------------------\n"$;

    while$fscanf(fp,"%d,%49[^,],%19[^,],%f",&s.id,s.name,s.dept,&s
.cgpa)==4$
    {
        printf$"%d\t%-10s\t%-10s\t%.2f\n",s.id,s.name,s.dept,s.cgp
a$;

        totalCGPA += s.cgpa;
        count++;

        if$s.cgpa > 3.50$
        {
            fprintf(newfp,"%d,%s,%s,%.2f\n",s.id,s.name,s.dept,s.c
gpa);
        }
    }

    averageCGPA = totalCGPA / count;

    printf$"\nAverage CGPA = %.2f\n",averageCGPA$;
    printf$"Total Students = %d\n",count$;

    fclose(fp);
    fclose(newfp);

    return 0;
}
```

Explanation:

a) The program first defines a structure named Student containing four fields: **id**, **name**, **department**, and **cgpa**.

b) It opens the file **students.txt** in read mode using fopen() and creates another file named **new.txt** in write mode.

c) Using a while loop and fscanf(), it reads each student record until the end of the file. Every record is displayed on the screen using printf().

d) During reading, the program adds each student's CGPA to totalCGPA and increments the student counter.

e) If a student's CGPA is greater than **3.50**, that student's record is written to **new.txt** using fprintf().

f) After all records are processed, the average CGPA is calculated by dividing the total CGPA by the number of students and is displayed along with the total number of students. Finally, both files are closed using fclose().

Sample Output:

```
Student Information
-----------------------------------------------
ID      Name        Department      CGPA
-----------------------------------------------
101     Rahim       CSE             3.75
102     Karim       EEE             3.20
103     Nila        CSE             3.90
104     Rafi        ICT             2.95
105     Ayesha      CSE             3.65
106     Sumaiya     EEE             3.55

Average CGPA = 3.50
Total Students = 6
```

Contents of new.txt:

```
101,Rahim,CSE,3.75
103,Nila,CSE,3.90
105,Ayesha,CSE,3.65
106,Sumaiya,EEE,3.55
```

Aptitude Question- 02

Consider the following two tables:

Department (dept_id, dept_name);
Employee (emp_id, name, dept_id, salary, joining_date);

a) Input some tuples into the tables,